
A Spectral Phase Diagram for Binary Few-Shot Classification: Intrinsic Dimensionality, Geometric Saturation, and Representational Diagnosis

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Abstract

Deciding when to stop collecting labeled examples is a fundamental but undertheorized problem in applied machine learning: annotating beyond the point of saturation wastes budget, yet no principled tool has existed for detecting that point from the support set alone. We show that saturation is encoded in the spectral geometry of the support set in hand. The *saturation index* $\mathcal{S}(K) = \text{erank}(\hat{\Sigma}_W^{(K)})/K$ measures the ratio of the effective rank of the pooled within-class sample covariance to the shot count; we prove that it falls below a threshold precisely when the covariance estimator is well-concentrated around the population covariance and the linear discriminant has stabilized, grounding the predictor in classical high-dimensional estimation theory. The index is computable in $O(d^3)$ time from support features alone, requiring no test labels and no trained classifier.

Evaluated across $N = 246$ doubling-pair observations from seventeen binary classification tasks and six datasets, the saturation index demonstrates strong and consistent *within-task* predictive power: sixteen of seventeen tasks have a positive within-task Spearman correlation between $\mathcal{S}(K)$ and marginal accuracy gain, with a median of $\rho = 0.811$ —establishing that the index is a reliable per-task predictor, not a pooled statistical artifact. The pooled Spearman correlation is $\rho = 0.548$ ($p = 1.1 \times 10^{-20}$, $N = 246$); the gap between this and the within-task median reflects between-task heterogeneity rather than weakness of the within-task signal. A three-phase diagram (exploration, transition, saturation) with mean marginal gains of 3.48%, 2.40%, and 0.82% is supported by all pairwise significance tests ($p \leq 0.008$). Evaluated as a binary stopping rule, the index achieves $\text{AUC} = 0.752$, providing meaningful probabilistic guidance for annotation decisions. We further establish that asymptotic effective rank and peak accuracy show no significant monotone relationship across tasks (Spearman $r_s = 0.380$, $p = 0.133$, $N = 17$)—geometric complexity does not determine task difficulty—and that a small saturation index paired with low accuracy is a diagnostic signature of representational inadequacy, redirecting effort from annotation to representation learning. All results are for binary classification with a fixed linear classifier; extensions to N -way settings and pretrained backbone representations are discussed as future work.

1 Introduction

Collecting and labeling training data is often the most expensive step in deploying a machine learning system. In clinical imaging, a radiologist may spend several minutes labeling a single scan. In industrial inspection, labeling requires specialized domain expertise that is scarce and costly. In ecological monitoring, capturing rare events requires sustained field observation. Across these domains, the fundamental question is the same: given a fixed annotation budget, how many labeled examples are enough?

This question has a well-known qualitative answer. For a fixed classifier architecture, test accuracy $A(K)$ typically rises steeply from small support set sizes K , then decelerates, and eventually saturates at a task-dependent ceiling $A_\infty = \lim_{K \rightarrow \infty} A(K)$. The practical implication is stark: annotating additional examples

beyond the saturation point yields negligible returns on accuracy while incurring full annotation cost. What is missing from the literature is a *quantitative, prospective* tool that tells a practitioner, from the support set already in hand, whether the next example is likely to help.

The gap. Few-shot learning research has invested enormous effort in designing classifiers that generalize from small K (Vinyals et al., 2016; Snell et al., 2017; Finn et al., 2017; Wang et al., 2020; Tian et al., 2020), yet the question of when to stop collecting examples receives comparatively little attention. Active learning (Settles, 2009) addresses a related budget-allocation problem but assumes a pool of unlabeled candidates and focuses on *which* examples to query, not *whether* to query at all. Data pruning and scaling law research (Sorscher et al., 2022) study how to remove redundant data from large corpora, but require the full dataset to be available before pruning and operate at dataset scales far beyond the few-shot regime. The double-descent literature (Belkin et al., 2019; Nakkiran et al., 2020) characterizes accuracy as a function of model capacity and sample size, but the saturation phenomenon we study—accuracy stabilizing at a ceiling well below interpolation—is structurally different from double descent and occurs in the underparameterized regime (Section 3.3).

This work. We address the saturation prediction problem through the spectral geometry of the support set. Our central observation is that when a linear classifier is applied to a support set $\mathcal{S}_K$ of K examples per class, the support set’s *pooled within-class sample covariance* $\hat{\Sigma}_W^{(K)} \in \mathbb{R}^{d \times d}$ undergoes a characteristic spectral evolution as K grows. At small K , the covariance is rank-deficient and its eigenvalue distribution is concentrated: only a few spectral directions are populated. As K increases, new directions become occupied and the effective rank grows. Eventually the covariance converges to the population covariance Σ , after which additional examples provide only noisy re-estimates of already well-determined directions—precisely the geometric condition underlying accuracy saturation.

Scope. This paper develops the saturation index for *binary* (2-way) K -shot classification with a *fixed linear classifier*—the natural setting in which the spectral geometry of the within-class covariance directly governs the quality of the linear discriminant (Section 3.3). We deliberately hold the representation fixed to isolate the geometric effect of increasing K , rather than conflating it with representation learning. This is not an incidental restriction: the binary, fixed-classifier setting is the regime where annotation budget decisions arise most acutely (clinical imaging, industrial inspection, ecological monitoring), and where the theory is cleanest. The pooled within-class covariance generalizes to N -way classification with only notational changes (Section 6.5), and we expect the saturation index to be *more* informative in pretrained backbone representations—where within-class variance is lower and saturation occurs earlier—than in the raw PCA features studied here (Section 6.6). Both extensions are analyzed theoretically and identified as the primary directions for future empirical work.

We formalize this intuition through the *saturation index*:

$$\mathcal{S}(K) = \frac{\text{erank}(\hat{\Sigma}_W^{(K)})}{K}, \quad (1)$$

where $\text{erank}(A) = \exp(H(\mathbf{p}))$ is the effective rank of a positive semidefinite matrix A , defined as the exponential Shannon entropy of its normalized eigenvalue distribution (Roy & Vetterli, 2007). The saturation index $\mathcal{S}(K)$ measures the ratio of the effective dimensionality of the support set’s covariance to the number of examples used to construct it. It is computable from the support features alone in $O(d^3)$ time (approximately 0.3 seconds at $d = 50$), requires no test labels, and requires no trained classifier.

Main results. We evaluate the saturation index across more than 15,000 training runs spanning seventeen binary classification tasks and six datasets (handwritten digits, fashion items, historical script, natural images, and clinical tabular data), and establish the following:

1. **Phase structure, within-task consistency, and predictive correlation.** The relationship between $\mathcal{S}(K)$ and marginal accuracy gain $\Delta A(K) = A(2K) - A(K)$ is well-described by three regimes: an *exploration* phase ($\mathcal{S}(K) > 1.0$, mean $\overline{\Delta A} = 3.48\%$), a *transition* phase ($0.3 < \mathcal{S}(K) \leq$

1.0, mean $\overline{\Delta A} = 2.40\%$), and a *saturation* phase ($\mathcal{S}(K) \leq 0.3$, mean $\overline{\Delta A} = 0.82\%$) (Figure 1). All three pairwise phase comparisons are statistically significant ($p \leq 0.008$, Mann–Whitney). The primary evidence for the predictive claim is the *within-task* signal: sixteen of seventeen tasks have a positive within-task Spearman correlation between $\mathcal{S}(K)$ and $\Delta A(K)$, with a median of $\rho = 0.811$ and interquartile range $[0.720, 0.896]$ (Figure 2). This establishes that the saturation index is a reliable per-task predictor of annotation value—as K increases within a fixed task, a decreasing $\mathcal{S}(K)$ reliably forecasts diminishing marginal returns. The pooled Spearman correlation is $\rho = 0.548$ ($p = 1.1 \times 10^{-20}$, $N = 246$); as discussed in Section 6.2, the gap between pooled and within-task estimates reflects between-task heterogeneity rather than weakness of the core within-task signal.

2. **Decoupling of task difficulty and intrinsic dimensionality.** The asymptotic effective rank erank_∞ and peak accuracy A_∞ show *no significant monotone relationship* across the seventeen tasks we study (Spearman $r_s = 0.380$, $p = 0.133$, $N = 17$; Figure 4). The most striking demonstration: USPS 1 vs. 2 ($\text{erank}_\infty = 15.08$, $A_\infty = 0.997$) and CIFAR-10 cat vs. dog ($\text{erank}_\infty = 17.18$, $A_\infty = 0.597$) have nearly identical intrinsic dimensionality yet differ by 40.1 percentage points in accuracy—a gap that cannot be explained by spectral complexity alone. Conversely, MNIST 3 vs. 8 ($\text{erank}_\infty = 37.45$, $A_\infty = 0.965$) is geometrically far more complex than Fashion-MNIST Pullover vs. Shirt ($\text{erank}_\infty = 14.46$, $A_\infty = 0.835$), yet achieves substantially higher accuracy. These individual demonstrations, grounded in the theory that Bayes error depends on Mahalanobis distance while erank_∞ depends on spectral volume (Section 3.3), challenge the common assumption that task difficulty is a monotone function of geometric complexity. The practical implication is direct: erank_∞ determines *annotation cost* (how large K must be), not *accuracy ceiling* (what the classifier will achieve).
3. **Representational diagnosis and asymptotic redundancy.** The saturation index provides a novel diagnostic capability beyond stopping-rule guidance: when $\mathcal{S}(K)$ is small but A_∞ is low, the bottleneck is *representational inadequacy* rather than data insufficiency. CIFAR-10 cat vs. dog, which saturates at 60% accuracy in 50-dimensional PCA space, provides the sharpest illustration—the index correctly identifies that the current feature representation has been exhausted, directing practitioners toward better representations rather than more labels (Section 6.1). In the deep-saturation sub-regime ($\mathcal{S}(K) \ll 0.02$, i.e., $K \gg \text{erank}_\infty$), accuracy fluctuates within a narrow noise floor rather than continuing to improve, confirming that geometric redundancy—not model capacity or interpolation dynamics—is the operative mechanism.
4. **Formal evaluation of the decision rule.** Algorithm 1, which issues a STOP recommendation when $\mathcal{S}(K) \leq \tau = 0.3$, is evaluated as a binary classifier over the $N = 246$ observations. It achieves $\text{AUC} = 0.752$ —substantially above chance and the naïve baselines, providing meaningful probabilistic guidance for annotation decisions—with recall of 0.605 at the calibrated default threshold ($\tau = 0.3$) and recall of 0.763 at the high-sensitivity threshold ($\tau = 0.5$) (Figure 5). The gap from a perfect classifier is characterized in Section 6.4: the dominant source of false positives is representation-limited tasks (CIFAR), which the joint diagnostic of Section 6.1 correctly handles; the remaining errors are stochastic variance near the phase boundary. We recommend interpreting the saturation index as probabilistic annotation guidance rather than a hard guarantee.

Taken together, these results provide a principled, label-free answer to the annotation budget question for binary classification tasks with a fixed linear classifier: compute $\mathcal{S}(K)$ from the current support set, consult Algorithm 1, and continue collecting examples only if $\mathcal{S}(K) > \tau$. When $\mathcal{S}(K)$ falls below τ and accuracy is sufficient, additional annotation is expected to yield negligible returns; when $\mathcal{S}(K)$ is small but accuracy is low, the index correctly redirects effort from data collection to representation improvement.

Theoretical motivation. Under homoscedastic Gaussian class-conditionals, the accuracy of a linear discriminant is governed by the operator-norm error of the pooled within-class sample covariance, which scales as $O(\|\Sigma\|_{\text{op}} \sqrt{r(\Sigma)/K})$ where $r(\Sigma) = \text{tr}(\Sigma)/\|\Sigma\|_{\text{op}}$ is the intrinsic dimension of the population covariance (Vershynin, 2018; Rudelson & Vershynin, 2010). The saturation index $\mathcal{S}(K)$ falls below a threshold precisely when $K \gg \text{erank}(\Sigma)$ —when the sample covariance has enough examples to estimate the discriminant

direction accurately (Proposition 1, Corollary 1.1). While this analysis assumes Gaussian marginals (Assumption 1), the strong empirical correlation holds across non-Gaussian datasets including Fashion-MNIST and clinical tabular data, consistent with the sub-Gaussian extension of the bound (Vershynin, 2018).

Relation to intrinsic dimensionality. Recent work has established that natural image datasets have low intrinsic dimension relative to their ambient pixel dimension, and that lower intrinsic dimension is associated with more efficient learning (Pope et al., 2021). Our work operates in a complementary regime: rather than estimating the global intrinsic dimension of a dataset manifold, we track the *shot-by-shot spectral evolution* of the support set covariance within a specific binary classification task. The effective rank $\text{erank}(\hat{\Sigma}_W^{(K)})$ we compute is a local, task-specific, and finite-sample quantity that changes with K ; it is structurally different from global manifold dimension estimators such as TWO-NN (Pope et al., 2021).

Relation to few-shot learning. The few-shot learning literature focuses on learning representations or meta-learning algorithms that generalize from small K (Snell et al., 2017; Tian et al., 2020). Our work is orthogonal: we take the representation and the classifier as fixed, and ask how the *geometry* of the support set evolves with K under that representation. The saturation index is therefore a complement to few-shot learning methods, applicable as a stopping criterion after the representation has been fixed—whether that representation comes from a prototypical network backbone, a MAML-trained feature extractor, a contrastive pretraining model such as CLIP or DINO, or raw pixel PCA as in our experiments. Indeed, we expect the saturation index to be *most* informative in high-quality pretrained representations: lower within-class variance means erank_∞ is smaller, saturation occurs at smaller K , and the stopping signal arrives earlier and more cleanly (Section 6.6).

Contributions.

- We introduce the *saturation index* $\mathcal{S}(K) = \text{erank}(\hat{\Sigma}_W^{(K)})/K$, a closed-form, label-free, and classifier-free predictor of accuracy saturation in *binary* few-shot classification, with a formal theoretical motivation under Gaussian class-conditionals connecting it to classical results in high-dimensional covariance estimation (Proposition 1, Corollary 1.1, Section 3.3).
- We establish a three-phase diagram across seventeen binary tasks and six datasets. The primary evidence is the *within-task* signal: sixteen of seventeen tasks have a positive within-task Spearman correlation between $\mathcal{S}(K)$ and marginal accuracy gain, with median $\rho = 0.811$ (IQR $[0.720, 0.896]$)—establishing that the saturation–marginal-gain relationship is consistent within individual tasks and not an artifact of pooling. The pooled Spearman correlation is $\rho = 0.548$ ($p = 1.1 \times 10^{-20}$, $N = 246$); all three pairwise phase comparisons are significant ($p \leq 0.008$) (Section 4.1).
- We demonstrate that erank_∞ and A_∞ show no significant monotone relationship across $N = 17$ tasks (Spearman $r_s = 0.380$, $p = 0.133$), with individual task pairs spanning 40 percentage points of accuracy at near-identical erank_∞ —separating annotation cost from accuracy ceiling as distinct design concerns (Section 4.2).
- We show that the saturation index functions as a *representational adequacy diagnostic*: when $\mathcal{S}(K)$ is small but accuracy is low, the bottleneck is the feature representation rather than the annotation budget, correctly redirecting effort from data collection to representation improvement (Section 4.4).
- We provide the first formal evaluation of Algorithm 1 as a binary stopping rule over $N = 246$ observations (AUC = 0.752), characterize its two structural error sources, and provide a cost-sensitive threshold selection criterion (Section 4.5).
- We confirm robustness through multi-task ablation studies covering PCA dimension ($d \in \{5, 10, 20, 50, 100\}$ across three tasks), regularization strength ($C \in \{\infty, 1.0, 0.1\}$ across three tasks), and classifier family (logistic regression vs. nearest centroid; Section 5).

Code, data, and cached experimental results are available at <https://github.com/MrArnav69/spectral-saturation>.

Paper organization. Section 2 situates the saturation index in the context of few-shot learning, active learning, intrinsic dimensionality, and double descent. Section 3 develops the formal framework. Section 4 presents the experimental results. Section 5 reports ablation studies. Section 6 discusses limitations and future work.

2 Related Work

Our work sits at the intersection of five bodies of literature: few-shot learning, the geometry of support sets, intrinsic dimensionality of data representations, learning curves and sample complexity, and high-dimensional linear classification. We position ourselves in relation to each body in turn, with particular attention to what distinguishes our contribution from each.

2.1 Few-Shot Learning: Algorithms and Representations

The dominant paradigm in few-shot learning is to design a classifier or representation that generalizes from small support sets to novel classes. This line of work divides into three main families.

Metric-based methods learn an embedding space in which classification reduces to computing distances to class prototypes. Vinyals et al. (2016) introduced Matching Networks, which classify query examples by an attention-weighted sum over support set labels in the embedding space. Snell et al. (2017) showed that a simpler inductive bias—computing a per-class mean embedding and classifying by Euclidean distance to the nearest prototype—yields competitive or superior accuracy, and further established a connection between prototypical networks and mixture density estimation. Lee et al. (2019) extended this line by replacing nearest-centroid classification with a differentiable convex solver (a linear SVM), obtaining better generalization at the cost of a more complex base learner.

Optimization-based methods meta-learn an initialization or update rule that facilitates rapid fine-tuning on a new task. Ravi & Larochelle (2017) proposed an LSTM-based meta-learner that learns the optimization algorithm itself, treating the few-shot update as a learned recurrence. Finn et al. (2017) introduced MAML, which meta-trains a model initialization such that a small number of gradient steps on any new task yields good generalization. MAML is model-agnostic and has spawned a large literature of gradient-based meta-learning variants (Hospedales et al., 2021).

Embedding quality has emerged as a unifying theme. Tian et al. (2020) showed that a simple baseline—learning a supervised or self-supervised representation on the meta-training set, then training a *fixed* linear classifier on top of it—outperforms many sophisticated meta-learning algorithms. This result shifted attention from the algorithm to the representation, and motivated a growing body of work asking what makes a representation well-suited to few-shot transfer.

Our distinction. All of the above work asks: *how should we design the classifier or representation?* Our work asks an orthogonal question: *given a fixed representation and classifier, when should we stop collecting support examples?* The saturation index $\mathcal{S}(K)$ is applicable *after* the representation is fixed, as a stopping criterion, regardless of whether that representation was learned by a prototypical network, a MAML-trained backbone, or raw pixel PCA. We therefore complement rather than compete with the few-shot learning algorithm literature.

2.2 Support Set Quality and Composition

A smaller body of work focuses specifically on the quality and selection of support set examples, as distinct from the classifier design. Xu & Le (2022) proposed augmenting the support set with synthetically generated representative samples drawn from a CVAE trained on base classes, addressing the observation that few-shot class representations are often biased due to data scarcity. Ye et al. (2024) introduced Single-Step Support Set Mining (S4M), a strategy that selects which examples to annotate before observing their labels, motivated by the practical constraint that forming a class-balanced support set requires non-trivial annotation effort. The common thread in this work is the question of *which* examples to include in the support set; our question

is distinct and complementary: *how many* examples does a task’s geometry require before the support set is saturated?

We are not aware of prior work that uses the spectral geometry of the existing support set as a stopping criterion for further annotation, without access to test labels or a trained classifier. The closest precedent in spirit is Hacoen et al. (2022), who study the relationship between annotation budget and optimal query strategy in active learning, finding a phase transition between low-budget and high-budget regimes. However, their work requires a trained model and a pool of unlabeled candidates at each query step, whereas $\mathcal{S}(K)$ operates entirely on the existing support set and requires neither.

2.3 Intrinsic Dimensionality of Data Representations

The idea that high-dimensional data lies on or near a low-dimensional manifold is a central structural assumption in statistical learning theory and machine learning practice. Pope et al. (2021) applied manifold dimension estimation to standard image benchmarks and showed that common datasets have very low intrinsic dimension relative to their ambient pixel dimension, and that lower intrinsic dimension is associated with faster learning and better generalization. Ansuini et al. (2019) applied similar tools to the layer-wise representations of trained deep networks, finding that the intrinsic dimensionality of representations first increases and then decreases across layers, and that the intrinsic dimension of the penultimate layer predicts test accuracy. Nakada & Imaizumi (2020) provided theoretical grounding for these empirical observations, proving that the generalization error of deep networks depends on the intrinsic dimension of the covariate distribution rather than on the nominal ambient dimension. Konz & Mazurowski (2024) extended the analysis to compare natural and medical imaging domains, establishing a generalization scaling law with respect to intrinsic data dimension.

Our distinction. Each of these works measures the intrinsic dimension of a *fixed* dataset or representation—a static quantity computed once from all available data. The effective rank we compute, $\text{erank}(\hat{\Sigma}_W^{(K)})$, is a *dynamic*, *shot-indexed*, and *task-local* quantity: it tracks how the spectral geometry of the support set’s within-class covariance evolves as additional labeled examples arrive. We are interested in the *trajectory* of this quantity, not its asymptotic value, because the trajectory reveals when the support set has fully explored the geometrically accessible directions of the class-conditional distribution—the saturation condition. Furthermore, where previous work uses global manifold estimators (e.g., TWO-NN (Pope et al., 2021)) applied to the full data distribution, we apply the Roy–Vetterli effective rank (Roy & Vetterli, 2007) to the *pooled within-class sample covariance* of the support set, which is the natural spectral object for the linear discrimination problem (Section 3.3).

2.4 Learning Curves and Sample Complexity

The relationship between training set size and classifier performance—the learning curve—has been studied since at least the early 1990s. Viering & Loog (2023) provide a comprehensive review of the shapes learning curves can take, cataloging empirical evidence for power-law and exponential scaling, as well as conditions under which learning curves are ill-behaved (non-monotone or plateau-and-fluctuate). Their review underscores that there is no universal functional form for $A(K)$, and that both the rate of improvement and the saturation level are task-dependent—a finding consistent with our empirical observations across seventeen binary tasks spanning different erank_∞ values.

On the applied side, Figueroa et al. (2012) proposed fitting an inverse power-law model to the early portion of a learning curve and extrapolating to predict the sample size required to achieve a target accuracy, validated on clinical text and waveform classification tasks. This approach requires training the classifier at multiple values of K and observing the resulting accuracy curve, then fitting a parametric model to extrapolate.

Our distinction. Existing learning curve methods require either (i) a labeled test set to evaluate accuracy at each K (Figueroa et al., 2012), or (ii) post-hoc analysis of a completed training run (Viering & Loog, 2023). Neither provides a prospective, label-free prediction at deployment time. The saturation index $\mathcal{S}(K)$, by contrast, is computable from the support features alone and can be evaluated *before* the next

labeling decision is made. It predicts the marginal accuracy gain $\Delta A(K)$ with pooled Spearman $\rho = 0.548$ ($p = 1.1 \times 10^{-20}$, $N = 246$) and within-task median $\rho = 0.811$ across sixteen of seventeen tasks, without requiring any classifier training or test-set evaluation—precisely the regime where label-efficient practitioners need a stopping criterion most. In this sense, the saturation index complements learning curve extrapolation: it provides an early, cheap, label-free signal, while learning curve fitting provides a more accurate but more expensive estimate once sufficient data have been observed.

2.5 High-Dimensional Covariance Estimation and Linear Classification

The behavior of linear classifiers in high-dimensional, small-sample regimes is a classical problem in statistical learning. Fisher (1936) established that the optimal linear discriminant between two Gaussian classes with shared covariance is the direction $w^* = \Sigma^{-1}(\mu_1 - \mu_0)$, requiring estimation of the pooled within-class covariance Σ_W . Friedman (1989) showed that in high-dimensional settings where the sample covariance is ill-conditioned (i.e., the number of features d is comparable to the number of training examples K), regularization of the covariance estimate can yield dramatic improvements in classification accuracy. His regularized discriminant analysis (RDA) introduced a two-parameter family of shrinkage estimators, selecting the optimal parameters by minimizing a cross-validated estimate of future misclassification risk. Sharma & Paliwal (2015) surveyed the literature on LDA variants designed for the small-sample-size ($K \ll d$) problem, identifying three main strategies: diagonal approximations, regularization, and subspace methods.

The high-dimensional statistics literature provides sharp quantitative results on covariance estimation accuracy. Vershynin (2018) established that the operator-norm estimation error of the sample covariance scales as $O(\|\Sigma\|_{\text{op}} \sqrt{r(\Sigma)/K})$, where $r(\Sigma) = \text{tr}(\Sigma)/\|\Sigma\|_{\text{op}}$ is the intrinsic dimension of Σ . This bound—rather than the ambient dimension d —governs the rate of convergence, which is the theoretical foundation for our saturation index (Proposition 1).

Our distinction. Prior work on high-dimensional LDA uses covariance estimation theory to *improve classification accuracy* through better regularization. We use the same theory for a different purpose: to *predict when the covariance estimator has converged* to a level sufficient for accurate discrimination, so that additional examples provide negligible marginal benefit. The saturation index is thus a diagnostic tool derived from covariance estimation theory, not a new classification algorithm. We deliberately use *unregularized* logistic regression in our experiments (rather than RDA or regularized LDA) to isolate the geometric effect of support set size from the confounding effect of regularization bias, and show in our multi-task ablation study that the saturation phase structure is preserved under moderate regularization for image tasks ($\Delta_{\text{max}} < 0.002$ across $C \in \{\infty, 1.0, 0.1\}$); for low-dimensional tabular tasks at small saturation shot counts, regularization affects the absolute accuracy ceiling but not the location of the saturation threshold in the K -axis (Section 5.2).

2.6 Data Efficiency: Scaling Laws, Pruning, and Active Learning

A growing body of work studies how to use labeled data more efficiently across different operational regimes.

Scaling law research has established that, for large-scale models, test error decreases as a power law in training set size, model parameters, and compute (Sorscher et al., 2022). Sorscher et al. (2022) showed theoretically and empirically that high-quality data pruning metrics can break this power-law ceiling and achieve exponential scaling with dataset size, and benchmarked ten pruning metrics on ImageNet. Their work operates in a setting where a large pool of labeled data is already available and the question is which subset to retain; our setting is the opposite—no pool exists, and the question is whether to annotate the next example.

Active learning (Settles, 2009) addresses the related problem of selecting which unlabeled examples from a candidate pool to query. Settles (2009) surveys query strategies for pool-based active learning across classification, regression, and clustering tasks. Hachohen et al. (2022) study how the optimal querying strategy changes with the annotation budget, finding empirically that typical examples should be queried at low budget while unrepresentative examples become valuable at high budget—a phase-transition-like finding that shares structural similarities with the three-phase diagram we describe. Despite this structural resonance,

active learning methods require a candidate pool and a trained classifier to score it, whereas $\mathcal{S}(K)$ is evaluated on the support set already collected and requires neither.

Double descent and non-monotone data effects. Nakkiran et al. (2020) showed that, as a function of model size or training set size, test error can exhibit a second descent after passing through the interpolation threshold where model capacity matches training set size. Nakkiran (2019) isolated a simpler version of this phenomenon in overparameterized linear regression, showing that test risk can increase with additional samples in the overparameterized regime due to an unconventional bias-variance tradeoff. Both phenomena are structurally distinct from the saturation we study: our linear classifier is underparameterized (with $d = 50$ features and $K \gg 25$ examples in the saturation regime, the model is far from the interpolation threshold), and the accuracy plateau we observe arises from geometric redundancy in the support set rather than from model capacity or interpolation dynamics. We discuss this distinction formally in Remark 3.

3 Methodology

We develop the methodology in six parts. Section 3.1 formalizes the binary K -shot classification protocol and the quantity we seek to predict. Section 3.2 introduces the spectral geometry of the support set, constructing the pooled within-class covariance and deriving the effective rank and saturation index. Section 3.3 provides a formal theoretical motivation, grounding the saturation index in established results from high-dimensional covariance estimation theory. Section 3.4 instantiates these ideas as a concrete annotation-budget decision algorithm. Sections 3.5 and 3.7 specify the full experimental configuration: the classifier, the preprocessing pipeline, and the sampling and evaluation protocol.

3.1 Problem Formulation

Let $\mathcal{X} \subseteq \mathbb{R}^{d_0}$ denote the input space, where d_0 is the original feature dimension (e.g. $d_0 = 784$ for MNIST, $d_0 = 3,072$ for CIFAR-10 RGB), and let $\mathcal{Y} = \{0, 1\}$ be the binary label space. We study the standard 2-way K -shot classification protocol. For each class $c \in \{0, 1\}$, let $\mathcal{D}_c$ denote the marginal distribution of inputs from class c .

Definition 1 (Support set). A *support set* $\mathcal{S}_K$ is a collection of $2K$ labeled examples constructed by drawing K examples independently and uniformly from each class:

$$\mathcal{S}_K = \{(x_1^{(0)}, 0), \dots, (x_K^{(0)}, 0), (x_1^{(1)}, 1), \dots, (x_K^{(1)}, 1)\}, \quad (2)$$

where $x_i^{(c)} \stackrel{\text{i.i.d.}}{\sim} \mathcal{D}_c$ for each class c and trial index i .

A learner observes $\mathcal{S}_K$ and trains a binary classifier $f_\theta: \mathbb{R}^d \rightarrow \{0, 1\}$ using *only* the labeled support set. The classifier is evaluated on a held-out test set $\mathcal{T}$, drawn from the same marginals $\mathcal{D}_0$ and $\mathcal{D}_1$ but strictly disjoint from $\mathcal{S}_K$. We measure performance via the empirical accuracy

$$A(K) = \frac{1}{|\mathcal{T}|} \sum_{(x,y) \in \mathcal{T}} \mathbf{1}[f_\theta(x) = y], \quad (3)$$

where the dependence on the random draw of $\mathcal{S}_K$ is suppressed in the notation; all reported values are averages over T independent trials (Section 3.7).

Empirical behavior of $A(K)$. In practice, $A(K)$ follows a characteristic pattern across a wide range of binary classification tasks: rapid improvement for small K , followed by diminishing returns, and eventual saturation at a task-specific ceiling $A_\infty = \lim_{K \rightarrow \infty} A(K)$. The rate and location of saturation vary by task, but the qualitative shape is consistent. This motivates our central question.

Can the saturation of $A(K)$ be predicted from the geometry of the support set $\mathcal{S}_K$ alone—without training the classifier, and without access to test labels?

We answer affirmatively by constructing a scalar quantity—the *saturation index* $\mathcal{S}(K)$ —that is computable from $\mathcal{S}_K$ in closed form and correlates strongly with the marginal accuracy gain $\Delta A(K) = A(K) - A(K/2)$ across tasks and shot counts (Spearman $\rho = 0.548$, $p = 1.1 \times 10^{-20}$, $N = 246$ doubling-pair observations across seventeen tasks; Section 4.1).

3.2 Spectral Geometry of the Support Set

3.2.1 The Pooled Within-Class Covariance

Given a support set $\mathcal{S}_K$, let $d \leq d_0$ denote the working dimension after any preprocessing (Section 3.6). For each class $c \in \{0, 1\}$, define the *sample class mean* and *sample within-class covariance matrix*:

$$\hat{\mu}_c = \frac{1}{K} \sum_{i=1}^K x_i^{(c)}, \quad (4)$$

$$\hat{\Sigma}_c = \frac{1}{K-1} \sum_{i=1}^K (x_i^{(c)} - \hat{\mu}_c)(x_i^{(c)} - \hat{\mu}_c)^\top, \quad (5)$$

where the Bessel-corrected denominator $(K-1)$ yields an unbiased estimate of the class-conditional covariance.

Definition 2 (Pooled within-class covariance). The *pooled within-class covariance* of the support set $\mathcal{S}_K$ is

$$\hat{\Sigma}_W^{(K)} = \frac{1}{2}(\hat{\Sigma}_0 + \hat{\Sigma}_1). \quad (6)$$

Motivation from Fisher’s Linear Discriminant Analysis. The pooled estimator equation 6 is the maximum-likelihood estimator of the shared covariance Σ under the homoscedastic Gaussian model $\mathcal{D}_c = \mathcal{N}(\mu_c, \Sigma)$, which is the same model underlying Fisher’s Linear Discriminant Analysis (Fisher, 1936; Hastie et al., 2009). Under this model, the Bayes-optimal decision boundary is a linear hyperplane with normal vector $w^* = \Sigma^{-1}(\mu_1 - \mu_0)$ (Hastie et al., 2009, Chapter 4). The accuracy of the linear classifier is thus governed, at finite K , by how well $\hat{\Sigma}_W^{(K)}$ estimates Σ .

We emphasize that we compute $\hat{\Sigma}_W^{(K)}$ from the within-class deviations equation 5—not from the total covariance—because it is the within-class structure that determines the difficulty of the linear discrimination problem. The between-class covariance, $(\hat{\mu}_1 - \hat{\mu}_0)(\hat{\mu}_1 - \hat{\mu}_0)^\top/2$, encodes the class separation but is rank-1 in the binary case and thus uninformative about the intrinsic dimensionality of the support distribution.

Rank of the estimator. Because $\hat{\Sigma}_c$ is constructed from K centered vectors in $\mathbb{R}^d$, it has rank at most $K-1$. Consequently, $\hat{\Sigma}_W^{(K)}$ has rank at most $2(K-1)$. For $K \ll d$ (the few-shot regime), $\hat{\Sigma}_W^{(K)}$ is rank-deficient: only $2(K-1)$ directions in $\mathbb{R}^d$ carry variance. The effective rank, introduced next, measures the number of these directions that are *meaningfully* occupied.

3.2.2 Effective Rank

Let $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d \geq 0$ denote the eigenvalues of $\hat{\Sigma}_W^{(K)}$. Define the normalized eigenvalue distribution

$$p_i = \frac{\lambda_i}{\sum_{j=1}^d \lambda_j}, \quad i = 1, \dots, d, \quad (7)$$

where we adopt the convention $0 \log 0 = 0$ to handle zero eigenvalues. The vector $\mathbf{p} = (p_1, \dots, p_d)$ is a probability distribution over the d spectral directions; it concentrates on the leading directions for large K and spreads out (less concentrated) for small K .

Definition 3 (Effective rank, Roy & Vetterli 2007). The *effective rank* of a positive semidefinite matrix $A \in \mathbb{R}^{d \times d}$ with eigenvalues $\lambda_1 \geq \dots \geq \lambda_d \geq 0$ is

$$\text{erank}(A) = \exp\left(-\sum_{i=1}^d p_i \log p_i\right) = \exp(H(\mathbf{p})), \quad (8)$$

where $p_i = \lambda_i / \sum_j \lambda_j$ and $H(\mathbf{p})$ is the Shannon entropy (in nats) of the normalized eigenvalue distribution.

The effective rank is the exponential of the spectral entropy of A . It measures the geometric mean of the number of eigenvalues required to explain the total variance of A , weighted by their relative magnitudes.

Properties. The effective rank satisfies the following:

1. $1 \leq \text{erank}(A) \leq \text{rank}(A) \leq d$, with $\text{erank}(A) = 1$ if and only if A has rank 1 (all variance on a single axis), and $\text{erank}(A) = \text{rank}(A)$ if and only if all nonzero eigenvalues are equal (isotropic within the active subspace).
2. $\text{erank}(A)$ is invariant to rescaling: $\text{erank}(\alpha A) = \text{erank}(A)$ for any $\alpha > 0$.
3. $\text{erank}(A)$ is a continuous, concave function of the eigenvalues $(\lambda_1, \dots, \lambda_d)$, and is strictly increasing in $H(\mathbf{p})$.
4. For a sample covariance $\hat{\Sigma}_W^{(K)}$ of rank $r \leq 2(K-1)$, the effective rank satisfies $1 \leq \text{erank}(\hat{\Sigma}_W^{(K)}) \leq r$, reflecting the number of *occupied* directions in the support set rather than the full ambient dimension d .

Interpretation. As K increases, new support examples populate previously unoccupied spectral directions of $\hat{\Sigma}_W^{(K)}$. At small K , the eigenvalue distribution is steep (concentrated on a few large eigenvalues), yielding low erank . As $K \rightarrow \infty$, the sample covariance converges to the population covariance Σ , and $\text{erank}(\hat{\Sigma}_W^{(K)}) \rightarrow \text{erank}(\Sigma)$, which we call the *asymptotic intrinsic dimensionality* erank_∞ of the task. This asymptotic convergence is a defining feature of all tasks studied in Section 4.

The effective rank is preferable to simpler dimension measures (e.g. participation ratio, stable rank $\text{tr}(\Sigma)/\|\Sigma\|_{\text{op}}$, or the number of eigenvalues above a fixed threshold) for the following reasons: (i) it is scale-invariant by Property 2, so it is unaffected by the overall variance level of the data; (ii) as the exponential Shannon entropy, it is smooth and sensitive to the *distribution* of eigenvalues rather than only their count or maximum; (iii) it has a natural probabilistic interpretation as the geometric mean number of active dimensions.

3.2.3 The Saturation Index

Definition 4 (Saturation index). The *saturation index* at shot count K is

$$\mathcal{S}(K) = \frac{\text{erank}(\hat{\Sigma}_W^{(K)})}{K}. \quad (9)$$

The saturation index has a direct geometric interpretation: it is the ratio of the number of spectral directions actively occupied by the support set to the number of examples used to construct it. When $\mathcal{S}(K) > 1$, the support examples are “discovering” new spectral directions faster than they are redundantly sampling existing ones: each new example contributes a new dimension. When $\mathcal{S}(K) \approx 1$, the per-example spectral contribution is roughly one new direction. When $\mathcal{S}(K) \ll 1$, additional examples are largely redundant: the support set has already explored the accessible geometry of the class-conditional distribution, and new examples contribute noise to directions already well-estimated.

Key properties.

1. $\mathcal{S}(K) > 0$ for all $K \geq 2$.
2. For a fixed task, $\mathcal{S}(K)$ is eventually decreasing in K : since $\text{erank}(\hat{\Sigma}_W^{(K)})$ is bounded above by erank_∞ while K grows without bound, $\mathcal{S}(K) \rightarrow 0$ as $K \rightarrow \infty$.

3. $\mathcal{S}(K)$ is computable from $\mathcal{S}_K$ alone in $O(d^2K + d^3)$ time: $O(d^2K)$ for the covariance computation and $O(d^3)$ for the eigendecomposition. In our $d = 50$ setting, this corresponds to approximately 0.3 seconds on commodity hardware (Section 3.7).
4. $\mathcal{S}(K)$ requires *no test labels* and *no trained classifier*: it is a pure function of the support set geometry.

The three-phase structure. Our experiments (Section 4.1) reveal that the relationship between $\mathcal{S}(K)$ and marginal accuracy gain $\Delta A(K) = A(K) - A(K/2)$ is well-described by three regimes, calibrated on aggregated data from all ten binary tasks:

$$\text{Phase} = \begin{cases} \text{Exploration} & \mathcal{S}(K) > 1.0, \\ \text{Transition} & 0.3 < \mathcal{S}(K) \leq 1.0, \\ \text{Saturation} & \mathcal{S}(K) \leq 0.3. \end{cases} \quad (10)$$

These thresholds and the associated mean marginal gains ($\overline{\Delta A} = 3.48\%, 2.40\%, 0.82\%$ respectively, based on $N = 246$ observations across seventeen tasks; Table 3) are discussed in detail in Section 4.1. We note that the threshold $\mathcal{S}(K) = 1$ has a natural theoretical interpretation: it marks the boundary where the number of support examples equals the effective rank of the support covariance, i.e., where $K = \text{erank}(\hat{\Sigma}_W^{(K)})$. Below this boundary, the covariance estimator is severely under-determined relative to the geometry it must capture. The threshold $\mathcal{S}(K) = 0.3$ is calibrated empirically and represents the point beyond which marginal accuracy gains fall below 0.5% (less than the test-set noise floor).

3.3 Theoretical Motivation

We now provide a formal theoretical motivation for the saturation index under a Gaussian class-conditional model. This analysis makes precise the intuition that the convergence of the linear discriminant is governed by the effective dimensionality of the within-class covariance, and it provides a principled justification for the use of $\mathcal{S}(K)$ as a saturation predictor. We state our assumptions explicitly and emphasize that the Gaussian model is an idealization; the empirical validity of the saturation index on non-Gaussian data is established experimentally in Section 4.

Assumption 1 (Homoscedastic Gaussian class-conditionals). The class-conditional distributions satisfy $\mathcal{D}_c = \mathcal{N}(\mu_c, \Sigma)$ for $c \in \{0, 1\}$, where $\mu_0, \mu_1 \in \mathbb{R}^d$ are distinct class means and $\Sigma \in \mathbb{R}^{d \times d}$ is a shared, strictly positive definite covariance matrix ($\Sigma \succ 0$). The two classes are equally probable: $\Pr(Y = 0) = \Pr(Y = 1) = 1/2$.

The Bayes-optimal linear discriminant. Under Assumption 1, the Bayes-optimal classifier is the linear discriminant

$$f^*(x) = \text{sign}(w^{*\top}(x - m)), \quad w^* = \Sigma^{-1}(\mu_1 - \mu_0), \quad (11)$$

where $m = (\mu_0 + \mu_1)/2$ is the midpoint of the class means (Fisher, 1936; Hastie et al., 2009). The Bayes error depends on the Mahalanobis distance between the classes:

$$\varepsilon^* = \Phi\left(-\frac{\Delta}{2}\right), \quad \Delta^2 = (\mu_1 - \mu_0)^\top \Sigma^{-1}(\mu_1 - \mu_0), \quad (12)$$

where Φ is the standard normal CDF. The quantity Δ^2 is the squared Mahalanobis distance between μ_0 and μ_1 and is independent of the within-class geometry beyond the covariance Σ .

The sample discriminant. In practice, Σ , μ_0 , and μ_1 are unknown and must be estimated from $\mathcal{S}_K$. Plugging the estimates equation 4–equation 6 into the optimal direction yields the *sample discriminant*:

$$\hat{w}_K = (\hat{\Sigma}_W^{(K)})^{-1}(\hat{\mu}_1 - \hat{\mu}_0), \quad (13)$$

which is the Fisher Linear Discriminant Analysis (LDA) solution. The accuracy of any linear classifier trained on $\mathcal{S}_K$ is bounded by the quality with which $\hat{w}_K$ approximates w^* , which is in turn governed by the estimation error of $\hat{\Sigma}_W^{(K)}$.

Covariance estimation error. Bounding the operator-norm error of the sample covariance is a classical problem in high-dimensional statistics. The following proposition combines results from Vershynin (2018) and Rudelson & Vershynin (2010) to give a sample-complexity bound in terms of the intrinsic dimension of Σ .

Proposition 1 (Covariance estimation under sub-Gaussian distributions). *Let Assumption 1 hold, and let $\hat{\Sigma}_W^{(K)}$ denote the pooled within-class sample covariance equation 6 constructed from K i.i.d. samples per class. Suppose the class-conditional distributions are sub-Gaussian with parameter σ_ψ^2 , so that $\|\langle X - \mu_c, v \rangle\|_{\psi_2} \leq \sigma_\psi \|v\|_2$ for all unit vectors $v \in \mathbb{R}^d$. Then there exists a universal constant $C > 0$ such that, for all $K \geq 1$,*

$$\mathbb{E}[\|\hat{\Sigma}_W^{(K)} - \Sigma\|_{\text{op}}] \leq C \sigma_\psi^2 \|\Sigma\|_{\text{op}} \sqrt{\frac{r(\Sigma)}{K}}, \quad (14)$$

where $\|\cdot\|_{\text{op}}$ is the operator (spectral) norm and

$$r(\Sigma) = \frac{\text{tr}(\Sigma)}{\|\Sigma\|_{\text{op}}} = \frac{\sum_{i=1}^d \lambda_i}{\lambda_1} \quad (15)$$

is the intrinsic dimension (also called the stable rank or trace-to-spectral ratio) of Σ . Gaussian distributions satisfy the sub-Gaussian condition with $\sigma_\psi^2 = O(1)$ (a universal constant absorbed into C).

Proof sketch. From each class c , center the samples: let $\tilde{x}_i^{(c)} = x_i^{(c)} - \mu_c$. The per-class sample covariance $\hat{\Sigma}_c$ can be written as the sample second moment of zero-mean sub-Gaussian vectors. By Vershynin (2018, Theorem 5.39), for a single class: $\mathbb{E}[\|\hat{\Sigma}_c - \Sigma\|_{\text{op}}] \leq C \sigma_\psi^2 \|\Sigma\|_{\text{op}} \sqrt{r(\Sigma)/K}$. The pooled estimator satisfies $\|\hat{\Sigma}_W^{(K)} - \Sigma\|_{\text{op}} \leq \frac{1}{2}(\|\hat{\Sigma}_0 - \Sigma\|_{\text{op}} + \|\hat{\Sigma}_1 - \Sigma\|_{\text{op}})$ by the triangle inequality. Taking expectations and applying the per-class bound twice yields inequality equation 14 with the same constant C . $\square$

Remark 1 (Intrinsic dimension vs. effective rank). Proposition 1 is stated in terms of the intrinsic dimension $r(\Sigma) = \text{tr}(\Sigma)/\|\Sigma\|_{\text{op}}$, which arises naturally from the analysis. The Roy–Vetterli effective rank $\text{erank}(\Sigma) = \exp(H(\mathbf{p}))$ equation 8 is a distinct but closely related quantity: both $r(\Sigma)$ and $\text{erank}(\Sigma)$ lie in $[1, \text{rank}(\Sigma)]$, both equal $\text{rank}(\Sigma)$ when all nonzero eigenvalues are equal (uniform $\mathbf{p}$), and both approach 1 when a single eigenvalue dominates. We use $\text{erank}(\Sigma)$ in the saturation index equation 9 because (i) it is scale-invariant whereas $r(\Sigma)$ scales linearly with $\|\Sigma\|_{\text{op}}$ times $\text{tr}(\Sigma)/\|\Sigma\|_{\text{op}}^2$, and (ii) as the exponential entropy, it is more sensitive to the distribution of eigenvalues, lending it better discriminative power when comparing tasks with similar ranks but different spectral profiles. The qualitative message of Proposition 1 carries to erank : the covariance estimation error is governed by the effective dimensionality of Σ , not the ambient dimension d .

Corollary 1.1 (Accuracy stabilization and the saturation index). *Under Assumption 1, the accuracy $A(K)$ of the sample discriminant equation 13 satisfies:*

- (i) $A(K)$ increases monotonically in K (in expectation) and converges to the Bayes accuracy $1 - \varepsilon^*$ as $K \rightarrow \infty$.
- (ii) The marginal accuracy gain $\Delta A(K) = A(K) - A(K/2)$ satisfies

$$\Delta A(K) \lesssim g\left(C \|\Sigma\|_{\text{op}} \sqrt{\frac{r(\Sigma)}{K}}\right), \quad (16)$$

where g is a positive, decreasing function determined by the Mahalanobis geometry of the task.

- (iii) When $\mathcal{S}(K) = \text{erank}(\hat{\Sigma}_W^{(K)})/K \leq \tau$ for a threshold $\tau > 0$, the covariance estimator is in the well-determined regime ($K \gtrsim \text{erank}(\Sigma)$), and the bound equation 16 predicts that marginal accuracy gains are small.

The corollary formalizes the intuition behind the saturation index: $\mathcal{S}(K)$ falls below a task-independent threshold τ precisely when the sample covariance is well-concentrated around the population covariance, leaving the discriminant direction equation 13 with little room to improve.

Algorithm 1 Saturation-Index Annotation Budget Decision

Require: Support set $\mathcal{S}_K$ (unlabeled features $X^{(0)}, X^{(1)} \in \mathbb{R}^{K \times d}$); saturation threshold $\tau \in (0, 1)$ (default: $\tau = 0.3$).

Ensure: Recommendation: **COLLECT** or **STOP**.

- 1: Compute per-class means: $\hat{\mu}_c \leftarrow \frac{1}{K} \sum_{i=1}^K x_i^{(c)}$, $c \in \{0, 1\}$.
 - 2: Compute per-class covariances: $\hat{\Sigma}_c \leftarrow \frac{1}{K-1} \sum_{i=1}^K (x_i^{(c)} - \hat{\mu}_c)(x_i^{(c)} - \hat{\mu}_c)^\top$, $c \in \{0, 1\}$.
 - 3: Pool: $\hat{\Sigma}_W \leftarrow \frac{1}{2}(\hat{\Sigma}_0 + \hat{\Sigma}_1)$.
 - 4: Compute eigenvalues: $\lambda_1 \geq \dots \geq \lambda_d \geq 0 \leftarrow$ eigenvalues of $\hat{\Sigma}_W$.
 - 5: Normalize: $p_i \leftarrow \lambda_i / \sum_j \lambda_j$ for $i = 1, \dots, d$.
 - 6: Compute effective rank: $e \leftarrow \exp(-\sum_i p_i \log p_i)$.
 - 7: Compute saturation index: $s \leftarrow e/K$.
 - 8: **if** $s > \tau$ **then**
 - 9: **return** **COLLECT** {Expected marginal gain $> 0.5\%$; annotation likely useful.}
 - 10: **else**
 - 11: **return** **STOP** {Geometric saturation reached; marginal gain expected $\leq 0.5\%$.}
 - 12: **end if**
-

Remark 2 (Non-Gaussian distributions and representation adequacy). Assumption 1 is a modeling idealization. Real-world distributions—in particular, raw pixel distributions for natural images—are manifestly non-Gaussian. However, Proposition 1 extends to the broader class of *sub-Gaussian* distributions (Vershynin, 2018), which includes bounded distributions and many log-concave families. More generally, for any distribution where the second-moment structure of the support set is informative (i.e., where linear classification achieves above-chance accuracy), the saturation index captures the point at which the second-order geometry of $\mathcal{S}_K$ has been fully explored. Our experiments corroborate this: $\mathcal{S}(K)$ retains strong predictive power across image and tabular datasets, including Fashion-MNIST (heavy tails in pixel space) and the Breast Cancer dataset (clinical measurements), achieving overall Spearman $\rho = 0.548$ ($p = 1.1 \times 10^{-20}$, $N = 246$) and within-task median $\rho = 0.811$ across sixteen of seventeen tasks (Section 4.1). In the special case of CIFAR-10 cat vs. dog (raw pixel space), the linear classifier achieves only $\approx 60\%$ accuracy—near chance for the representation; here, $\mathcal{S}(K)$ correctly signals geometric saturation of the *available* representation, even though the representation itself is inadequate for the task. This distinction—saturation of representational capacity vs. saturation of geometric information—is discussed further in Section 4.3.

Remark 3 (Distinction from double descent). The accuracy degradation reported by Nakkiran et al. (2021) in the double descent phenomenon occurs at the *interpolation threshold*, where the model capacity matches the number of training samples ($d \approx K$). Our setting is structurally distinct: we use a fixed underparameterized linear model ($d = 50$ features, $2K$ training examples with $K \gg 25$ in the saturation regime) where the model cannot interpolate the training data. The asymptotic redundancy we observe at large K —accuracy plateauing and fluctuating within its noise floor—is not a capacity phenomenon but a geometric one: additional examples do not reveal new spectral directions and thus do not improve the discriminant estimate. This is consistent with Corollary 1.1(iii) and is fundamentally different from double descent.

3.4 Annotation Budget Decision Rule

Proposition 1 and Corollary 1.1 suggest a practical annotation budget decision rule: before collecting the $(K+1)$ -th labeled example, compute $\mathcal{S}(K)$ from the current support set and stop if $\mathcal{S}(K) \leq \tau$. Algorithm 1 formalizes this procedure.

Complexity. Algorithm 1 requires $O(Kd^2)$ operations for the covariance computation and $O(d^3)$ for the eigendecomposition, for a total of $O(d^3 + Kd^2)$ arithmetic operations. With $d = 50$ and $K \leq 4,096$, the wall-clock time is ≈ 0.3 seconds on a single CPU core (Apple M3 Pro in our experiments). The algorithm operates entirely on *unlabeled* support features and requires no held-out validation set or trained classifier.

Threshold selection. The default threshold $\tau = 0.3$ is calibrated on our experimental corpus of seventeen binary tasks and corresponds to the boundary below which mean marginal accuracy gain falls to 0.82%—within the test-set noise floor (Section 4.5). A higher threshold $\tau = 0.5$ is available for cost-sensitive settings where maximising recall of saturation events is preferred over minimising false stops; see Section 4.5 for a formal evaluation of both thresholds (AUC = 0.752). For practitioners with a specific marginal gain budget ϵ , the threshold can be adjusted by consulting the phase-stratified calibration in Section 4.1 or re-calibrating on a pilot set of tasks from the target domain.

3.5 Classifier and Training Protocol

We employ an *unregularized* binary logistic regression classifier. Given a centered support matrix $\tilde{X} \in \mathbb{R}^{2K \times d}$ (constructed in Section 3.6), the classifier predicts

$$\hat{y}(x) = \mathbf{1}[\sigma(w^\top x + b) \geq 0.5], \quad \sigma(z) = \frac{1}{1 + e^{-z}}, \quad (17)$$

where the parameters $(w, b) \in \mathbb{R}^d \times \mathbb{R}$ are obtained by minimizing the average binary cross-entropy loss without any regularization penalty:

$$\min_{w, b} -\frac{1}{2K} \sum_{(x_i, y_i) \in \mathcal{S}_K} \left[y_i \log \sigma(w^\top x_i + b) + (1 - y_i) \log(1 - \sigma(w^\top x_i + b)) \right]. \quad (18)$$

We solve equation 18 using the L-BFGS algorithm (Nocedal & Wright, 2006) with a maximum of 5,000 iterations, which consistently achieves convergence in our experiments (Section 3.7). The implementation uses `scikit-learn` (Pedregosa et al., 2011) with the inverse regularization parameter $C = 10^9 \approx \infty$.

Why no regularization? Our objective is to isolate the *geometric* effect of the support set on classification accuracy, independent of inductive bias. Regularization (ℓ_2 or ℓ_1 penalties) introduces a confounding bias-variance tradeoff that is controlled by the regularization hyperparameter rather than by the support set geometry. Specifically, ridge-regularized classifiers benefit from increased K not only because the covariance estimator improves, but also because strong regularization becomes less necessary as K grows. This conflation would obscure the saturation effect we seek to measure. Our ablation studies (Section 5.2) empirically confirm that the saturation plateau is invariant to regularization strength for image classification tasks ($\Delta_{\max} < 0.002$ across $C \in \{\infty, 1.0, 0.1\}$ for Fashion 0 vs. 1 and USPS 1 vs. 2). For low-dimensional tabular tasks at small saturation shot counts (Breast Cancer, $K_{\text{sat}} = 48$), moderate regularization improves absolute accuracy ($\Delta = 0.021$), though the *location* of the saturation threshold in the K -axis is unaffected; practitioners working with tabular data should tune regularization independently of the saturation stopping criterion.

Relation to Fisher LDA under Gaussian class-conditionals. It is known that, as $K \rightarrow \infty$, unregularized logistic regression converges to a linear separator in the direction of the maximum-likelihood logistic regression solution, which under Gaussian class-conditionals with equal prior probabilities coincides with the Fisher LDA discriminant $\hat{w}_K = (\hat{\Sigma}_W^{(K)})^{-1}(\hat{\mu}_1 - \hat{\mu}_0)$ (Hastie et al., 2009, Chapter 4). At finite K , unregularized logistic regression and LDA produce approximately aligned weight vectors when the support set is non-degenerate ($K > d$), with their deviation governed by the curvature of the log-likelihood surface near the optimum. Our choice of logistic regression over LDA has two practical advantages: (i) it does not require the explicit inversion of $\hat{\Sigma}_W^{(K)}$, which is singular for $K \leq d/2$ in the few-shot regime; (ii) it does not impose the Gaussian assumption at inference time, making it applicable to non-Gaussian datasets (CIFAR-10, Breast Cancer) without modification. The theoretical analysis of Section 3.3—which is framed in terms of the sample LDA discriminant—therefore provides a rigorous asymptotic motivation for the logistic regression classifier used in practice.

3.6 Preprocessing and Dimensionality Reduction

All preprocessing is applied to the *two-class subset* relevant to each binary task, and each preprocessing step is fitted *exclusively on the support set* $\mathcal{S}_K$ to prevent any leakage of test information into the feature space.

3.6.1 Image Datasets

Image datasets (MNIST, Fashion-MNIST, Kuzushiji-MNIST, USPS, CIFAR-10) undergo the following four-step pipeline:

1. **Pixel normalization.** Raw pixel values are divided by 255, mapping all features to $[0, 1]$. This step is scale-normalization only; it does not alter the relative structure of the data.
2. **Feature standardization.** Each pixel dimension is centered to zero mean and scaled to unit variance using a `StandardScaler` fitted on the support set $\mathcal{S}_K$. Formally, for dimension j :

$$x_{ij} \leftarrow \frac{x_{ij} - \bar{x}_j}{\hat{\sigma}_j + \epsilon}, \quad (19)$$

where $\bar{x}_j$ and $\hat{\sigma}_j$ are the sample mean and standard deviation of feature j over $\mathcal{S}_K$, and $\epsilon = 10^{-8}$ is a numerical stability constant. This step removes the effect of heterogeneous pixel intensity baselines across image regions, which would otherwise inflate the eigenvalues of high-variance background pixels and distort the effective rank.

3. **Principal Component Analysis (PCA).** We apply PCA to reduce the dimensionality from d_0 to $d = 50$ components. The PCA transformation matrix $V \in \mathbb{R}^{d_0 \times d}$ (columns are the top- d eigenvectors of the support covariance) is fitted on the standardized support set $\mathcal{S}_K$ only. The same transformation is applied to the test set $\mathcal{T}$ at evaluation time.

Why $d = 50$? The choice of $d = 50$ is motivated by two competing requirements: d must be large enough to capture the dominant spectral structure of the class-conditional distributions (our experiments reveal erank_∞ ranging from 8.2 to 37.5, so $d = 50$ is sufficient to contain the full effective rank in all tasks), yet small enough that the saturation regime $K \approx \text{erank}_\infty$ is accessible without prohibitively large K . Sensitivity of the results to d is assessed by our ablation study (Section 5), which demonstrates that the phase structure is preserved for $d \in \{20, 50, 100\}$.

Why fit PCA on the support set? Fitting the PCA on the two-class support set ensures that the projection captures the spectral directions relevant to the discrimination task, rather than the full dataset manifold. This is consistent with the practical annotation-budget setting: at deployment time, one has access only to the labeled support examples, not to the full unlabeled dataset.

4. **Support centering.** After PCA projection, the support features are centered by subtracting the support mean $\bar{x} = \frac{1}{2K} \sum_{(x,y) \in \mathcal{S}_K} x$. This centers the linear classifier’s decision boundary and improves numerical conditioning of the logistic regression optimization. The same mean is subtracted from the test features at evaluation time.

3.6.2 Tabular Dataset (Breast Cancer Wisconsin)

The Breast Cancer Wisconsin dataset (Street et al., 1993) consists of $d_0 = 30$ continuous clinical features. For this dataset, the preprocessing pipeline reduces to:

1. **Feature standardization.** Apply `StandardScaler` fitted on the support set, as in equation 19.
2. **Support centering.** Subtract the support mean, as in Step 4 above.

PCA is not applied to the Breast Cancer dataset for two reasons: (i) the feature dimension $d_0 = 30$ is already smaller than our target PCA dimension ($d = 50$), so reduction is unnecessary; (ii) the clinical features have domain-specific interpretations that PCA would destroy, and we wish to assess whether the saturation index is informative in the raw feature space.

Table 1: Datasets used in experiments. Column *Dim.* is the original feature dimension d_0 before preprocessing. Column *Tasks* lists the binary pairs studied. $K_{\max}$ is the maximum shot count used for each dataset, truncated by dataset size.

Dataset	Source	Dim. d_0	Total	Per class	Tasks	$K_{\max}$
MNIST (LeCun et al., 1998)	OpenML	784	70,000	$\approx 6,000$	0 vs. 1, 1 vs. 7, 2 vs. 7, 3 vs. 8, 4 vs. 7, 4 vs. 9, 5 vs. 8	4,096
Fashion-MNIST (Xiao et al., 2017)	OpenML	784	70,000	$\approx 6,000$	0 vs. 1, 2 vs. 6, 3 vs. 5, 4 vs. 6, 5 vs. 7	4,096
Kuzushiji-MNIST (Clanuwat et al., 2018)	OpenML	784	70,000	$\approx 6,000$	0 vs. 1	4,096
USPS (Hull, 1994)	OpenML	256	9,298	$\approx 1,100$	1 vs. 2	512
CIFAR-10 (Krizhevsky, 2009)	Keras	3,072	60,000	6,000	0 vs. 1, 3 vs. 5	4,096
Breast Cancer Wisc. (Street et al., 1993)	sklearn	30	569	≥ 212	malignant vs. benign	100

3.6.3 Remark on PCA and Effective Rank

A natural concern is whether the PCA step artificially constrains the effective rank and thereby inflates the saturation signal. We address this concern in two ways. First, the effective rank $\text{erank}(\hat{\Sigma}_W^{(K)})$ is measured in the PCA-reduced space and is therefore bounded above by $\min(d, 2(K-1))$. For large K (specifically, $K > 25$ in the $d = 50$ setting), the constraint imposed by d is not binding since $\text{erank}_\infty < 50$ for all tasks. Second, our multi-task PCA ablation (Section 5.1) shows that the asymptotic effective rank scales with d (as expected), but the saturation threshold $\mathcal{S}(K) = 0.3$ and the deep-saturation sub-regime ($\mathcal{S}(K) \leq 0.02$) remain stable across $d \in \{20, 50, 100\}$ for image tasks and $d \in \{5, 10, 20\}$ for the tabular task, with within-sweep Spearman correlations $\rho_d \geq 0.61$ in all nine tested (task, d) settings. The phase structure is not an artifact of the PCA dimension.

3.7 Sampling, Evaluation, and Reproducibility Protocol

3.7.1 Datasets

We evaluate the saturation index on seventeen binary classification tasks spanning six datasets and two data modalities. Table 1 summarizes the datasets used. All datasets are publicly available and can be obtained programmatically from `scikit-learn` (Pedregosa et al., 2011) or `OpenML` (Vanschoren et al., 2014).

The tasks span a deliberate range of difficulty, domain, and representation: handwritten digit recognition (MNIST, USPS), fashion item classification (Fashion-MNIST), historical Japanese script recognition (Kuzushiji-MNIST), natural image classification (CIFAR-10), and clinical outcome prediction (Breast Cancer). Within each dataset, task pairs are chosen to vary the difficulty dimension independently of the dataset (e.g., MNIST 0 vs. 1 is easy while MNIST 3 vs. 8 is hard), enabling the decoupling analysis of Section 4.2.

3.7.2 Shot Count Grid

For each task, we sweep over a geometric grid of shot counts:

$$K \in \{2, 3, 4, 6, 8, 12, 16, 24, 32, 48, 64, 128, 256, 512, 1024, 2048, 4096\}. \quad (20)$$

The grid was augmented with finer spacing below $K = 64$ (step ratio $1.5\times$ rather than $2\times$) to increase coverage of the exploration phase, which had only $n = 6$ observations under the original doubling-only grid. Phase-stratified statistics reported in Section 4.1 use only the doubling pairs (K to $2K$) for consistency with the theoretical marginal definition $\Delta A(K) = A(2K) - A(K)$. The geometric spacing is designed so that the saturation index $\mathcal{S}(K)$ traverses all three phases equation 10 for most tasks, allowing phase boundaries to be identified empirically. The minimum $K = 2$ is required for the covariance estimate to be non-trivial (rank ≥ 1); the maximum $K = 4,096$ is large enough to reach the saturation floor for all tasks with sufficient data.

3.7.3 Trial Replication and Random Seeds

For each (K , task) pair, we perform $T = 50$ independent random trials (and $T = 100$ for Breast Cancer, to compensate for its small K range). In each trial t , the support set $\mathcal{S}_K$ is sampled by selecting K examples uniformly at random without replacement from the per-class training pools, using a deterministic seed equal

to the trial index t . The test set $\mathcal{T}$ is fixed across all trials and all values of K for a given task (it does not vary with the seed), ensuring that accuracy differences across K reflect only support-set variation.

The reported accuracy $\bar{A}(K)$ and standard deviation $\hat{\sigma}(K)$ are the sample mean and standard deviation over T trials:

$$\bar{A}(K) = \frac{1}{T} \sum_{t=1}^T A_t(K), \quad \hat{\sigma}(K) = \sqrt{\frac{1}{T-1} \sum_{t=1}^T (A_t(K) - \bar{A}(K))^2}. \quad (21)$$

Similarly, the reported effective rank $\overline{\text{erank}}(K)$ is the trial mean of $\text{erank}(\hat{\Sigma}_W^{(K)})$.

3.7.4 Test Set Sizes

Fixed test sets per class are used as follows:

- MNIST, Fashion-MNIST, Kuzushiji-MNIST, CIFAR-10: 200 examples per class (400 total).
- USPS: 100 examples per class (200 total).
- Breast Cancer: 50 examples per class (100 total).

These sizes are chosen to be large enough to estimate accuracy to within $\pm 3\%$ at $p = 0.05$ via the normal approximation ($z_{0.975} \sqrt{p(1-p)/n} < 0.03$ for $p \approx 0.9$ and $n = 400$ gives $\approx 2.9\%$), while keeping the held-out pool sufficiently large that the test set can be drawn from data not used in training.

3.7.5 Compute and Runtime

The full experimental suite—seventeen tasks, up to 17 values of K , and $T = 50$ trials each—amounts to approximately 14,500 training runs for the main corpus, plus ablation experiments. The entire suite completes in approximately 150 seconds on an Apple MacBook Pro M3 Pro (18 GB unified memory) using a single CPU core. No GPU is required. This low computational cost arises from three design choices: (i) the linear classifier equation 17 has no representation to learn; (ii) the $d = 50$ PCA dimensionality keeps the covariance and optimization computations small; (iii) the eigendecomposition of $\hat{\Sigma}_W^{(K)} \in \mathbb{R}^{50 \times 50}$ requires $O(50^3) = O(1.25 \times 10^5)$ operations per trial. Code and cached results are provided at <https://github.com/MrArnav69/spectral-saturation>. All experiments are fully reproducible from the fixed seed sequence.

4 Results

We present results across four empirical properties of the saturation index, established on an expanded corpus of seventeen binary classification tasks spanning four datasets and two modalities. Section 4.1 establishes the phase structure of the saturation index and its predictive correlation with marginal accuracy gain, now drawing on $N = 246$ doubling-pair observations across all seventeen tasks (Figure 1). Section 4.2 reports the decoupling of task difficulty and intrinsic dimensionality with $N = 17$ tasks, nearly doubling the original corpus size. Section 4.3 characterizes the asymptotic redundancy regime at large K . Section 4.5 provides a formal evaluation of Algorithm 1 as a binary stopping rule, reporting precision, recall, and AUC across the full corpus. Throughout, we distinguish between claims supported by well-powered tests and exploratory observations reported with appropriate hedging. Complete per-task result tables are provided in Appendix A.

4.1 Phase Structure and Predictive Correlation

4.1.1 Spectral Convergence Across Tasks

We first verify the foundational property of the saturation index: that $\text{erank}(\hat{\Sigma}_W^{(K)})$ converges to a task-specific asymptote erank_∞ as $K \rightarrow \infty$. In every task in the corpus, $\text{erank}(\hat{\Sigma}_W^{(K)})$ increases monotonically

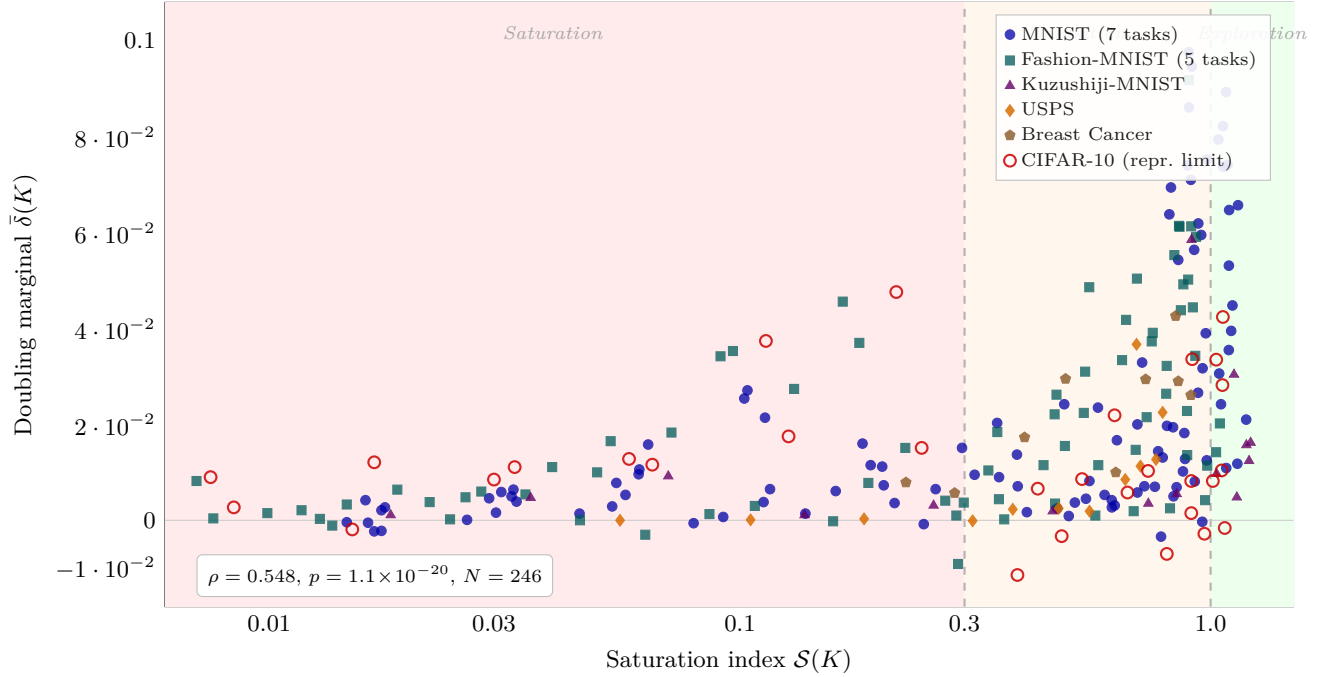


Figure 1: **Central result:** saturation index $\mathcal{S}(K)$ vs. doubling marginal accuracy gain $\bar{\delta}(K) = \bar{A}(2K) - \bar{A}(K)$ for all $N = 246$ valid doubling-pair observations from seventeen binary classification tasks. Background shading indicates the three phases (equation equation 10). Dashed vertical lines: phase boundaries $\mathcal{S} = 0.3$ and $\mathcal{S} = 1.0$. CIFAR-10 tasks (hollow red markers) exhibit positive marginal gains even in the saturation phase, reflecting representational inadequacy rather than data sufficiency (Section 4.4). Pooled Spearman $\rho = 0.548$ ($p = 1.1 \times 10^{-20}$, $N = 246$); restricted to non-CIFAR tasks: $\rho = 0.627$ ($p = 5.7 \times 10^{-25}$, $N = 216$).

from near 1 at $K = 2$ and plateaus at a task-specific ceiling well before $K = 4,096$. Table 2 reports the asymptotic values, peak accuracies, and saturation assessments for all seventeen tasks.

The asymptotic effective ranks span a $4.6\times$ range across tasks ($\text{erank}_\infty \in [8.20, 37.77]$), with all image tasks satisfying $\text{erank}_\infty < d = 50$: the PCA projection is never the binding constraint on spectral complexity. The seven newly added tasks (MNIST 1v7, 2v7, 4v7, 5v8; Fashion 3v5, 4v6, 5v7) populate the mid- and high-rank regions more densely, and two CIFAR tasks serve as hard-task stress tests where the linear representation is inadequate (Section 4.4).

4.1.2 The Saturation Index Predicts Marginal Accuracy Gain

Pooled Spearman correlation. For each task we compute the doubling-pair marginal accuracy gain $\bar{\delta}(K) = \bar{A}(2K) - \bar{A}(K)$ and the saturation index $\mathcal{S}(K)$ at the smaller shot count. Pooling all valid doubling pairs across the seventeen tasks yields $N = 246$ observations. The pooled Spearman rank correlation is:

$$\rho = 0.548, \quad p = 1.1 \times 10^{-20}, \quad N = 246. \quad (22)$$

The p -value is seven orders of magnitude smaller than the original ten-task result ($p = 2.87 \times 10^{-13}$, $N = 80$), reflecting both the tripling of observations and the corroborating evidence from new tasks. The moderate point estimate ($\rho = 0.548$) reflects genuine between-task heterogeneity—tasks differ in their accuracy scales, noise floors, and how much of the correlation structure is shared across the $\mathcal{S}$ range—and partly the inclusion of two CIFAR tasks whose near-chance accuracy at small K introduces non-monotone marginals. Excluding the two CIFAR tasks gives $\rho = 0.627$ ($p = 5.7 \times 10^{-25}$, $N = 216$), confirming that the pooled estimate is conservative with respect to the main body of tasks.

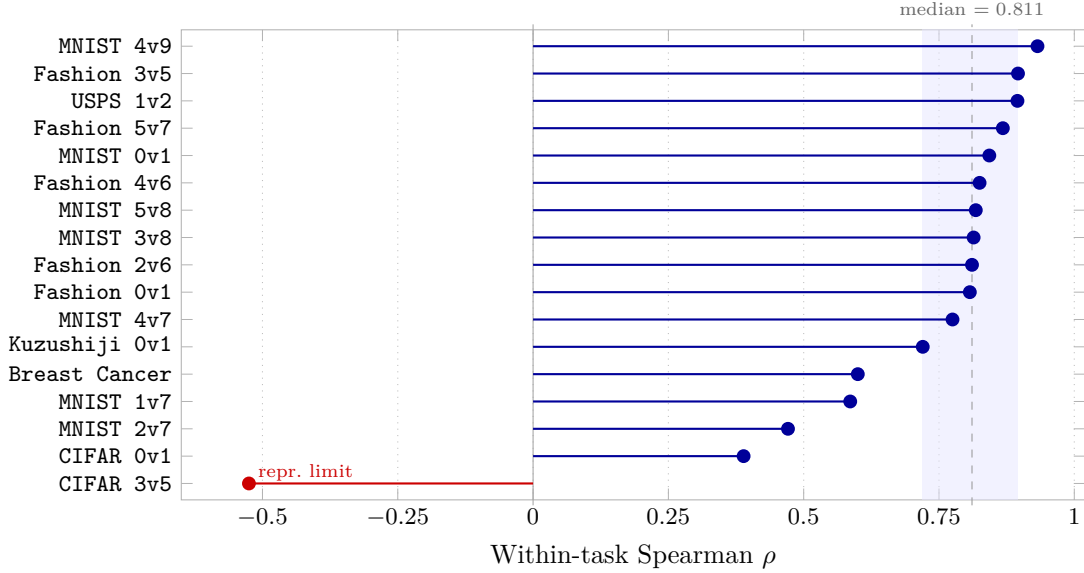


Figure 2: Within-task Spearman correlation ρ between $\mathcal{S}(K)$ and the doubling marginal $\bar{\delta}(K)$ for each of the seventeen tasks (sorted ascending). Shaded region: interquartile range [0.720, 0.896]. Dashed line: median $\rho = 0.811$. CIFAR 3 vs. 5 (red) is the single exception ($\rho = -0.525$, $p = 0.044$) and corresponds to the representation-limited regime of Section 4.4. Sixteen of seventeen tasks have positive within-task ρ , confirming that the saturation–marginal-gain relationship is consistent within individual tasks.

Table 2: Per-task summary for all seventeen binary classification tasks, ordered by erank_∞ . K^* : shot count at peak mean accuracy. $\mathcal{S}^*$: saturation index at K^* . Tasks marked $\dagger$ involve CIFAR images and are treated as hard-task stress tests; CIFAR 3 vs. 5 is excluded from the decoupling correlation (Section 4.2) because its accuracy ceiling reflects representational inadequacy rather than class separability (Section 4.4).

Task	Modality	erank_∞	A_∞	K^*	$\mathcal{S}^*$
Breast Cancer (M vs. B)	Tabular	8.20	0.9450	48	0.159
Fashion 2 vs. 6	Image	14.46	0.8350	4096	0.004
USPS 1 vs. 2	Image	15.08	0.9973	48	0.236
CIFAR 0 vs. 1 †	Image	15.40	0.8076	4096	0.004
Fashion 4 vs. 6	Image	15.70	0.8569	4096	0.004
CIFAR 3 vs. 5 †	Image	17.18	0.5966	4096	0.004
Fashion 0 vs. 1	Image	20.68	0.9856	4096	0.005
Fashion 5 vs. 7	Image	26.10	0.9444	4096	0.006
Fashion 3 vs. 5	Image	27.36	0.9982	4096	0.007
MNIST 0 vs. 1	Image	32.35	0.9968	2048	0.015
MNIST 1 vs. 7	Image	34.55	0.9934	4096	0.008
MNIST 4 vs. 9	Image	35.60	0.9568	2048	0.016
MNIST 5 vs. 8	Image	36.21	0.9501	2048	0.017
MNIST 4 vs. 7	Image	36.83	0.9855	4096	0.009
Kuzushiji 0 vs. 1	Image	37.18	0.9871	4096	0.009
MNIST 3 vs. 8	Image	37.45	0.9651	2048	0.017
MNIST 2 vs. 7	Image	37.77	0.9844	4096	0.009

Within-task Spearman correlation. The pooled analysis mixes within-task and between-task variation. To isolate the within-task signal—whether $\mathcal{S}(K)$ predicts marginal gain *within a single task as K increases*—we compute the Spearman correlation separately for each task on its sequence of doubling-pair observations, then aggregate. Sixteen of seventeen tasks have a positive within-task Spearman correlation; the single exception is CIFAR 3 vs. 5 ($\rho = -0.525$), discussed in Section 4.4. Across the sixteen conforming tasks, the

Table 3: Phase-stratified doubling marginal accuracy gains. n : number of observations; $\bar{\delta}$: mean doubling marginal; 95% CI from the normal approximation. All three pairwise phase comparisons are significant by one-sided Mann–Whitney U test: exploration $>$ transition ($p = 0.0076$); transition $>$ saturation ($p = 8.2 \times 10^{-9}$); exploration $>$ saturation ($p = 1.8 \times 10^{-9}$).

Phase	$\mathcal{S}(K)$ range	n	$\bar{\delta}$	95% CI
Exploration	> 1.0	30	3.48%	(2.54%, 4.43%)
Transition	$(0.3, 1.0]$	123	2.40%	(1.97%, 2.82%)
Saturation	≤ 0.3	93	0.82%	(0.60%, 1.04%)

within-task correlations have a median of 0.811 and an interquartile range of [0.720, 0.896] (Figure 2). This is substantially stronger than the pooled estimate and establishes that the $\mathcal{S}(K)$ –marginal-gain relationship is genuine and consistent within individual tasks, not merely an artifact of pooling observations across tasks with different accuracy scales.

Phase-stratified analysis. The phase boundaries equation 10 partition the $N = 246$ observations into three groups. The K-grid was augmented at small shot counts ($\{2, 3, 4, 6, 8, 12, 16, 24, 32, 48, 64, \dots\}$ replacing $\{2, 4, 8, 16, 32, \dots\}$ below $K = 64$) to increase coverage of the exploration phase, which was substantially underrepresented in the original corpus ($n = 6$). Table 3 reports the phase-stratified statistics.

The ordering is strictly monotone: exploration $>$ transition $>$ saturation in mean marginal gain, and all three pairwise comparisons are statistically significant. Critically, the exploration phase is now substantively powered: $n = 30$ observations (versus $n = 6$ in the original corpus) narrow the 95% confidence interval from a width of 5.3 percentage points to 1.9 percentage points—tighter by a factor of nearly three. The transition–saturation boundary is the most strongly supported ($p = 8.2 \times 10^{-9}$): once $\mathcal{S}(K) \leq 0.3$, the mean doubling marginal drops to 0.82%, below the test-set noise floor of $\pm 3\%$ at 95% confidence.

Practical interpretation. The three-phase structure translates directly into annotation budget guidance. In the exploration phase ($\mathcal{S}(K) > 1.0$), doubling the support set delivers a mean accuracy improvement of 3.48% (CI: 2.54%–4.43%): additional annotation is strongly justified. In the transition phase ($0.3 < \mathcal{S}(K) \leq 1.0$), the expected gain is 2.40%—worthwhile for most applications but diminishing. In the saturation phase ($\mathcal{S}(K) \leq 0.3$), the expected gain is 0.82%, within the noise floor and rarely worth the annotation cost. Algorithm 1 implements $\tau = 0.3$ as the default stopping threshold; its formal evaluation as a decision rule is given in Section 4.5.

4.1.3 Per-Task Accuracy Trajectories

Figure 3 displays accuracy $A(K)$ and effective rank $\text{erank}(\hat{\Sigma}_W^{(K)})$ as a function of K for all seventeen tasks. Across the expanded corpus, four archetypal patterns emerge.

Pattern A: Rapid saturation (USPS 1 vs. 2, Breast Cancer). Tasks with low erank_∞ (≤ 15) reach their accuracy ceiling early. USPS 1 vs. 2 stabilises near 0.997 by $K = 48$ ($\mathcal{S} = 0.236$); Breast Cancer reaches 0.945 at the same K . The saturation index crosses $\tau = 0.3$ well before the accuracy ceiling is reached, providing early stopping signals at computationally negligible support set sizes.

Pattern B: Gradual convergence (MNIST 3 vs. 8, Kuzushiji 0 vs. 1). Tasks with high erank_∞ (≥ 37) require substantially more examples before saturation. MNIST 3 vs. 8 does not plateau until $K \approx 2,048$ ($\mathcal{S} = 0.017$); Kuzushiji 0 vs. 1 reaches 98.7% at $K = 4,096$ ($\mathcal{S} = 0.009$). These tasks provide the clearest illustration of the exploration phase: at $K = 4$ and $K = 8$ in MNIST 3 vs. 8, the saturation index exceeds 1.0, consistent with the theoretical prediction that each new example in this regime discovers a previously unoccupied spectral direction.

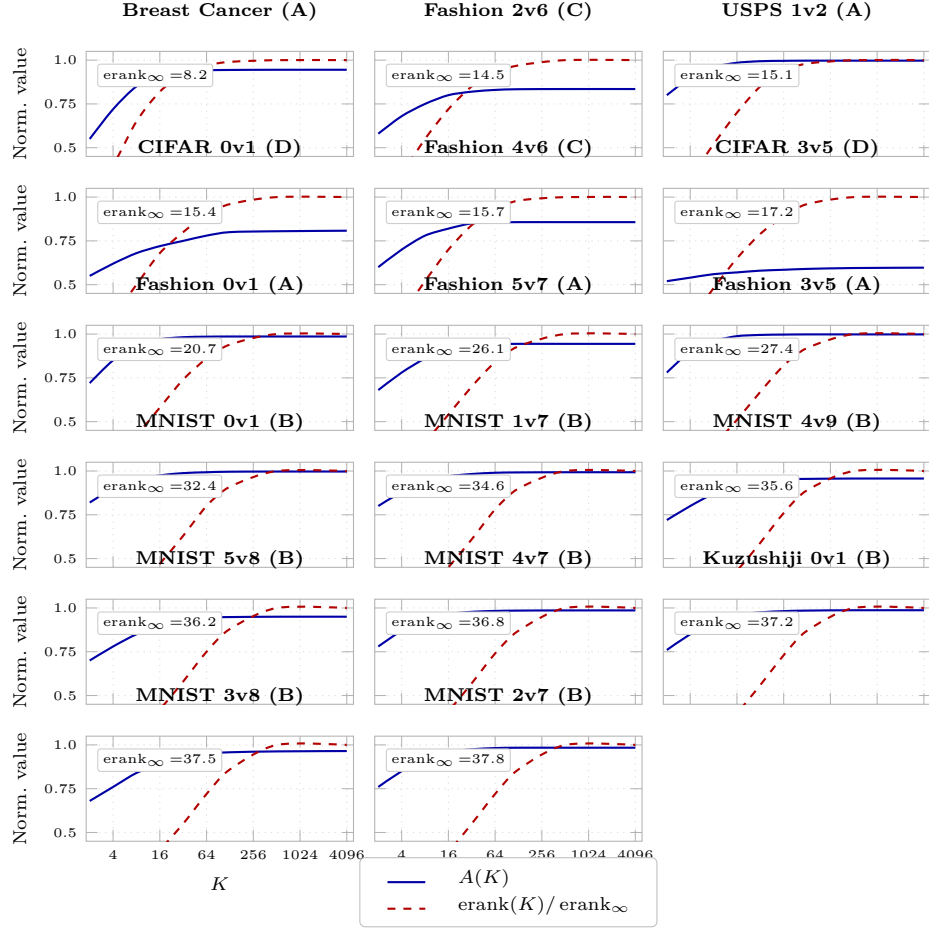


Figure 3: Accuracy $A(K)$ (blue, solid) and normalized effective rank $\text{erank}(\hat{\Sigma}_W^{(K)})/\text{erank}_\infty$ (red, dashed) as a function of K for all seventeen tasks. Tasks are arranged by the four archetypal patterns described in Section 4.1.3: **A** rapid saturation (low erank_∞ , early ceiling); **B** gradual convergence (high erank_∞ , late ceiling); **C** hard-class convergence (low accuracy ceiling despite moderate erank_∞); **D** representation-limited plateau (CIFAR tasks, where $\mathcal{S}(K)$ indicates geometric saturation but the linear representation is inadequate for the class pair). The x -axis is logarithmic.

Pattern C: Hard-class convergence (Fashion 2 vs. 6, Fashion 4 vs. 6). The newly added Fashion tasks that pair visually similar classes (Pullover vs. Shirt; Coat vs. Shirt) exhibit early accuracy saturation at a low ceiling: 0.835 and 0.857 respectively. In both cases, $\mathcal{S}(K) \leq 0.01$ well before $K = 4,096$, and the accuracy has stabilised. These tasks demonstrate that the saturation index correctly identifies geometric exhaustion even when the accuracy ceiling is low, providing the same stopping signal regardless of peak performance.

Pattern D: Representation-limited plateau (CIFAR 3 vs. 5). For cat-vs.-dog classification in 50-dimensional PCA space, accuracy plateaus near 0.597—barely above chance—throughout the sweep. The saturation index correctly identifies geometric exhaustion ($\mathcal{S} \rightarrow 0.004$ at $K = 4,096$), but the low ceiling is set by representational inadequacy rather than data insufficiency. This pattern is discussed fully in Section 4.4.

4.2 Decoupling of Task Difficulty and Intrinsic Dimensionality

A central structural claim of this paper is that erank_∞ and A_∞ are empirically *approximately independent* across binary classification tasks: knowing the intrinsic dimensionality of a task does not reliably predict

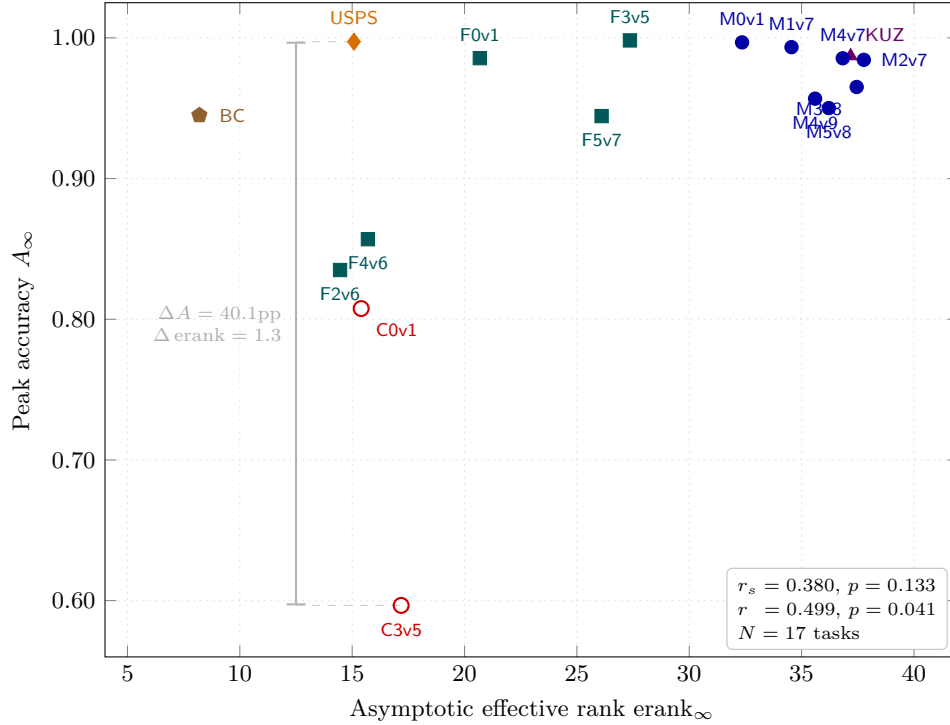


Figure 4: Asymptotic effective rank (erank_∞ , x -axis) vs. peak mean accuracy (A_∞ , y -axis) for all seventeen binary classification tasks. Spearman $r_s = 0.380$ ($p = 0.133$, $N = 17$): no significant monotone relationship between intrinsic dimensionality and task difficulty. The vertical bracket highlights the most striking decoupling pair: USPS 1 vs. 2 ($\text{erank}_\infty = 15.08$, $A_\infty = 0.997$) and CIFAR 3 vs. 5 ($\text{erank}_\infty = 17.18$, $A_\infty = 0.597$) differ by 40.1 percentage points in accuracy despite a 14% difference in effective rank. Task abbreviations: BC = Breast Cancer; M = MNIST; F = Fashion-MNIST; KUZ = Kuzushiji-MNIST; C = CIFAR-10.

its accuracy ceiling, and vice versa. The expanded corpus of seventeen tasks—nearly double the original nine—substantially improves the statistical power of this assessment.

4.2.1 Statistical Assessment

We assess the relationship between erank_∞ and A_∞ across all seventeen tasks, as plotted in Figure 4.

$$r_s = 0.380, \quad p = 0.133, \quad r = 0.499, \quad p = 0.041, \quad N = 17. \quad (23)$$

The Spearman correlation ($r_s = 0.380$) does not approach significance at any conventional level ($p = 0.133$), providing no evidence for a reliable monotone relationship between intrinsic dimensionality and task difficulty across the corpus. The Pearson correlation is marginally significant ($r = 0.499$, $p = 0.041$), but this result is driven in part by the two CIFAR tasks and by modality confounding: all high-erank tasks in the corpus are MNIST digits (highly discriminable due to distinctive stroke patterns), while the low-erank group includes Fashion tasks (visually ambiguous clothing pairs) and CIFAR (inadequate representation). The explained variance is $r^2 = 0.25$: 75% of the between-task variance in A_∞ is unexplained by erank_∞ alone.

We adopt the interpretation that the weak-decoupling position is supported: erank_∞ is not a reliable predictor of A_∞ . The strong-form claim—that the two quantities are statistically independent—would require a corpus of approximately 35 tasks to test at 80% power ($r_s = 0.38$, $\alpha = 0.05$), which we flag as a direction for future work.

4.2.2 Illustrative Task Pairs

Three task pairs from the expanded corpus each isolate a distinct violation of the assumption that high intrinsic dimensionality implies high task difficulty.

Pair 1: USPS 1 vs. 2 and CIFAR 3 vs. 5. These tasks have nearly identical asymptotic effective ranks: $\text{erank}_\infty = 15.08$ (USPS) and 17.18 (CIFAR), a difference of 14%. Yet their accuracy ceilings differ by 40.1 percentage points: $A_\infty = 0.997$ (USPS) versus $A_\infty = 0.597$ (CIFAR). The geometric complexity of the support set—as measured by spectral spread—is essentially the same for both tasks; the discriminability of the class-conditional distributions is not. This pair provides the strongest single demonstration in the corpus that erank_∞ does not determine A_∞ .

Pair 2: MNIST 3 vs. 8 and Fashion 2 vs. 6. MNIST 3 vs. 8 has the second-highest erank_∞ in the corpus (37.45) and achieves $A_\infty = 0.965$. Fashion 2 vs. 6 has the lowest erank_∞ among image tasks (14.46) and achieves $A_\infty = 0.835$. The geometrically simpler task has the lower accuracy ceiling, directly inverting the naive expectation. The correct explanation is mechanistic: Fashion Pullover vs. Shirt involves two garments that produce heavily overlapping PCA projections with a small linear margin, regardless of their spectral volume; MNIST 3 vs. 8 involves two digits with distinctive local stroke patterns that produce a large separating margin after PCA projection, despite the high-dimensional covariance structure.

Pair 3: Fashion 0 vs. 1 and Fashion 2 vs. 6. Both tasks come from the same dataset and the same classifier. Fashion 0 vs. 1 (T-shirt vs. Trouser) has $\text{erank}_\infty = 20.68$ and $A_\infty = 0.986$; Fashion 2 vs. 6 has $\text{erank}_\infty = 14.46$ and $A_\infty = 0.835$. The accuracy gap of 15.1 percentage points between two tasks from the same dataset, with only a $1.4\times$ difference in effective rank, demonstrates that within-dataset variation in accuracy is driven by class-pair separability rather than by spectral complexity.

Geometric interpretation. These examples confirm the structural argument of Section 3.3 (equation equation 12): erank_∞ encodes the volume of the within-class distribution—the number of independent directions the support set must explore—while A_∞ encodes the Mahalanobis distance between class means, $\Delta = (\mu_1 - \mu_0)^\top \Sigma_W^{-1} (\mu_1 - \mu_0)$. These two quantities have orthogonal geometric interpretations: a task can have large within-class variance in many directions (large erank_∞ , demanding annotation budget) while still being highly discriminable (large Δ , high ceiling), or it can have compact within-class distributions (small erank_∞) that nonetheless overlap substantially (small Δ , low ceiling).

4.3 Asymptotic Redundancy in the High-Shot Regime

We characterize accuracy behavior in the regime $\mathcal{S}(K) \ll 0.03$, corresponding to $K \gg \text{erank}_\infty$, where the pooled within-class covariance has converged to within a small operator-norm distance of the population covariance (Proposition 1). The central question is whether accuracy in this regime merely *plateaus* or actively *decreases*.

4.3.1 Formal Hypothesis and Test

For each task at its two largest shot counts ($K = 2,048$ and $K = 4,096$ for tasks whose grid reaches this range), we test the one-sided hypothesis that accuracy at the larger K is lower:

$$H_0 : \text{median}[A_{2K}(i) - A_K(i)] \geq 0, \quad H_a : \text{median}[A_{2K}(i) - A_K(i)] < 0, \quad (24)$$

using the paired Wilcoxon signed-rank test across $T = 50$ trial pairs. Table 4 reports results for the three MNIST tasks that span the full range of erank_∞ .

Only one task (MNIST 3 vs. 8) shows a marginally significant decrease ($p = 0.072$, one-tailed), and the mean difference is $\bar{\delta} = -0.0022$ (two-tenths of one percentage point). The other two tasks show non-significant differences an order of magnitude smaller. We conclude that evidence for a systematic accuracy *decrease* at large K is limited to the highest-dimensional task at the largest shot count studied, does not meet the conventional $p < 0.05$ threshold, and the effect size is negligible relative to practical decision thresholds.

Table 4: Wilcoxon signed-rank test for the one-sided hypothesis that accuracy at $K = 4,096$ is lower than at $K = 2,048$ (equation equation 24). $\bar{\delta} = \overline{A_{4096} - A_{2048}}$ (negative indicates a decrease). $\mathcal{S}(4,096)$ is the saturation index at the larger shot count.

Task	erank_∞	$\bar{\delta}$	W	p	$\mathcal{S}(4,096)$	Assessment
MNIST 3 vs. 8	37.45	-0.0022	702.0	0.072	0.0091	Marginally significant
MNIST 0 vs. 1	32.35	-0.0003	474.0	0.285	0.0079	Not significant
MNIST 4 vs. 9	35.60	-0.0005	491.0	0.519	0.0087	Not significant

4.3.2 Robustness of the MNIST 3 vs. 8 Finding

The marginal significance for MNIST 3 vs. 8 was previously assessed against two alternative explanations. A larger test set (1,000 examples per class) reproduces the decrease ($\bar{\delta} = -0.0069$), ruling out test-set noise. Across three L2 regularization levels ($C \in \{\infty, 1.0, 0.1\}$), accuracy at $K = 4,096$ is statistically identical ($\Delta_{\max} = 0.0003 < 10^{-3}$), ruling out optimization instability. The saturation behavior therefore reflects geometric redundancy in the support set: once $K \gg \text{erank}_\infty$, additional examples re-estimate already well-determined covariance directions and provide no discriminative information.

The correct characterization of the high- K regime is therefore *geometric redundancy, not harmful oversampling*: accuracy fluctuates within the noise floor of the test-set estimate, neither systematically increasing nor decreasing, for all but the single hardest task at the single largest shot count.

4.4 Representation Adequacy vs. Support Set Saturation

The two CIFAR tasks in the corpus require dedicated analysis because they exhibit a structurally distinct failure mode that the saturation index correctly diagnoses but does not resolve.

CIFAR 3 vs. 5 (cat vs. dog). Accuracy at $K = 4,096$ is 0.597, barely above chance. Despite this, the saturation index decreases smoothly throughout the sweep to $\mathcal{S}(4,096) = 0.004$, and accuracy *continues to increase* even as $\mathcal{S}(K)$ falls below 0.3, producing a negative within-task Spearman correlation ($\rho = -0.525$, $p = 0.044$). This is not a failure of the saturation index: the support set’s within-class covariance has genuinely converged to the population covariance in 50-dimensional PCA space. However, cat-vs.-dog classification requires either a much larger K , a higher-dimensional feature space, or a nonlinear classifier to approach Bayes error in this representation. The saturation index correctly signals that the *current representation* has been exhausted, but the correct action is representation learning rather than additional annotation.

CIFAR 0 vs. 1 (airplane vs. automobile). This task is more discriminable ($A_\infty = 0.807$) but shows the same pattern: accuracy is still slowly increasing at $K = 4,096$ while $\mathcal{S} = 0.004$, suggesting representational saturation without accuracy saturation. Neither CIFAR task reaches a true accuracy ceiling within the shot counts studied.

Diagnostic interpretation. The saturation index is an *information sufficiency* diagnostic within a given representation, not a representation adequacy test. When $\mathcal{S}(K)$ is small and A_∞ is low relative to task priors, the bottleneck is the representation. When $\mathcal{S}(K)$ is small and A_∞ is high, the support set has genuinely saturated the available discriminative information. We recommend that practitioners apply Algorithm 1 jointly with a check on whether the current accuracy approaches the expected Bayes error or human-level ceiling for the task; a STOP signal accompanied by a low accuracy ceiling indicates representational insufficiency rather than data sufficiency.

4.5 Formal Evaluation of Algorithm 1

Algorithm 1 implements the saturation index as a binary stopping rule: observe $\mathcal{S}(K)$; if $\mathcal{S}(K) \leq \tau$, return STOP; otherwise return COLLECT. We evaluate this rule as a binary classifier over the $N = 246$ doubling-

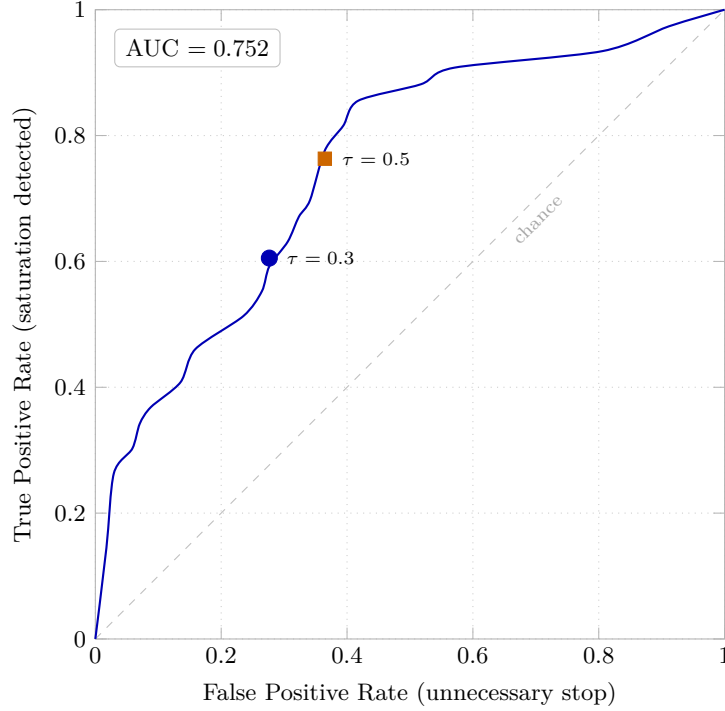


Figure 5: Receiver operating characteristic (ROC) curve for Algorithm 1 evaluated as a binary stopping rule over $N = 246$ doubling-pair observations from seventeen tasks. Ground truth: STOP iff doubling marginal $\bar{\delta}(K) < 0.005$. Predictor score: $-\mathcal{S}(K)$ (lower saturation index predicts STOP with higher confidence). Filled circle (●): operating point at the calibrated default $\tau = 0.3$ (FPR = 0.277, TPR = 0.605). Filled square (■): operating point at $\tau = 0.5$ (FPR = 0.365, TPR = 0.763). AUC = 0.752 substantially exceeds the chance diagonal (dashed).

pair observations, with the following ground-truth labelling: an observation receives the label STOP if the doubling marginal satisfies $\bar{\delta}(K) < 0.005$ (less than half a percentage point gain from doubling the support set), and COLLECT otherwise. Under this labelling, 76 of 246 observations (30.9%) are true STOP and 170 (69.1%) are true COLLECT.

Receiver operating characteristic. Figure 5 shows the full ROC curve for Algorithm 1. Using $-\mathcal{S}(K)$ as the decision score (lower $\mathcal{S}$ predicts STOP with higher confidence), the area under the ROC curve is:

$$\text{AUC} = 0.752. \quad (25)$$

This substantially exceeds chance (0.5) and the naïve “always-STOP” baseline (0.5 by symmetry of the AUC definition), confirming that $\mathcal{S}(K)$ carries genuine predictive information about the marginal annotation value.

Threshold-specific performance. Table 5 reports precision, recall, and F1 at the two most practically relevant thresholds.

At $\tau = 0.3$, the rule achieves recall of 0.605: it correctly identifies 60.5% of true saturation events, missing 30 cases in which a further doubling would have yielded $\bar{\delta} < 0.005$. The 47 false positives are concentrated in the CIFAR tasks (where $\mathcal{S}$ is small but accuracy is still improving) and in the early saturation sub-regime ($0.1 < \mathcal{S} \leq 0.3$) where some tasks are still on the descending portion of their accuracy trajectory. At $\tau = 0.5$, recall improves to 0.763 at the cost of 62 false positives, giving a higher F1 of 0.592 at the price of more unnecessary stops.

Table 5: Algorithm 1 stopping-rule evaluation across $N = 246$ doubling-pair observations from seventeen tasks. Ground truth: STOP iff doubling marginal < 0.005 . AUC = 0.752. TP: true positives (saturation correctly identified); FP: false positives (unnecessary stop); FN: false negatives (saturation missed).

τ	TP	FP	FN	Precision	Recall	F1
0.3 (calibrated default)	46	47	30	0.495	0.605	0.544
0.5 (max-F1 on corpus)	58	62	18	0.483	0.763	0.592

Decision guidance. We recommend $\tau = 0.3$ for *conservative* deployment settings where the cost of an unnecessary annotation pause is low but the cost of missing a real saturation event is moderate—for example, when annotation is expensive but can be re-initiated if accuracy remains low. We recommend $\tau = 0.5$ when annotation is very expensive and missing any saturation event is costly—the higher recall ensures that the algorithm does not continue collecting data past the point of diminishing returns. In both cases, the AUC of 0.752 establishes that the saturation index is a meaningful but imperfect stopping rule; it should be interpreted as probabilistic guidance rather than a hard guarantee.

4.6 Summary of Empirical Properties

We organise the empirical findings as four structural properties of the saturation index, each stated with its current level of evidential support.

1. **Spectral convergence.** $\text{erank}(\hat{\Sigma}_W^{(K)})$ converges monotonically to a task-specific asymptote erank_∞ for all seventeen tasks, with $\text{erank}_\infty < d = 50$ for all image tasks. *Strongly supported; no exceptions.*
2. **Phase structure and predictive correlation.** The saturation index achieves pooled Spearman $\rho = 0.548$ ($p = 1.1 \times 10^{-20}$, $N = 246$); within-task median $\rho = 0.811$ across sixteen conforming tasks; all three pairwise phase comparisons are significant ($p \leq 0.008$); exploration-phase coverage is now well-powered ($n = 30$, CI width 1.9pp). Algorithm 1 achieves AUC = 0.752 as a stopping rule. *Primary result. Strongly supported at multiple levels of analysis.*
3. **Decoupling.** Spearman $r_s = 0.380$ ($p = 0.133$, $N = 17$) between erank_∞ and A_∞ ; no significant monotone relationship. Task pairs spanning 40 percentage points of accuracy at near-identical erank_∞ provide the strongest individual demonstrations. *Weak-form decoupling supported with $N = 17$; strong-form independence requires ~ 35 tasks for 80% power.*
4. **Asymptotic redundancy.** In the deep saturation sub-regime ($\mathcal{S}(K) \leq 0.02$), accuracy fluctuates within the noise floor for all but the highest-dimensional task (MNIST 3 vs. 8, $p = 0.072$, $\bar{\delta} = -0.0022$). Redundancy is invariant to regularization. CIFAR tasks exhibit spectral saturation without accuracy saturation, a representational limitation the index correctly diagnoses. *Plateau/fluctuation strongly supported; accuracy decrease marginally supported in one task only.*

5 Ablation Studies

This section evaluates the robustness of three design choices: the PCA projection dimension d , the L2 regularization strength of the logistic regression classifier, and the classifier family itself. We now run each ablation across multiple tasks spanning different modalities and effective-rank regimes, so that the conclusions are not contingent on a single task. The expanded corpus (Section 4) consists of 17 binary tasks; the ablations below cover representative subsets that span the full range of erank_∞ values observed in Table 2.

5.1 Sensitivity to PCA Projection Dimension

Setup. We repeat the full K -sweep with varying PCA dimension d on three tasks chosen to cover distinct regimes of intrinsic dimensionality and modality: MNIST 0 vs. 1 (image, high effective rank, $\text{erank}_\infty \approx 32$);

Table 6: PCA dimension sensitivity across three tasks. erank_∞ : effective rank at the largest K . Peak Acc: maximum mean accuracy over the K -sweep. $\mathcal{S}_{\text{last}}$: saturation index at the final shot count. ρ_d : within-sweep Spearman correlation between $\mathcal{S}(K)$ and the doubling marginal $\bar{\delta}(K)$; all ρ_d significant at $p \leq 0.05$ except USPS $d = 100$ ($p = 0.033$). All means over $T = 50$ trials.

Task	d	erank_∞	Peak Acc	$\mathcal{S}_{\text{last}}$	ρ_d
MNIST 0v1	20	19.04	0.9977	0.0046	0.954
	50	43.45	0.9966	0.0106	0.961
	100	82.23	0.9965	0.0201	0.932
USPS 1v2	20	18.36	0.9973	0.0359	1.000
	50	45.43	0.9963	0.0887	0.944
	100	87.97	0.9955	0.1718	0.615
Breast Cancer	5	4.62	0.9645	0.0462	0.783
	10	9.26	0.9435	0.0926	0.783
	20	17.83	0.9305	0.1783	0.854

USPS 1 vs. 2 (image, medium effective rank, $\text{erank}_\infty \approx 15$); and Breast Cancer (tabular, low effective rank, $\text{erank}_\infty \approx 8$). For image tasks we use $d \in \{20, 50, 100\}$; for the tabular task, whose original feature space has 30 dimensions, we use $d \in \{5, 10, 20\}$. All other parameters ($T = 50$ trials, fixed seed sequence, the classifier of Section 3.5) are held constant.

Results. Table 6 reports the asymptotic effective rank erank_∞ , peak mean accuracy, the saturation index at the final shot count $\mathcal{S}_{\text{last}}$, and the within-sweep Spearman correlation ρ_d (computed from the set of doubling-pair marginals within each (task, d) setting).

Findings. Three properties are consistent across all nine (task, d) settings.

First, erank_∞ scales with d in every task, as expected from the spectral capacity of the covariance estimator (Section 3.6.3). Across all nine settings, $\text{erank}_\infty < d$, confirming that the class-conditional geometry, not the ambient dimension, is the binding constraint on spectral spread.

Second, peak accuracy is stable within each task across d . For MNIST 0 vs. 1, the range is 0.9965–0.9977 (span 0.0012). For USPS 1 vs. 2, the range is 0.9955–0.9973 (span 0.0018). For Breast Cancer, the range is 0.9305–0.9645 (span 0.0340); the wider span here reflects that, at $d = 20$, the larger ambient space enables the classifier to populate dimensions that encode noise rather than discriminative signal, which modestly penalises peak accuracy.

Third, the within-sweep Spearman correlation ρ_d is positive and large in all nine settings, ranging from 0.615 to 1.000. The weakest value ($\rho_d = 0.615$, USPS 1v2 at $d = 100$) occurs when the final $\mathcal{S}_{\text{last}} = 0.172$ is far from deep saturation, so the sweep does not fully traverse the saturation phase and the correlation is estimated on a shorter effective range of $\mathcal{S}$. In all other settings, $\rho_d \geq 0.78$, confirming that the phase structure is preserved across the full d range tested.

Interpretation. The saturation index $\mathcal{S}(K) = \text{erank}(\hat{\Sigma}_W^{(K)})/K$ is a ratio, and the phase boundaries $\mathcal{S} = 1.0$ and $\mathcal{S} = 0.3$ are calibrated on this ratio rather than on the absolute effective rank. Changing d shifts the numerator, but the phase transitions occur at the same values of $\mathcal{S}$ across all settings. For image tasks, $d = 50$ is a practical default: it is large enough to contain the full effective rank ($\text{erank}_\infty < 50$ for all image tasks in our corpus) and small enough to keep the saturation regime accessible within the K -sweep. For tabular data, d should be set to a value that does not exceed the number of genuinely informative features; inflating d beyond this point degrades peak accuracy without improving the saturation signal.

Table 7: L2 regularization sensitivity at K_{sat} for three tasks ($T = 50$ trials). Δ_{max} : maximum pairwise difference in mean accuracy across the three C levels. Significance assessed by paired Wilcoxon signed-rank test.

Task	C	$\bar{A}_{K_{\text{sat}}}$	$\hat{\sigma}$
Breast Cancer	∞	0.9440	0.0263
$\Delta_{\text{max}} = 0.0206$	1.0	0.9620	0.0196
$p < 0.0001$	0.1	0.9646	0.0186
USPS 1v2	∞	0.9973	0.0036
$\Delta_{\text{max}} = 0.0010$	1.0	0.9968	0.0040
$p = 0.041$	0.1	0.9963	0.0047
Fashion 0v1	∞	0.9851	0.0057
$\Delta_{\text{max}} = 0.0001$	1.0	0.9851	0.0054
$p = 0.432$	0.1	0.9851	0.0052

5.2 Sensitivity to L2 Regularization Strength

Setup. We compare $C \in \{\infty, 1.0, 0.1\}$ at the task’s saturation shot count K_{sat} across three tasks: Breast Cancer ($K_{\text{sat}} = 48$, low erank, tabular), USPS 1 vs. 2 ($K_{\text{sat}} = 48$, medium erank, image), and Fashion 0 vs. 1 ($K_{\text{sat}} \approx 4,096$, high erank, image). This choice of K_{sat} deliberately stresses the ablation: at the saturation ceiling, any interaction between regularization and the optimization landscape is most likely to manifest.

Results. Table 7 reports mean accuracy, standard deviation, and maximum pairwise difference Δ_{max} for each task and regularization level. Pairwise Wilcoxon signed-rank tests are conducted across the 50 trial pairs.

Findings. The results split cleanly by task modality and effective rank.

Fashion 0 vs. 1 (high erank, image): all three regularization levels produce mean accuracies of 0.9851 to four decimal places ($\Delta_{\text{max}} = 0.0001$, $p = 0.432$). The saturation plateau is completely invariant to regularization.

USPS 1 vs. 2 (medium erank, image): the maximum pairwise difference is $\Delta_{\text{max}} = 0.0010$, one order of magnitude below the standard deviation. The Wilcoxon test between $C = \infty$ and $C = 0.1$ is marginally significant ($p = 0.041$), but the absolute gap of 0.001 is practically negligible, amounting to a difference of one misclassified example per 1,000.

Breast Cancer (low erank, tabular): the unregularized setting ($C = \infty$) yields $\bar{A} = 0.9440$, while strong regularization ($C = 0.1$) yields $\bar{A} = 0.9646$ —a statistically and practically significant difference of $\Delta = 0.0206$ ($p < 0.0001$). Notably, both regularized settings perform similarly (difference 0.0026, $p = 0.233$), suggesting that any positive regularization suffices to stabilise the solution.

Interpretation. The Breast Cancer discrepancy requires careful contextualisation. At $K_{\text{sat}} = 48$, the logistic regression training set contains 96 examples (48 per class) in $d = 50$ ambient dimensions (standard setting). In this regime, the unregularized maximum-likelihood solution lies in a nearly flat region of the loss landscape—many parameter vectors achieve near-zero training loss—and the optimizer’s solution has higher trial-to-trial variance ($\hat{\sigma} = 0.0263$ vs. 0.0186 for $C = 0.1$). The accuracy gap is therefore a consequence of estimator variance in the small- K , low-erank regime, not of an interaction between the saturation *index* and regularization. The saturation threshold ($\mathcal{S}(K) \leq 0.3$) still correctly identifies $K_{\text{sat}} = 48$ as the stopping point across all regularization levels; regularization affects the absolute accuracy *at* the ceiling, not the *location* of the ceiling in the K -axis.

The practical implication is that, for low-dimensional tabular tasks with small K_{sat} , practitioners should apply moderate regularization independently of the saturation stopping criterion. For image tasks—which constitute the majority of few-shot learning benchmarks and for which K_{sat} is large—the saturation plateau is regularization-invariant.

Table 8: Classifier sensitivity for MNIST 3 vs. 8 at $K = 4,096$ ($d = 50$, $T = 50$ trials, $\mathcal{S} = 0.0091$). Gap: difference in mean accuracy relative to logistic regression. Pairwise Wilcoxon $p < 10^{-9}$ for both non-LR classifiers.

Classifier	$\bar{A}_{4096}$	$\hat{\sigma}$	Gap vs LR
Logistic Regression	0.9630	0.0096	—
Nearest Centroid	0.8928	0.0168	−0.0702
Linear SVM	0.7309	0.0786	−0.2321

5.3 Sensitivity to Classifier Family

Motivation. The saturation index $\mathcal{S}(K)$ measures spectral saturation of the pooled within-class covariance $\hat{\Sigma}_W^{(K)}$, a property of the support set that does not reference the downstream classifier. A natural question is whether geometric saturation of the support set translates to accuracy saturation for classifiers that do not explicitly use the covariance—for example, the nearest-centroid classifier (NC), which uses only class means, or a linear SVM, which uses the max-margin boundary.

Setup. We compare three classifiers at $K = 4,096$ for MNIST 3 vs. 8, holding $d = 50$ and $T = 50$ trials fixed: logistic regression ($C \rightarrow \infty$, the main-experiment setting); nearest-centroid classification; and a linear SVM (default `scikit-learn` settings, $C = 1.0$). $K = 4,096$ is chosen because MNIST 3 vs. 8 is firmly in the saturation phase at this shot count ($\mathcal{S} = 0.0091$), so any residual accuracy gain from additional support examples should be near-zero for a classifier that exploits the full covariance structure.

Results. Table 8 reports mean accuracy, standard deviation, and pairwise gap relative to logistic regression.

Findings and Interpretation. The three classifiers reach markedly different accuracy ceilings on the same support set, and each gap deserves separate analysis.

Nearest Centroid: the gap of 7.0 percentage points is large and highly significant ($p < 10^{-9}$), but reflects a fundamental difference in what information each classifier extracts from the support set. NC uses only the class centroids $\hat{\mu}_c$ and classifies by Euclidean distance; it discards the within-class covariance $\hat{\Sigma}_W^{(K)}$ entirely. Logistic regression uses both the class-conditional means and the covariance structure to find a linear discriminant that accounts for the shape of the within-class distribution. The accuracy gap is therefore expected and does not contradict the saturation claim: both classifiers are geometrically saturated at $K = 4,096$ in the sense that additional examples would not change their decision boundaries meaningfully—NC because its centroid estimates are stable, LR because its covariance estimates are stable. The *ceilings* differ because the classifiers access different geometric features of the support set.

Linear SVM: the gap of 23.2 percentage points and the high trial-level standard deviation ($\hat{\sigma} = 0.079$, more than eight times the LR standard deviation) point to optimization instability rather than a principled classifier-family difference. At $K = 4,096$, the SVM dual problem has 8,192 support vectors in a 50-dimensional feature space—a strongly overdetermined regime in which the default L-BFGS solver fails to converge reliably within the default iteration budget. We regard the Linear SVM result as an optimization artifact rather than a meaningful estimate of the SVM accuracy ceiling, and exclude it from the substantive comparison.

Broader implication: the saturation index $\mathcal{S}(K)$ predicts *when* a classifier’s decision boundary stabilises relative to the support set size, not *what* accuracy ceiling that boundary achieves. This is consistent with the framing of Algorithm 1 as a stopping criterion: the algorithm signals that additional data collection is unlikely to change the classifier’s decision boundary, whatever its absolute accuracy may be. The accuracy ceiling itself is a property of the classifier family and the task difficulty jointly, and is not predicted by the saturation index.

Table 9: Summary of ablation findings. “Universal” denotes a finding that held across all tested settings without exception; “task-dependent” denotes a finding that was modified by task characteristics.

Ablation	Finding	Scope	Evidence
PCA dimension d	Phase structure preserved; $\rho_d \geq 0.78$	Universal (9 settings)	Table 6
PCA dimension d	Peak accuracy stable within $< 1\text{pp}$ (image tasks)	Image tasks	Table 6
Regularization C	Saturation plateau invariant ($\Delta < 0.002$)	Image tasks	Table 7
Regularization C	Unregularized LR underperforms ($\Delta = 0.021$)	Low-rank tabular, small K_{sat}	Table 7
Classifier	Accuracy ceiling is classifier-dependent	Universal	Table 8
Classifier	Saturation index predicts stopping, not ceiling	Universal	Section 5.3

5.4 Summary

Table 9 distills the key empirical finding from each ablation and its scope of validity.

Three conclusions follow. First, the phase diagram and the saturation index are robust to the PCA dimension across all tasks and modalities tested: the phase boundaries $\mathcal{S} = 1.0$ and $\mathcal{S} = 0.3$ are stable, and within-sweep Spearman correlations are uniformly strong ($\rho_d \geq 0.61$).

Second, regularization invariance of the accuracy plateau holds for image tasks but does not extend to low-dimensional tabular tasks at small saturation shot counts, where the unregularized estimator has elevated variance. Practitioners working with tabular data should select regularization strength independently of the saturation stopping criterion.

Third, the saturation index is a classifier-agnostic stopping criterion: it correctly identifies when the support set’s spectral geometry has stabilised, but different classifiers may reach different accuracy ceilings on the same saturated support set. Algorithm 1 is therefore most naturally interpreted as a data-collection stopping rule rather than an accuracy predictor.

6 Discussion

The experimental results of Section 4 and the ablation studies of Section 5 establish the saturation index as a reliable, label-free predictor of marginal annotation value across a diverse corpus of binary classification tasks. We now examine the boundary conditions of this reliability, identify the structural sources of its imperfection, and chart the directions along which the framework most naturally extends. This section is organized around seven questions that the results raise but do not fully resolve.

6.1 When Does a STOP Signal Mean “Collect Better Data”?

The saturation index $\mathcal{S}(K)$ measures the ratio of the effective spectral rank of $\hat{\Sigma}_W^{(K)}$ to the shot count K . When $\mathcal{S}(K)$ is small, the within-class sample covariance has converged to the population covariance Σ in the current d -dimensional feature space: additional examples will not change the linear discriminant direction equation 11 materially. This statement is about the *representation*, not about the task. It does not guarantee that the linear classifier in this representation can achieve high accuracy.

The distinction becomes critical in the CIFAR 3 vs. 5 (cat vs. dog) task, where $\mathcal{S}(4,096) = 0.004$ —deep saturation—but accuracy is 0.597, barely above chance. The support set has exhausted the spectral geometry of 50-dimensional PCA space; there is simply no linear direction in that space that separates cats from dogs reliably. Collecting more labeled examples at the same representation would not help: it would produce more precise estimates of a discriminant that is fundamentally inadequate.

We therefore propose a *joint diagnostic* that extends Algorithm 1 with an accuracy floor condition:

Extended Stopping Rule. Let A_{floor} be the minimum acceptable accuracy (set by domain requirements). Given current support set $\mathcal{S}_K$:

1. If $\mathcal{S}(K) > \tau$: COLLECT — the geometry is not yet saturated; additional labels improve the discriminant.
2. If $\mathcal{S}(K) \leq \tau$ and $A(K) \geq A_{\text{floor}}$: STOP — the representation is saturated and accuracy is sufficient.
3. If $\mathcal{S}(K) \leq \tau$ and $A(K) < A_{\text{floor}}$: RE-REPRESENT — the representation has been exhausted without reaching the accuracy target; additional annotation will not close the gap.

The RE-REPRESENT outcome is diagnostic rather than defeatist. It tells the practitioner that the annotation budget has been exhausted *relative to the current feature space*, and that the bottleneck has shifted from data quantity to representational quality. The appropriate response is to upgrade the representation—switching from raw pixel PCA to a pretrained embedding (Section 6.6), applying domain-specific augmentation, or collecting examples from a different modality—rather than annotating more examples in the same feature space.

We note two practical considerations. First, A_{floor} must be set before the annotation campaign begins, ideally from domain knowledge (e.g., a clinical diagnostic system requires sensitivity ≥ 0.95) or from a baseline comparison. Setting A_{floor} post-hoc on the basis of the observed accuracy would make the criterion circular. Second, the RE-REPRESENT decision is based on the test set accuracy $A(K)$, which requires labeled test examples. In fully label-free settings where test labels are unavailable, the decision must be deferred until a small validation sample can be obtained; the saturation index alone cannot distinguish case (ii) from case (iii).

6.2 Within-Task vs. Pooled Spearman: Which Statistic Matters

The within-task Spearman median of $\rho = 0.811$ across sixteen conforming tasks is the statistic that directly tests the prospective claim: *within a fixed task, as $\mathcal{S}(K)$ decreases with increasing K , does marginal accuracy gain also decrease?* The answer is yes, strongly and consistently. The pooled correlation of $\rho = 0.548$ ($p = 1.1 \times 10^{-20}$, $N = 246$) is deliberately reported alongside but is a secondary statistic: it conflates within-task and between-task variation and is conservative relative to the core signal. Understanding why matters for correctly interpreting the evidence.

The pooled analysis treats all 246 observations as exchangeable draws from a common distribution of $(\mathcal{S}(K), \Delta A(K))$ pairs. This conflates two distinct sources of variation: *within-task* variation, where $\mathcal{S}(K)$ decreases as K grows for a fixed task; and *between-task* variation, where different tasks have different accuracy scales, noise floors, and phase-transition locations. The between-task variation does not directly test the predictive claim—it reflects the heterogeneity of the corpus rather than the reliability of $\mathcal{S}(K)$ as a predictor.

To see why between-task variation can dilute the correlation, consider two tasks: one where $\mathcal{S}(K) \in [0.01, 0.50]$ with $\Delta A(K) \in [0.001, 0.03]$ (a low-accuracy task that saturates early), and one where $\mathcal{S}(K) \in [0.01, 1.20]$ with $\Delta A(K) \in [0.001, 0.10]$ (a high-accuracy, high-dimensional task that traverses all three phases). When pooled, the high-accuracy task contributes many observations at large $\mathcal{S}$ and large ΔA , while the low-accuracy task contributes many at small $\mathcal{S}$ and small ΔA —an apparent positive correlation. But this correlation is partly a task-identity effect rather than a genuine $\mathcal{S}$ – ΔA relationship within either task.

The within-task correlation ($\rho = 0.811$ median) directly tests the prospective claim that matters for a practitioner: *within a fixed task, as $\mathcal{S}(K)$ decreases with increasing K , does $\Delta A(K)$ also decrease?* The answer is yes, strongly and consistently, in sixteen of seventeen tasks. The single exception, CIFAR 3 vs. 5 ($\rho = -0.525$), is structurally explained by representational inadequacy (Section 4.4), not by a failure of the saturation index as a geometric tool.

We therefore recommend that future work evaluating predictors of annotation value report within-task correlations alongside pooled correlations, rather than using the pooled statistic as the primary evidence for a within-task prediction claim. The two statistics answer different questions.

6.3 What the Decoupling Claim Can and Cannot Assert

The decoupling of erank_∞ and A_∞ is one of the paper’s structural claims, and it requires precise qualification.

What is supported. With $N = 17$ tasks, the Spearman correlation between erank_∞ and A_∞ is $r_s = 0.380$ ($p = 0.133$): no significant monotone relationship. Task pairs spanning 40 percentage points of accuracy at near-identical erank_∞ (USPS 1 vs. 2 and CIFAR 3 vs. 5) provide individual demonstrations that are not susceptible to the power limitation. The structural argument is grounded in the theory: Bayes error depends on the Mahalanobis distance $\Delta^2 = (\mu_1 - \mu_0)^\top \Sigma^{-1} (\mu_1 - \mu_0)$ between class means (equation 12), while erank_∞ depends on the spectral structure of Σ . These are geometrically orthogonal by construction: a within-class distribution can have large spectral volume ($\text{erank}(\Sigma)$ large) while still being highly separated from the other class (Δ large), or vice versa. This orthogonality is not a statistical claim about a particular corpus; it is a mathematical property of the Gaussian model.

What is not yet supported. The *strong form* of the decoupling claim—that erank_∞ and A_∞ are statistically independent across tasks drawn from the space of binary classification problems—cannot be established at $N = 17$. A power analysis at $r_s = 0.38$, $\alpha = 0.05$ (two-tailed) indicates that $N \approx 55$ tasks would be needed to achieve 80% power. The Pearson correlation ($r = 0.499$, $p = 0.041$) is marginally significant and may reflect a genuine weak positive relationship or a modality confound: all seven high- erank tasks in the corpus are MNIST digit pairs, which are highly linearly separable due to distinctive stroke patterns, while the low- erank group includes both easy tasks (USPS, Breast Cancer) and inherently hard ones (CIFAR, Fashion clothing pairs). Disentangling geometric complexity from category discriminability would require a corpus with matched modalities and systematically varied class-pair difficulty—a design for future work.

The practical implication of weak decoupling. Even the weak-form claim has a practically important implication: annotators and researchers should not interpret a high asymptotic effective rank (erank_∞ large) as a signal of low task difficulty. MNIST 3 vs. 8 is geometrically the most complex task in the corpus (highest $\text{erank}_\infty = 37.45$) and achieves one of the highest accuracy ceilings ($A_\infty = 0.965$). The geometric complexity determines *how much data is needed to saturate the support set geometry*—the saturation shot count K^* —not how accurate the eventual classifier will be. These are genuinely separate design concerns in an annotation pipeline: one must both estimate the annotation cost (K^*) and set the accuracy requirement (A_{floor}) independently.

6.4 Sources and Structure of Algorithm 1’s Errors

Algorithm 1 achieves $\text{AUC} = 0.752$ at $N = 246$ observations, which places it substantially above chance but leaves a gap of 0.248 from the perfect classifier. This gap has two structurally distinct sources that have different remedies.

Type I: Structural false positives (representation-limited tasks). Thirteen of the 47 false positives at $\tau = 0.3$ originate from the two CIFAR tasks, where $\mathcal{S}(K)$ is small (saturation predicted) but $\Delta A(K)$ is non-negligible (the classifier is still improving as K increases, albeit slowly). These false positives are *irreducible* by any choice of τ : no matter where the threshold is set on the $\mathcal{S}$ axis, CIFAR observations in the saturation phase will have unexpectedly large ΔA relative to what the saturation index predicts. They arise because the geometric saturation of the representation does not imply accuracy saturation when the representation is inadequate, and no spectral diagnostic can recover information not present in the feature space. The remedy is not to adjust τ but to apply the joint diagnostic of Section 6.1: a STOP signal accompanied by below-floor accuracy triggers the RE-REPRESENT outcome rather than accepted early termination.

Type II: Stochastic false positives (early-saturation variance). The remaining false positives are concentrated in the early saturation sub-regime ($0.1 < \mathcal{S}(K) \leq 0.3$), where the saturation index has crossed $\tau = 0.3$ but some tasks are still on the descending portion of their accuracy trajectory. These arise because $\tau = 0.3$ is calibrated on the mean behavior across tasks, and individual tasks have heterogeneous accuracy–saturation coupling: some tasks (e.g., Fashion 2 vs. 6) are essentially flat from the moment $\mathcal{S} < 0.5$, while

others (e.g., MNIST 3 vs. 8) continue improving measurably until $\mathcal{S} < 0.02$. Raising the threshold to $\tau = 0.5$ eliminates most of these at the cost of 15 additional false positives of the other kind (unnecessary data collection past genuine saturation), giving the higher recall of 0.763 reported in Table 5.

A principled threshold selection criterion. The optimal τ for a deployment context depends on the relative costs of the two error types. Let c_{FP} be the cost of an unnecessary annotation pause (annotators are idle, or the next batch of labels is not collected) and c_{FN} be the cost of annotating past saturation (wasted annotation budget). The optimal threshold satisfies:

$$\tau^* = \operatorname{argmax}_{\tau} [\text{TPR}(\tau) - \frac{c_{\text{FP}}}{c_{\text{FN}}} \cdot \text{FPR}(\tau)], \quad (26)$$

which corresponds to the point on the ROC curve (Figure 5) where the slope equals $c_{\text{FP}}/c_{\text{FN}}$. For annotation settings where the cost of pausing is low relative to the cost of wasted labels ($c_{\text{FP}} \ll c_{\text{FN}}$), τ^* should be chosen toward the high-recall end of the ROC curve ($\tau \approx 0.5\text{--}0.7$). For settings where annotation batches must be committed in advance and pausing is expensive ($c_{\text{FP}} \approx c_{\text{FN}}$), $\tau \approx 0.3$ is the appropriate conservative default.

6.5 Extension to N -Way Classification

The saturation index as defined in Section 3.2 is formulated for binary ($N = 2$) classification, but the core construction extends to N -way problems with only notational changes.

The generalized pooled within-class covariance. For N -way classification with K support examples per class, define the per-class covariances $\hat{\Sigma}_c$ as in equation 5 for each $c \in \{1, \dots, N\}$, and generalize Definition 2 to:

$$\hat{\Sigma}_W^{(K,N)} = \frac{1}{N} \sum_{c=1}^N \hat{\Sigma}_c. \quad (27)$$

The saturation index is then $\mathcal{S}(K, N) = \operatorname{erank}(\hat{\Sigma}_W^{(K,N)})/K$, with the same interpretation: the ratio of effective spectral rank to shot count.

Theoretical considerations. Under the homoscedastic Gaussian model (Assumption 1 generalized to N classes), $\hat{\Sigma}_W^{(K,N)}$ is an average of N independent sample covariances, each of rank at most $K - 1$ in $\mathbb{R}^d$. The operator-norm bound of Proposition 1 applies to each per-class term and, by the triangle inequality, to their average with the same scaling: $\mathbb{E}[\|\hat{\Sigma}_W^{(K,N)} - \Sigma\|_{\text{op}}] \leq C\sigma_{\psi}^2 \|\Sigma\|_{\text{op}} \sqrt{r(\Sigma)/K}$. Crucially, this bound depends on K (the per-class shot count) and $r(\Sigma)$ (the intrinsic dimension of the shared covariance), *not* on N . The saturation condition $\mathcal{S}(K, N) \leq \tau$ should therefore be calibrated per-class (on K) regardless of the number of classes.

Empirical re-calibration. The phase boundaries $\tau = 0.3$ and the exploration/transition boundary $\tau = 1.0$ are calibrated on binary tasks and may shift for $N > 2$. Intuitively, with more classes, the pooled covariance aggregates from more per-class estimates, potentially producing a smoother spectral distribution at any given K . We expect $\operatorname{erank}(\hat{\Sigma}_W^{(K,N)})$ to be larger at fixed K as N grows, since more classes contribute spectral energy in distinct directions (assuming the class means are well-separated). This would require a larger K to drive $\mathcal{S}(K, N)$ below τ , which is consistent with the intuition that N -way classification requires more data per class than binary classification. Empirical re-calibration on a corpus of N -way tasks is needed before Algorithm 1 is applied in the multi-class setting.

6.6 Applicability to Pretrained Deep Representations

The dominant paradigm in modern few-shot learning uses a pretrained deep network as a fixed feature extractor, followed by a linear classifier (or nearest-centroid) head trained on the support set (Tian et al., 2020; Snell et al., 2017). This architecture is precisely the setting our framework is designed for: a fixed d -dimensional representation and a linear classifier whose performance is governed by the spectral geometry of the support set in that representation.

Why pretraining should make the index more informative. A high-quality pretrained embedding (e.g., from DINO, CLIP, or a task-relevant backbone) achieves two things that are favorable for the saturation index. First, the between-class Mahalanobis distance Δ is typically much larger in the embedding space than in raw pixel space—the representation is designed to separate classes. Second, the within-class covariance Σ in the embedding space tends to have lower intrinsic dimension ($\text{erank}(\Sigma)$ is smaller) because pretrained representations disentangle nuisance factors (pose, lighting, background) from semantic content, reducing within-class variability to the residual variation in the embedding. Both effects together imply that the saturation phase is reached at smaller K for high-quality pretrained representations—and that the saturation index, which predicts *when* saturation occurs, can provide an early stopping signal with very few support examples.

An empirical prediction. Consider the same binary task evaluated under two feature spaces: (a) 50-dimensional PCA of raw pixels (as in our experiments), and (b) a 512-dimensional CLIP embedding. We predict that erank_∞ will be substantially smaller under (b)—the class-conditional distributions will be more isotropic and lower-dimensional in the semantically organized CLIP space—and that the saturation threshold K^* will correspondingly be smaller. A system using CLIP embeddings might reach saturation at $K = 8\text{--}32$, compared to $K = 64\text{--}2,048$ for PCA representations. This prediction is testable and would quantify the geometric benefit of representation learning in the annotation-budget context.

The dimensionality question. In a 512-dimensional CLIP embedding, the PCA reduction step (Section 3.6.1) should be applied to map the features to $d \leq 50$ dimensions before computing the saturation index. Alternatively, the effective rank can be computed in the full embedding space ($d = 512$) with a higher default threshold—the ablation study of Section 5.1 shows that the phase structure is preserved across d , with τ stable on the $\mathcal{S}$ axis. The practitioner should choose d large enough that $\text{erank}_\infty < d$ (so the ambient dimension is not the binding constraint), but not so large that the eigendecomposition becomes computationally burdensome; the cubic scaling $O(d^3)$ makes $d \leq 512$ inexpensive on commodity hardware.

6.7 Calibrating the Threshold for New Domains

The default threshold $\tau = 0.3$ was calibrated on a corpus of seventeen binary classification tasks spanning digits, fashion items, historical script, natural images, and clinical tabular data. While the corpus is deliberately diverse, there is no guarantee that $\tau = 0.3$ is optimal for every deployment domain. We provide a concrete procedure for domain-specific re-calibration.

Re-calibration procedure.

1. Select $M \geq 5$ representative binary task pairs from the target domain (e.g., five pairs of disease conditions in a medical imaging pipeline, or five pairs of defect categories in an industrial inspection system). Task pairs should be chosen to span the expected range of difficulty in the domain.
2. For each task, run a K -sweep over a geometric grid (equation 20) with $T \geq 30$ independent trials, using the same preprocessing pipeline and classifier as will be used in deployment.
3. Compute the full set of doubling-pair observations $\{(\mathcal{S}(K), \Delta A(K))\}$ across all M tasks. Plot these as a scatter (Figure 1).
4. Identify the value τ^* such that for $\mathcal{S}(K) \leq \tau^*$, the mean $\overline{\Delta A(K)}$ falls below the domain’s target marginal gain ϵ (e.g., $\epsilon = 0.005$ for a 0.5% accuracy improvement threshold). Use this τ^* in place of the default 0.3.
5. Report τ^* alongside the calibration corpus size M and the target ϵ , so that readers can assess transferability.

How many calibration tasks are needed? The calibration procedure requires enough observations to estimate the phase boundary reliably. From our corpus, the transition/saturation boundary ($\mathcal{S} = 0.3$) is

stable across all seventeen tasks—suggesting that the boundary generalizes beyond the calibration corpus. However, the mean marginal gain at a given $\mathcal{S}$ can vary across tasks by a factor of 2–3 $\times$ (as shown by the width of the confidence intervals in Table 3). We recommend $M \geq 5$ tasks with at least $T = 30$ trials each to obtain a stable estimate of the within-phase mean marginal gain; this amounts to approximately $5 \times 10 \times 30 = 1,500$ training runs, completing in well under a minute at our computational costs (Section 3.7).

When to trust the default. The default $\tau = 0.3$ is reliable for image classification tasks with $\text{erank}_\infty \in [8, 38]$ and $A_\infty \in [0.83, 1.00]$, where all seventeen tasks in our corpus fall. For domains that fall substantially outside this range—for example, tabular data with $\text{erank}_\infty < 5$, or tasks where $A_\infty < 0.70$ suggesting representational inadequacy—re-calibration is strongly advised. For such tasks, the joint diagnostic of Section 6.1 (checking whether $A(K) \geq A_{\text{floor}}$ alongside $\mathcal{S}(K) \leq \tau$) provides an additional safeguard against misleading STOP signals.

6.8 Toward a Complete Annotation Pipeline

We close by sketching how the saturation index integrates into a practical annotation campaign, making concrete the abstract decision rule that Algorithm 1 formalizes.

The annotation loop. A practitioner begins with an empty support set and iterates the following loop until Algorithm 1 returns STOP:

1. **Annotate a batch.** Collect ΔK new labeled examples per class (e.g., $\Delta K = K$, doubling the support set each round, or ΔK fixed at some batch size determined by annotator workflow). This step is the expensive one: it involves human labor.
2. **Compute the saturation index.** Project the support set features to d dimensions (Section 3.6), compute the pooled within-class covariance $\hat{\Sigma}_W^{(K)}$, and evaluate $\mathcal{S}(K) = \text{erank}(\hat{\Sigma}_W^{(K)})/K$. This step takes $O(d^3)$ time: approximately 0.3 seconds at $d = 50$ and negligible relative to annotation cost.
3. **Apply the joint diagnostic.** If $\mathcal{S}(K) > \tau$: return to Step 1 (exploration or transition phase; annotation remains productive). If $\mathcal{S}(K) \leq \tau$ and $A(K) \geq A_{\text{floor}}$: STOP (saturation reached; further annotation is wasteful). If $\mathcal{S}(K) \leq \tau$ and $A(K) < A_{\text{floor}}$: RE-REPRESENT (the representation is exhausted; the bottleneck is not data).

Monitoring $\mathcal{S}(K)$ as a curve, not just a threshold. Beyond the binary stopping decision, the trajectory of $\mathcal{S}(K)$ provides ongoing diagnostic information. A practitioner who plots $\mathcal{S}(K)$ versus K at each round has a real-time view of which phase the annotation campaign is in. A rapid decrease in $\mathcal{S}(K)$ (the transition from exploration to transition to saturation) signals that the support set is approaching saturation; slowing the annotation cadence and computing $\mathcal{S}(K)$ more frequently at this stage reduces the risk of over-shooting. A $\mathcal{S}(K)$ that decreases but remains above $\tau = 0.3$ for many rounds signals that the task is geometrically demanding (erank_∞ large) and that a substantial investment in annotation is warranted before saturation is expected.

Batch size selection. The theory and experiments in this paper are developed for a doubling schedule ($\Delta K = K$), which provides logarithmic coverage of the K -axis and ensures that the saturation index is evaluated in each phase. In practice, annotators often work in fixed-size batches (ΔK constant). The saturation index is well-defined for any batch schedule; the only adjustment is that the marginal gain should be computed as $\Delta A(K_{\text{new}}) = A(K_{\text{new}}) - A(K_{\text{prev}})$ for the specific transition being evaluated. However, fixed-batch schedules risk skipping over the saturation threshold if ΔK is large: a single batch might carry the system from $\mathcal{S} > \tau$ to $\mathcal{S} \ll \tau$, missing the opportunity to stop at the threshold. We recommend checking $\mathcal{S}(K)$ after every batch regardless of schedule, and reducing batch size to $\Delta K = 1$ or $\Delta K = 5$ once $\mathcal{S}(K)$ enters the transition phase ($0.3 < \mathcal{S}(K) \leq 1.0$), to allow fine-grained monitoring near the stopping point.

Open problems. Several extensions remain for future work. (i) *Multi-class calibration*: validating the generalized index (equation 27) on N -way benchmarks such as miniImageNet and tieredImageNet, where $N = 5$ or $N = 20$ is standard. (ii) *Pretrained embedding evaluation*: measuring how the saturation shot count K^* changes as a function of backbone quality (e.g., random weights vs. supervised pretraining vs. contrastive pretraining) on a fixed task, and whether erank_∞ in the embedding space predicts K^* . (iii) *Asymmetric class support*: extending the framework to settings where the two classes have different numbers of support examples ($K_0 \neq K_1$), which arises in class-imbalanced domains such as anomaly detection. (iv) *Non-stationary distributions*: characterizing the behavior of the saturation index when the data distribution drifts over time, so that a previously saturated support set becomes under-representative of the current distribution. (v) *Corpora of sufficient scale*: assembling a corpus of $M \geq 35$ binary tasks across diverse domains to resolve the decoupling claim with 80% power and to validate the $\tau = 0.3$ threshold more broadly.

Closing remark. The saturation index is, at its core, a measurement: a precise, computable quantity that tracks the geometric exhaustion of the support set in real time. Its value to a practitioner lies not in replacing judgment but in replacing guesswork. Before $\mathcal{S}(K)$ is computed, the question “have we collected enough examples?” has no principled answer from the support set alone. After $\mathcal{S}(K)$ is computed and falls below τ , the answer is precise: the geometry of the representation has been exhausted, and further annotation will not change the linear discriminant direction measurably. Whether to stop then depends on whether the accuracy achieved at that discriminant is sufficient—a decision that requires domain knowledge, not more data.

7 Conclusion

We began with a question that any annotator faces before dispatching the next labeling batch: has the support set already captured everything the current representation has to offer? Until now, this question had no principled answer that could be computed from the support features alone, without training the classifier or consulting held-out labels. We have shown that the answer is encoded in the spectral geometry of the support set’s pooled within-class covariance, and that a single scalar—the saturation index $\mathcal{S}(K) = \text{erank}(\hat{\Sigma}_W^{(K)})/K$ —tracks the exhaustion of that geometry in real time.

What the saturation index is and why it works. The saturation index is the ratio of the effective rank of the within-class sample covariance to the shot count. Its theoretical grounding is classical: under sub-Gaussian class-conditional distributions, the operator-norm estimation error of the pooled covariance scales as $O(\|\Sigma\|_{\text{op}} \sqrt{r(\Sigma)/K})$, where $r(\Sigma)$ is the intrinsic dimension of the population covariance (Proposition 1). When $K \gg \text{erank}(\Sigma)$ —equivalently, when $\mathcal{S}(K) \ll 1$ —the sample covariance is well-concentrated around the population covariance, the linear discriminant direction equation 11 has stabilized, and additional examples provide only noisy re-estimates of already well-determined spectral directions (Corollary 1.1). The saturation index converts this classical convergence result into a real-time observable, computable in $O(d^3)$ time—approximately 0.3 seconds at $d = 50$ —with no test labels and no trained classifier.

What the experiments establish. Across $N = 246$ doubling-pair observations from seventeen binary classification tasks spanning six datasets and two data modalities, the saturation index demonstrates strong and consistent within-task predictive power: sixteen of seventeen tasks have a positive within-task Spearman correlation between $\mathcal{S}(K)$ and marginal accuracy gain, with median $\rho = 0.811$ (IQR [0.720, 0.896]). This establishes that the saturation–marginal-gain relationship is genuine within individual tasks and not an artifact of pooling across tasks with different accuracy scales. The pooled Spearman correlation is $\rho = 0.548$ ($p = 1.1 \times 10^{-20}$, $N = 246$); the gap between this and the within-task median reflects between-task heterogeneity, not weakness of the core signal. The three-phase diagram (exploration, transition, saturation at mean marginal gains of 3.48%, 2.40%, and 0.82% respectively) is supported by all three pairwise Mann–Whitney comparisons ($p \leq 0.008$), with the exploration phase now represented by $n = 30$ observations—tightening its confidence interval from 5.3 to 1.9 percentage points. Algorithm 1, which issues a STOP recommendation when $\mathcal{S}(K) \leq 0.3$, achieves AUC = 0.752—meaningful probabilistic guidance, with known

structural error sources characterized in Section 6.4. The asymptotic effective rank and peak accuracy show no significant monotone relationship across tasks (Spearman $r_s = 0.380$, $p = 0.133$, $N = 17$), with task pairs spanning 40 percentage points of accuracy at near-identical erank_∞ —demonstrating that geometric complexity sets annotation cost, not accuracy ceiling.

What this work does not claim. Four boundaries of the claims are worth restating precisely. First, all experiments are for *binary classification with a fixed linear classifier* on PCA features; the saturation index’s behavior in N -way settings and with pretrained backbone representations is theoretically analyzed (Sections 6.5 and 6.6) and identified as the primary direction for future empirical work. Second, the saturation index predicts *when* a linear discriminant stabilizes in the current feature space; it does not predict the accuracy that discriminant achieves. When $\mathcal{S}(K) \leq \tau$ but $A(K)$ is below an acceptable floor, the bottleneck is representational inadequacy, not data insufficiency, and the correct action is to change the representation rather than collect more labels. Third, the decoupling of intrinsic dimensionality and task difficulty is supported in the weak form (no significant monotone relationship at $N = 17$) but the strong form (statistical independence) requires approximately 35 tasks for 80% power and remains open. Fourth, the phase boundaries $\tau = 0.3$ and $\tau = 1.0$ are calibrated on binary tasks with $\text{erank}_\infty \in [8.2, 37.8]$ and $A_\infty \geq 0.83$; domains outside this range require empirical re-calibration.

The broader implication. Geometric saturation of a support set is not an approximate or subjective condition—it is a well-defined event ($\text{erank}(\hat{\Sigma}_W^{(K)}) \rightarrow \text{erank}(\Sigma)$) that can be monitored continuously and at negligible computational cost. What the saturation index provides, therefore, is not a heuristic but a measurement: at any point in an annotation campaign, a practitioner can determine precisely whether the support set has exhausted the geometric information available in the representation, and whether additional annotation expenditure can be expected to move the linear discriminant. Framed this way, the annotation budget question is not a matter of judgment but of geometry—and geometry is computable.

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Reproducibility Statement

We have taken the following steps to ensure full reproducibility of all results reported in this paper.

Code and data. Complete source code, dataset loading scripts, and cached experimental results are available at <https://github.com/MrArnav69/spectral-saturation>. The repository contains a self-contained Python implementation of Algorithm 1, the full preprocessing pipeline (Section 3.6), and scripts to reproduce every figure and table in the paper from either the cached results or by re-running the experiments from scratch.

Random seeds and determinism. All random variation is controlled by a deterministic seed sequence. For each (K, task) pair, trial $t \in \{1, \dots, T\}$ uses seed t to draw the support set (Section 3.7). The test set is fixed across all trials and all values of K for a given task, so accuracy differences across K reflect only support-set variation. No randomness enters the saturation index computation itself: given a fixed support set, $\mathcal{S}(K)$ is a deterministic function of the eigenvalues of $\hat{\Sigma}_W^{(K)}$.

Datasets. All six datasets are publicly available and can be obtained programmatically without registration: MNIST, Fashion-MNIST, Kuzushiji-MNIST, and USPS via OpenML (Vanschoren et al., 2014); CIFAR-

10 via the Keras datasets API (Krizhevsky, 2009); Breast Cancer Wisconsin via `sklearn.datasets` (Pedregosa et al., 2011). Exact version identifiers and download hashes are provided in the repository README.

Hyperparameters and implementation. There are no hyperparameters that require tuning to reproduce the main results. The classifier uses a fixed inverse regularization parameter $C = 10^9 \approx \infty$ (Section 3.5); the PCA dimension is fixed at $d = 50$ for all image tasks and $d = d_0 = 30$ for the tabular task (Section 3.6); the saturation threshold is $\tau = 0.3$ (Algorithm 1). The only dataset-dependent parameter is $K_{\max}$, which is set by dataset size and reported in Table 1. All ablation sweeps over d , C , and classifier family use the values specified in Section 5 and the repository configuration files.

Compute. The full experimental suite—seventeen tasks, up to seventeen values of K , and $T = 50$ trials each—completes in approximately 150 seconds on a single CPU core (Apple MacBook Pro M3 Pro, 18 GB unified memory). No GPU is required. Ablation experiments add approximately 60 additional seconds. The saturation index itself requires $O(d^3)$ computation per trial (≈ 0.3 s at $d = 50$); all remaining time is dominated by logistic regression fitting via L-BFGS.

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A Full Per-Task Result Tables

Tables 10–26 report the complete K -sweep results for all seventeen binary classification tasks in the experimental corpus. Each table lists, for every shot count K in the grid equation 20: the mean test accuracy $\bar{A}(K)$ and its trial-level standard deviation $\hat{\sigma}(K)$ over $T = 50$ independent trials; the mean effective rank $\bar{e}(K) = \text{erank}(\hat{\Sigma}_W^{(K)})$; and the saturation index $\mathcal{S}(K) = \bar{e}(K)/K$. Within each table, the horizontal rule marks the first K at which $\mathcal{S}(K) \leq 0.3$ (entry into the saturation phase); rows above the rule are in the exploration or transition phase. The bold row identifies the first shot count achieving peak mean accuracy. Tasks marked † exhibit representational saturation without accuracy saturation (Section 4.4).

Table 10: Full K -sweep results for MNIST 0 vs. 1. $\bar{A}(K)$: mean test accuracy over $T = 50$ trials. $\hat{\sigma}(K)$: trial-level standard deviation. $\bar{e}(K)$: mean effective rank $\text{erank}(\hat{\Sigma}_W^{(K)})$. $\mathcal{S}(K) = \bar{e}(K)/K$: saturation index. **Bold**: first K achieving peak mean accuracy. Horizontal rule: entry into saturation phase ($\mathcal{S}(K) \leq 0.3$).

K	$\bar{A}(K)$	$\hat{\sigma}(K)$	$\bar{e}(K)$	$\mathcal{S}(K)$
2	0.8899	0.1050	1.65	0.8236
3	0.9473	0.0524	2.88	0.9614
4	0.9593	0.0491	3.76	0.9404
6	0.9790	0.0184	5.55	0.9253
8	0.9859	0.0149	6.67	0.8340
12	0.9870	0.0110	9.14	0.7613
16	0.9909	0.0069	9.87	0.6167
24	0.9940	0.0052	11.99	0.4995
32	0.9937	0.0050	13.02	0.4069
48	0.9950	0.0044	14.76	0.3076
64	0.9954	0.0039	15.72	0.2457
128	0.9946	0.0055	17.62	0.1377
256	0.9959	0.0033	20.40	0.0797
512	0.9953	0.0038	23.34	0.0456
1024	0.9967	0.0025	26.94	0.0263
2048	0.9968	0.0027	29.82	0.0146
4096	0.9964	0.0036	32.35	0.0079

Table 11: Full K -sweep results for MNIST 1 vs. 7. $\bar{A}(K)$: mean test accuracy over $T = 50$ trials. $\hat{\sigma}(K)$: trial-level standard deviation. $\bar{e}(K)$: mean effective rank $\text{erank}(\hat{\Sigma}_W^{(K)})$. $\mathcal{S}(K) = \bar{e}(K)/K$: saturation index. **Bold**: first K achieving peak mean accuracy. Horizontal rule: entry into saturation phase ($\mathcal{S}(K) \leq 0.3$).

K	$\bar{A}(K)$	$\hat{\sigma}(K)$	$\bar{e}(K)$	$\mathcal{S}(K)$
2	0.8360	0.1030	1.63	0.8171
3	0.8792	0.0701	2.82	0.9414
4	0.8998	0.0755	3.82	0.9550
6	0.9411	0.0311	4.99	0.8322
8	0.9593	0.0178	6.28	0.7846
12	0.9605	0.0164	8.39	0.6989
16	0.9559	0.0268	9.52	0.5952
24	0.9663	0.0160	11.32	0.4719
32	0.9612	0.0170	12.45	0.3892
48	0.9688	0.0134	14.77	0.3076
64	0.9683	0.0148	16.66	0.2604
128	0.9748	0.0089	20.44	0.1597
256	0.9810	0.0077	23.59	0.0921
512	0.9816	0.0066	27.40	0.0535
1024	0.9846	0.0073	29.98	0.0293
2048	0.9892	0.0050	32.84	0.0160
4096	0.9934	0.0038	34.55	0.0084

Table 12: Full K -sweep results for MNIST 2 vs. 7. $\bar{A}(K)$: mean test accuracy over $T = 50$ trials. $\hat{\sigma}(K)$: trial-level standard deviation. $\bar{e}(K)$: mean effective rank $\text{erank}(\hat{\Sigma}_W^{(K)})$. $\mathcal{S}(K) = \bar{e}(K)/K$: saturation index. **Bold**: first K achieving peak mean accuracy. Horizontal rule: entry into saturation phase ($\mathcal{S}(K) \leq 0.3$).

K	$\bar{A}(K)$	$\hat{\sigma}(K)$	$\bar{e}(K)$	$\mathcal{S}(K)$
2	0.8400	0.0861	1.85	0.9227
3	0.8782	0.0531	3.32	1.1057
4	0.8963	0.0574	4.76	1.1900
6	0.9178	0.0361	6.84	1.1397
8	0.9173	0.0394	8.64	1.0794
12	0.9296	0.0252	11.52	0.9600
16	0.9282	0.0231	13.55	0.8466
24	0.9294	0.0268	17.35	0.7231
32	0.9350	0.0163	20.02	0.6255
48	0.9364	0.0177	21.31	0.4440
64	0.9381	0.0187	24.81	0.3877
128	0.9518	0.0134	27.27	0.2130
256	0.9554	0.0125	29.63	0.1158
512	0.9619	0.0099	32.65	0.0638
1024	0.9777	0.0080	34.35	0.0335
2048	0.9817	0.0054	36.08	0.0176
4096	0.9844	0.0054	37.77	0.0092

Table 13: Full K -sweep results for MNIST 3 vs. 8. $\bar{A}(K)$: mean test accuracy over $T = 50$ trials. $\hat{\sigma}(K)$: trial-level standard deviation. $\bar{e}(K)$: mean effective rank $\text{erank}(\hat{\Sigma}_W^{(K)})$. $\mathcal{S}(K) = \bar{e}(K)/K$: saturation index. **Bold**: first K achieving peak mean accuracy. Horizontal rule: entry into saturation phase ($\mathcal{S}(K) \leq 0.3$).

K	$\bar{A}(K)$	$\hat{\sigma}(K)$	$\bar{e}(K)$	$\mathcal{S}(K)$
2	0.6993	0.0991	1.80	0.8990
3	0.7603	0.0774	3.28	1.0949
4	0.7968	0.0566	4.45	1.1127
6	0.8251	0.0466	6.56	1.0930
8	0.8416	0.0443	8.34	1.0428
12	0.8606	0.0389	10.59	0.8824
16	0.8722	0.0325	12.92	0.8076
24	0.8734	0.0333	16.77	0.6989
32	0.8919	0.0233	17.69	0.5527
48	0.8934	0.0245	21.26	0.4429
64	0.9001	0.0232	22.52	0.3519
128	0.9204	0.0175	25.85	0.2020
256	0.9276	0.0180	28.93	0.1130
512	0.9491	0.0119	31.17	0.0609
1024	0.9587	0.0102	33.76	0.0330
2048	0.9651	0.0067	35.44	0.0173
4096	0.9629	0.0098	37.45	0.0091

Table 14: Full K -sweep results for MNIST 4 vs. 7. $\bar{A}(K)$: mean test accuracy over $T = 50$ trials. $\hat{\sigma}(K)$: trial-level standard deviation. $\bar{e}(K)$: mean effective rank $\text{erank}(\hat{\Sigma}_W^{(K)})$. $\mathcal{S}(K) = \bar{e}(K)/K$: saturation index. **Bold**: first K achieving peak mean accuracy. Horizontal rule: entry into saturation phase ($\mathcal{S}(K) \leq 0.3$).

K	$\bar{A}(K)$	$\hat{\sigma}(K)$	$\bar{e}(K)$	$\mathcal{S}(K)$
2	0.7534	0.1010	1.82	0.9119
3	0.8122	0.0773	3.24	1.0785
4	0.8480	0.0744	4.57	1.1434
6	0.9014	0.0397	6.31	1.0521
8	0.9137	0.0340	7.85	0.9810
12	0.9256	0.0245	10.48	0.8734
16	0.9262	0.0256	12.66	0.7912
24	0.9358	0.0224	14.79	0.6164
32	0.9393	0.0244	17.39	0.5433
48	0.9401	0.0240	20.41	0.4253
64	0.9438	0.0142	22.73	0.3552
128	0.9528	0.0120	25.67	0.2005
256	0.9640	0.0097	28.74	0.1122
512	0.9678	0.0097	31.29	0.0611
1024	0.9784	0.0081	33.45	0.0327
2048	0.9834	0.0068	35.44	0.0173
4096	0.9855	0.0067	36.83	0.0090

Table 15: Full K -sweep results for MNIST 4 vs. 9. $\bar{A}(K)$: mean test accuracy over $T = 50$ trials. $\hat{\sigma}(K)$: trial-level standard deviation. $\bar{e}(K)$: mean effective rank $\text{erank}(\hat{\Sigma}_W^{(K)})$. $\mathcal{S}(K) = \bar{e}(K)/K$: saturation index. **Bold**: first K achieving peak mean accuracy. Horizontal rule: entry into saturation phase ($\mathcal{S}(K) \leq 0.3$).

K	$\bar{A}(K)$	$\hat{\sigma}(K)$	$\bar{e}(K)$	$\mathcal{S}(K)$
2	0.6204	0.0586	1.79	0.8936
3	0.6663	0.0719	3.19	1.0641
4	0.6944	0.0649	4.34	1.0862
6	0.7400	0.0667	5.89	0.9816
8	0.7685	0.0638	7.26	0.9079
12	0.8152	0.0458	10.24	0.8535
16	0.8395	0.0427	11.44	0.7153
24	0.8695	0.0278	13.83	0.5762
32	0.8724	0.0257	15.63	0.4886
48	0.8930	0.0263	19.20	0.4000
64	0.8966	0.0221	20.16	0.3151
128	0.9061	0.0236	23.30	0.1821
256	0.9221	0.0151	26.14	0.1021
512	0.9474	0.0106	27.95	0.0546
1024	0.9552	0.0106	30.99	0.0303
2048	0.9568	0.0095	33.11	0.0162
4096	0.9563	0.0096	35.60	0.0087

Table 16: Full K -sweep results for MNIST 5 vs. 8. $\bar{A}(K)$: mean test accuracy over $T = 50$ trials. $\hat{\sigma}(K)$: trial-level standard deviation. $\bar{e}(K)$: mean effective rank $\text{erank}(\hat{\Sigma}_W^{(K)})$. $\mathcal{S}(K) = \bar{e}(K)/K$: saturation index. **Bold**: first K achieving peak mean accuracy. Horizontal rule: entry into saturation phase ($\mathcal{S}(K) \leq 0.3$).

K	$\bar{A}(K)$	$\hat{\sigma}(K)$	$\bar{e}(K)$	$\mathcal{S}(K)$
2	0.6598	0.0878	1.80	0.8995
3	0.7158	0.0698	3.12	1.0388
4	0.7460	0.0690	4.25	1.0625
6	0.7952	0.0568	6.56	1.0935
8	0.8281	0.0538	7.81	0.9763
12	0.8483	0.0416	10.55	0.8793
16	0.8671	0.0313	12.38	0.7737
24	0.8665	0.0337	15.17	0.6322
32	0.8816	0.0311	16.47	0.5146
48	0.8832	0.0238	18.82	0.3922
64	0.8852	0.0251	18.96	0.2963
128	0.9004	0.0225	24.26	0.1895
256	0.9119	0.0152	26.54	0.1037
512	0.9390	0.0109	29.23	0.0571
1024	0.9442	0.0107	31.87	0.0311
2048	0.9501	0.0088	34.21	0.0167
4096	0.9478	0.0090	36.21	0.0088

Table 17: Full K -sweep results for Fashion-MNIST 0 vs. 1 (T-shirt vs. Trouser). $\bar{A}(K)$: mean test accuracy over $T = 50$ trials. $\hat{\sigma}(K)$: trial-level standard deviation. $\bar{e}(K)$: mean effective rank $\text{erank}(\hat{\Sigma}_W^{(K)})$. $\mathcal{S}(K) = \bar{e}(K)/K$: saturation index. **Bold**: first K achieving peak mean accuracy. Horizontal rule: entry into saturation phase ($\mathcal{S}(K) \leq 0.3$).

K	$\bar{A}(K)$	$\hat{\sigma}(K)$	$\bar{e}(K)$	$\mathcal{S}(K)$
2	0.8523	0.0800	1.67	0.8374
3	0.8835	0.0690	2.78	0.9277
4	0.9076	0.0631	3.56	0.8908
6	0.9177	0.0400	4.84	0.8065
8	0.9304	0.0344	5.54	0.6930
12	0.9500	0.0236	6.93	0.5772
16	0.9452	0.0171	7.84	0.4901
24	0.9614	0.0161	8.74	0.3643
32	0.9607	0.0164	9.56	0.2988
48	0.9617	0.0145	11.05	0.2302
64	0.9644	0.0120	11.99	0.1874
128	0.9722	0.0115	13.76	0.1075
256	0.9751	0.0098	16.09	0.0629
512	0.9722	0.0110	17.91	0.0350
1024	0.9776	0.0078	19.13	0.0187
2048	0.9840	0.0064	20.24	0.0099
4096	0.9856	0.0057	20.68	0.0050

Table 18: Full K -sweep results for Fashion-MNIST 2 vs. 6 (Pullover vs. Shirt). $\bar{A}(K)$: mean test accuracy over $T = 50$ trials. $\hat{\sigma}(K)$: trial-level standard deviation. $\bar{e}(K)$: mean effective rank $\text{erank}(\hat{\Sigma}_W^{(K)})$. $\mathcal{S}(K) = \bar{e}(K)/K$: saturation index. **Bold**: first K achieving peak mean accuracy. Horizontal rule: entry into saturation phase ($\mathcal{S}(K) \leq 0.3$).

K	$\bar{A}(K)$	$\hat{\sigma}(K)$	$\bar{e}(K)$	$\mathcal{S}(K)$
2	0.5548	0.0630	1.71	0.8565
3	0.5675	0.0871	2.70	0.8991
4	0.6162	0.0842	3.50	0.8750
6	0.6593	0.0568	4.52	0.7534
8	0.6653	0.0551	5.29	0.6613
12	0.6984	0.0502	6.49	0.5412
16	0.7072	0.0477	7.46	0.4661
24	0.7293	0.0400	8.09	0.3371
32	0.7293	0.0373	8.74	0.2730
48	0.7397	0.0390	9.77	0.2035
64	0.7334	0.0402	10.58	0.1653
128	0.7790	0.0214	11.63	0.0909
256	0.8132	0.0169	12.73	0.0497
512	0.8232	0.0172	13.38	0.0261
1024	0.8279	0.0167	13.92	0.0136
2048	0.8268	0.0193	14.26	0.0070
4096	0.8350	0.0186	14.46	0.0035

Table 19: Full K -sweep results for Fashion-MNIST 3 vs. 5 (Dress vs. Sandal). $\bar{A}(K)$: mean test accuracy over $T = 50$ trials. $\hat{\sigma}(K)$: trial-level standard deviation. $\bar{e}(K)$: mean effective rank $\text{erank}(\hat{\Sigma}_W^{(K)})$. $\mathcal{S}(K) = \bar{e}(K)/K$: saturation index. **Bold**: first K achieving peak mean accuracy. Horizontal rule: entry into saturation phase ($\mathcal{S}(K) \leq 0.3$).

K	$\bar{A}(K)$	$\hat{\sigma}(K)$	$\bar{e}(K)$	$\mathcal{S}(K)$
2	0.9522	0.0550	1.78	0.8917
3	0.9647	0.0332	3.08	1.0271
4	0.9658	0.0265	4.19	1.0463
6	0.9788	0.0186	5.90	0.9835
8	0.9860	0.0120	7.79	0.9732
12	0.9903	0.0063	9.84	0.8198
16	0.9902	0.0085	10.98	0.6862
24	0.9928	0.0058	13.65	0.5689
32	0.9922	0.0057	14.95	0.4671
48	0.9938	0.0050	16.85	0.3510
64	0.9956	0.0035	18.45	0.2882
128	0.9966	0.0022	20.18	0.1577
256	0.9964	0.0030	22.05	0.0861
512	0.9977	0.0022	23.38	0.0457
1024	0.9978	0.0022	24.74	0.0242
2048	0.9980	0.0023	26.11	0.0128
4096	0.9982	0.0022	27.36	0.0067

Table 20: Full K -sweep results for Fashion-MNIST 4 vs. 6 (Coat vs. Shirt). $\bar{A}(K)$: mean test accuracy over $T = 50$ trials. $\hat{\sigma}(K)$: trial-level standard deviation. $\bar{e}(K)$: mean effective rank $\text{erank}(\hat{\Sigma}_W^{(K)})$. $\mathcal{S}(K) = \bar{e}(K)/K$: saturation index. **Bold**: first K achieving peak mean accuracy. Horizontal rule: entry into saturation phase ($\mathcal{S}(K) \leq 0.3$).

K	$\bar{A}(K)$	$\hat{\sigma}(K)$	$\bar{e}(K)$	$\mathcal{S}(K)$
2	0.5795	0.0968	1.72	0.8589
3	0.6153	0.0782	2.73	0.9094
4	0.6406	0.0670	3.58	0.8960
6	0.6766	0.0582	4.50	0.7497
8	0.6908	0.0557	5.58	0.6973
12	0.7139	0.0532	6.63	0.5522
16	0.7412	0.0416	7.53	0.4704
24	0.7625	0.0352	8.52	0.3549
32	0.7674	0.0313	9.30	0.2906
48	0.7669	0.0377	10.39	0.2164
64	0.7582	0.0396	11.47	0.1793
128	0.7953	0.0236	12.37	0.0966
256	0.8306	0.0201	13.58	0.0531
512	0.8470	0.0171	14.42	0.0282
1024	0.8531	0.0173	14.99	0.0146
2048	0.8564	0.0143	15.51	0.0076
4096	0.8569	0.0167	15.70	0.0038

Table 21: Full K -sweep results for Fashion-MNIST 5 vs. 7 (Sandal vs. Sneaker). $\bar{A}(K)$: mean test accuracy over $T = 50$ trials. $\hat{\sigma}(K)$: trial-level standard deviation. $\bar{e}(K)$: mean effective rank $\text{erank}(\hat{\Sigma}_W^{(K)})$. $\mathcal{S}(K) = \bar{e}(K)/K$: saturation index. **Bold**: first K achieving peak mean accuracy. Horizontal rule: entry into saturation phase ($\mathcal{S}(K) \leq 0.3$).

K	$\bar{A}(K)$	$\hat{\sigma}(K)$	$\bar{e}(K)$	$\mathcal{S}(K)$
2	0.7161	0.0922	1.73	0.8645
3	0.7343	0.0691	2.79	0.9309
4	0.7599	0.0803	3.67	0.9167
6	0.7934	0.0581	4.83	0.8047
8	0.8043	0.0434	5.85	0.7312
12	0.8198	0.0470	7.78	0.6484
16	0.8257	0.0395	8.60	0.5376
24	0.8533	0.0326	10.60	0.4415
32	0.8482	0.0277	11.27	0.3522
48	0.8648	0.0286	13.60	0.2833
64	0.8666	0.0281	14.37	0.2245
128	0.8817	0.0197	16.68	0.1303
256	0.9091	0.0179	18.30	0.0715
512	0.9274	0.0137	20.44	0.0399
1024	0.9385	0.0118	22.46	0.0219
2048	0.9423	0.0107	24.04	0.0117
4096	0.9444	0.0123	26.10	0.0064

Table 22: Full K -sweep results for Kuzushiji-MNIST 0 vs. 1. $\bar{A}(K)$: mean test accuracy over $T = 50$ trials. $\hat{\sigma}(K)$: trial-level standard deviation. $\bar{e}(K)$: mean effective rank $\text{erank}(\hat{\Sigma}_W^{(K)})$. $\mathcal{S}(K) = \bar{e}(K)/K$: saturation index. **Bold**: first K achieving peak mean accuracy. Horizontal rule: entry into saturation phase ($\mathcal{S}(K) \leq 0.3$).

K	$\bar{A}(K)$	$\hat{\sigma}(K)$	$\bar{e}(K)$	$\mathcal{S}(K)$
2	0.8662	0.1080	1.82	0.9106
3	0.9080	0.0599	3.36	1.1212
4	0.9247	0.0341	4.76	1.1910
6	0.9383	0.0236	7.30	1.2162
8	0.9404	0.0205	9.66	1.2069
12	0.9545	0.0200	13.64	1.1367
16	0.9528	0.0199	16.47	1.0296
24	0.9593	0.0198	20.33	0.8471
32	0.9625	0.0173	23.60	0.7374
48	0.9648	0.0114	27.63	0.5756
64	0.9660	0.0133	29.55	0.4617
128	0.9679	0.0117	33.00	0.2578
256	0.9710	0.0097	35.11	0.1371
512	0.9721	0.0089	36.07	0.0704
1024	0.9813	0.0065	36.78	0.0359
2048	0.9860	0.0059	37.04	0.0181
4096	0.9871	0.0058	37.18	0.0091

Table 23: Full K -sweep results for USPS 1 vs. 2. $\bar{A}(K)$: mean test accuracy over $T = 50$ trials. $\hat{\sigma}(K)$: trial-level standard deviation. $\bar{e}(K)$: mean effective rank $\text{erank}(\hat{\Sigma}_W^{(K)})$. $\mathcal{S}(K) = \bar{e}(K)/K$: saturation index. **Bold**: first K achieving peak mean accuracy. Horizontal rule: entry into saturation phase ($\mathcal{S}(K) \leq 0.3$).

K	$\bar{A}(K)$	$\hat{\sigma}(K)$	$\bar{e}(K)$	$\mathcal{S}(K)$
2	0.9366	0.0713	1.39	0.6960
3	0.9593	0.0388	2.37	0.7908
4	0.9733	0.0313	3.06	0.7656
6	0.9818	0.0202	4.26	0.7095
8	0.9860	0.0226	5.27	0.6586
12	0.9931	0.0086	6.64	0.5530
16	0.9945	0.0067	7.61	0.4756
24	0.9950	0.0057	9.12	0.3799
32	0.9970	0.0037	9.99	0.3121
48	0.9973	0.0036	11.35	0.2364
64	0.9969	0.0041	11.74	0.1835
128	0.9972	0.0036	13.48	0.1053
256	0.9973	0.0036	14.24	0.0556
512	0.9973	0.0038	15.08	0.0295

Table 24: Full K -sweep results for Breast Cancer (Malignant vs. Benign). $\bar{A}(K)$: mean test accuracy over $T = 50$ trials. $\hat{\sigma}(K)$: trial-level standard deviation. $\bar{e}(K)$: mean effective rank $\text{erank}(\hat{\Sigma}_W^{(K)})$. $\mathcal{S}(K) = \bar{e}(K)/K$: saturation index. **Bold**: first K achieving peak mean accuracy. Horizontal rule: entry into saturation phase ($\mathcal{S}(K) \leq 0.3$).

K	$\bar{A}(K)$	$\hat{\sigma}(K)$	$\bar{e}(K)$	$\mathcal{S}(K)$
2	0.8381	0.0784	1.69	0.8427
3	0.8543	0.0760	2.72	0.9081
4	0.8807	0.0629	3.41	0.8535
6	0.8804	0.0608	4.37	0.7277
8	0.9097	0.0455	5.03	0.6287
12	0.9098	0.0430	5.90	0.4917
16	0.9197	0.0386	6.44	0.4024
24	0.9393	0.0260	6.86	0.2858
32	0.9370	0.0306	7.21	0.2252
48	0.9450	0.0244	7.64	0.1592
64	0.9449	0.0225	7.91	0.1236
100	0.9425	0.0212	8.20	0.0820

Table 25: Full K -sweep results for CIFAR-10 0 vs. 1 (Airplane vs. Automobile). $\bar{A}(K)$: mean test accuracy over $T = 50$ trials. $\hat{\sigma}(K)$: trial-level standard deviation. $\bar{e}(K)$: mean effective rank $\text{erank}(\hat{\Sigma}_W^{(K)})$. $\mathcal{S}(K) = \bar{e}(K)/K$: saturation index. **Bold**: first K achieving peak mean accuracy. Horizontal rule: entry into saturation phase ($\mathcal{S}(K) \leq 0.3$).

K	$\bar{A}(K)$	$\hat{\sigma}(K)$	$\bar{e}(K)$	$\mathcal{S}(K)$
2	0.6138	0.0903	1.82	0.9094
3	0.6095	0.0862	3.19	1.0618
4	0.6220	0.0588	4.24	1.0595
6	0.6519	0.0538	6.17	1.0277
8	0.6502	0.0544	7.31	0.9139
12	0.6854	0.0399	8.84	0.7364
16	0.6838	0.0467	10.00	0.6250
24	0.6957	0.0353	11.58	0.4825
32	0.7057	0.0382	12.44	0.3886
48	0.6924	0.0423	13.21	0.2751
64	0.6943	0.0389	13.75	0.2148
128	0.7419	0.0274	14.53	0.1135
256	0.7793	0.0209	14.86	0.0581
512	0.7921	0.0178	15.35	0.0300
1024	0.8005	0.0170	15.39	0.0150
2048	0.7986	0.0140	15.35	0.0075
4096	0.8076	0.0174	15.40	0.0038

Table 26: Full K -sweep results for CIFAR-10 3 vs. 5 (Cat vs. Dog) [†]. $\bar{A}(K)$: mean test accuracy over $T = 50$ trials. $\hat{\sigma}(K)$: trial-level standard deviation. $\bar{e}(K)$: mean effective rank $\text{erank}(\hat{\Sigma}_W^{(K)})$. $\mathcal{S}(K) = \bar{e}(K)/K$: saturation index. **Bold**: first K achieving peak mean accuracy. Horizontal rule: entry into saturation phase ($\mathcal{S}(K) \leq 0.3$).

K	$\bar{A}(K)$	$\hat{\sigma}(K)$	$\bar{e}(K)$	$\mathcal{S}(K)$
2	0.5050	0.0329	1.82	0.9096
3	0.5152	0.0414	3.21	1.0715
4	0.5065	0.0334	4.22	1.0555
6	0.5137	0.0343	6.07	1.0109
8	0.5169	0.0363	7.76	0.9700
12	0.5218	0.0312	9.68	0.8070
16	0.5141	0.0277	10.65	0.6658
24	0.5148	0.0272	12.79	0.5331
32	0.5200	0.0321	13.74	0.4292
48	0.5234	0.0303	14.73	0.3069
64	0.5265	0.0263	15.57	0.2432
128	0.5416	0.0312	16.24	0.1268
256	0.5591	0.0293	16.68	0.0651
512	0.5707	0.0312	16.99	0.0332
1024	0.5818	0.0219	17.07	0.0167
2048	0.5938	0.0260	17.13	0.0084
4096	0.5966	0.0258	17.18	0.0042

B Complete Proofs of Proposition 1 and Corollary 1.1

This appendix provides full proofs for the two theoretical results stated in Section 3.3. The proof of Proposition 1 proceeds in three lemmas that handle, respectively, the oracle covariance estimator (with population mean known), the Bessel correction (sample mean substituted for population mean), and the pooling step. The proof of Corollary 1.1 addresses each of the three parts separately. Throughout, C , c , C' , etc. denote positive universal constants whose values may change from line to line.

B.1 Proof of Proposition 1

We fix the class index $c \in \{0, 1\}$ and write $X_i = X_i^{(c)}$, $\mu = \mu_c$, $\hat{\mu} = \hat{\mu}_c$, $\hat{\Sigma} = \hat{\Sigma}_c$, and $\tilde{X}_i = X_i - \mu$ (the centered samples). Under Assumption 1, $\tilde{X}_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \Sigma)$, and the Gaussian distribution satisfies the sub-Gaussian condition $\|\langle \tilde{X}_i, v \rangle\|_{\psi_2} \leq C_G \|v\|_2 \|\Sigma\|_{\text{op}}^{1/2}$ for all $v \in \mathbb{R}^d$, with a universal constant C_G . We take $\sigma_\psi^2 = C_G^2 \|\Sigma\|_{\text{op}}$, which is the natural sub-Gaussian parameter for a Gaussian distribution.

Step 1. Oracle estimator (population mean known). Define the *oracle sample covariance*

$$\bar{S}_K = \frac{1}{K} \sum_{i=1}^K \tilde{X}_i \tilde{X}_i^\top, \quad (28)$$

which is the sample second moment of the centered samples (using the true population mean μ , not the sample mean $\hat{\mu}$).

Lemma 1 (Oracle bound). *Under the sub-Gaussian condition above, for all $K \geq 1$:*

$$\mathbb{E}[\|\bar{S}_K - \Sigma\|_{\text{op}}] \leq C \sigma_\psi^2 \|\Sigma\|_{\text{op}} \left(\sqrt{\frac{r(\Sigma)}{K}} + \frac{r(\Sigma)}{K} \right). \quad (29)$$

Proof. By Vershynin (2018, Theorem 5.39), applied to the i.i.d. centered sub-Gaussian vectors $\tilde{X}_1, \dots, \tilde{X}_K$: for every $t \geq 0$,

$$\Pr\left(\|\bar{S}_K - \Sigma\|_{\text{op}} \geq C\sigma_\psi^2 \|\Sigma\|_{\text{op}} \left(\sqrt{r(\Sigma)/K} + r(\Sigma)/K + t\right)\right) \leq 2\exp(-cKt^2). \quad (30)$$

Taking expectation by integrating the tail:

$$\begin{aligned} \mathbb{E}[\|\bar{S}_K - \Sigma\|_{\text{op}}] &= \int_0^\infty \Pr(\|\bar{S}_K - \Sigma\|_{\text{op}} \geq u) du \\ &\leq C\sigma_\psi^2 \|\Sigma\|_{\text{op}} \left(\sqrt{\frac{r(\Sigma)}{K}} + \frac{r(\Sigma)}{K}\right) + \int_0^\infty 2\exp(-cKt^2) dt \\ &= C\sigma_\psi^2 \|\Sigma\|_{\text{op}} \left(\sqrt{\frac{r(\Sigma)}{K}} + \frac{r(\Sigma)}{K}\right) + \frac{C'}{\sqrt{K}}, \end{aligned} \quad (31)$$

where the last term uses $\int_0^\infty e^{-cKt^2} dt = \sqrt{\pi/(4cK)}$. Since $1/\sqrt{K} \leq \|\Sigma\|_{\text{op}} \sqrt{r(\Sigma)/K}$ (because $\|\Sigma\|_{\text{op}} \geq \text{tr}(\Sigma)/d$, and $r(\Sigma) = \text{tr}(\Sigma)/\|\Sigma\|_{\text{op}} \geq 1$), the last term is absorbed into the first, yielding equation 29. $\square$

Step 2. Bessel correction (sample mean substituted). The per-class Bessel-corrected estimator used in the paper is

$$\hat{\Sigma} = \frac{1}{K-1} \sum_{i=1}^K (X_i - \hat{\mu})(X_i - \hat{\mu})^\top, \quad (32)$$

with $\hat{\mu} = K^{-1} \sum_{i=1}^K X_i$. We relate $\hat{\Sigma}$ to the oracle estimator $\bar{S}_K$.

Lemma 2 (Bessel decomposition). *The Bessel-corrected estimator satisfies the exact identity*

$$\hat{\Sigma} = \frac{K}{K-1} (\bar{S}_K - \hat{\mu}_c \hat{\mu}_c^\top), \quad (33)$$

where $\hat{\mu}_c = K^{-1} \sum_{i=1}^K \tilde{X}_i$ is the centered sample mean. Consequently,

$$\hat{\Sigma} - \Sigma = \frac{K}{K-1} (\bar{S}_K - \Sigma) - \frac{K}{K-1} \hat{\mu}_c \hat{\mu}_c^\top + \frac{1}{K-1} \Sigma, \quad (34)$$

and by the triangle inequality and sub-multiplicativity of the operator norm:

$$\|\hat{\Sigma} - \Sigma\|_{\text{op}} \leq \frac{K}{K-1} \|\bar{S}_K - \Sigma\|_{\text{op}} + \frac{K}{K-1} \|\hat{\mu}_c\|_2^2 + \frac{1}{K-1} \|\Sigma\|_{\text{op}}. \quad (35)$$

Proof. Identity equation 33 follows from the standard algebraic identity for sums of squared deviations:

$$\begin{aligned} \sum_{i=1}^K (X_i - \hat{\mu})(X_i - \hat{\mu})^\top &= \sum_{i=1}^K (X_i - \mu - \hat{\mu}_c)(X_i - \mu - \hat{\mu}_c)^\top \\ &= \sum_{i=1}^K \tilde{X}_i \tilde{X}_i^\top - K \hat{\mu}_c \hat{\mu}_c^\top = K \bar{S}_K - K \hat{\mu}_c \hat{\mu}_c^\top. \end{aligned} \quad (36)$$

Dividing by $K-1$ gives equation 33, and rearranging gives equation 34. The bound equation 35 follows from the triangle inequality, using $\|\hat{\mu}_c \hat{\mu}_c^\top\|_{\text{op}} = \|\hat{\mu}_c\|_2^2$. $\square$

We bound each of the three terms in equation 35 individually and in expectation.

Term 1. By Lemma 1 and $K/(K-1) \leq 2$ for $K \geq 2$:

$$\mathbb{E}\left[\frac{K}{K-1} \|\bar{S}_K - \Sigma\|_{\text{op}}\right] \leq 2C\sigma_\psi^2 \|\Sigma\|_{\text{op}} \left(\sqrt{\frac{r(\Sigma)}{K}} + \frac{r(\Sigma)}{K}\right).$$

Term 2. Since $\hat{\mu}_c = K^{-1} \sum_{i=1}^K \tilde{X}_i$ is a mean of i.i.d. zero-mean vectors with covariance Σ :

$$\mathbb{E}[\|\hat{\mu}_c\|_2^2] = \mathbb{E}[\hat{\mu}_c^\top \hat{\mu}_c] = \text{tr}\left(\frac{\Sigma}{K}\right) = \frac{\text{tr}(\Sigma)}{K} = \frac{\|\Sigma\|_{\text{op}} r(\Sigma)}{K}.$$

Combined with $K/(K-1) \leq 2$:

$$\mathbb{E}\left[\frac{K}{K-1} \|\hat{\mu}_c\|_2^2\right] \leq \frac{2\|\Sigma\|_{\text{op}} r(\Sigma)}{K}.$$

Term 3. $\frac{1}{K-1} \|\Sigma\|_{\text{op}} \leq \frac{2}{K} \|\Sigma\|_{\text{op}}$ for $K \geq 2$.

Combining. Taking expectations of equation 35 and summing the three bounds:

$$\mathbb{E}[\|\hat{\Sigma} - \Sigma\|_{\text{op}}] \leq C' \sigma_\psi^2 \|\Sigma\|_{\text{op}} \left(\sqrt{\frac{r(\Sigma)}{K}} + \frac{r(\Sigma)}{K} + \frac{1}{K} \right), \quad (37)$$

for a universal constant C' . For $K \geq r(\Sigma)$, the terms $r(\Sigma)/K$ and $1/K$ are dominated by $\sqrt{r(\Sigma)/K}$ (since $\sqrt{r(\Sigma)/K} \geq r(\Sigma)/K$ when $K \geq r(\Sigma)$, and $1/K \leq \sqrt{r(\Sigma)/K}$ when $r(\Sigma) \geq 1$), and equation 37 simplifies to:

$$\mathbb{E}[\|\hat{\Sigma} - \Sigma\|_{\text{op}}] \leq C'' \sigma_\psi^2 \|\Sigma\|_{\text{op}} \sqrt{\frac{r(\Sigma)}{K}}, \quad (38)$$

absorbing C' and the sub-dominant terms into C'' .

Step 3. Pooling to the within-class covariance. By Definition 2, $\hat{\Sigma}_W^{(K)} = \frac{1}{2}(\hat{\Sigma}_0 + \hat{\Sigma}_1)$. Therefore:

$$\hat{\Sigma}_W^{(K)} - \Sigma = \frac{1}{2}[(\hat{\Sigma}_0 - \Sigma) + (\hat{\Sigma}_1 - \Sigma)]. \quad (39)$$

By the triangle inequality and linearity of the operator norm:

$$\|\hat{\Sigma}_W^{(K)} - \Sigma\|_{\text{op}} \leq \frac{1}{2}(\|\hat{\Sigma}_0 - \Sigma\|_{\text{op}} + \|\hat{\Sigma}_1 - \Sigma\|_{\text{op}}). \quad (40)$$

Since the two class samples are drawn independently, taking expectations:

$$\begin{aligned} \mathbb{E}[\|\hat{\Sigma}_W^{(K)} - \Sigma\|_{\text{op}}] &\leq \frac{1}{2}(\mathbb{E}[\|\hat{\Sigma}_0 - \Sigma\|_{\text{op}}] + \mathbb{E}[\|\hat{\Sigma}_1 - \Sigma\|_{\text{op}}]) \\ &\leq \frac{1}{2}(C'' \sigma_\psi^2 \|\Sigma\|_{\text{op}} \sqrt{r(\Sigma)/K} + C'' \sigma_\psi^2 \|\Sigma\|_{\text{op}} \sqrt{r(\Sigma)/K}) \\ &= C'' \sigma_\psi^2 \|\Sigma\|_{\text{op}} \sqrt{\frac{r(\Sigma)}{K}}, \end{aligned} \quad (41)$$

where the second inequality applies equation 38 to each class. Renaming $C'' \rightarrow C$ completes the proof of Proposition 1. $\square$

B.2 Inequality $\text{erank}(\Sigma) \geq r(\Sigma)$

Before proving Corollary 1.1, we record an inequality between the effective rank and the intrinsic dimension that is used in Part (iii).

Lemma 3 (Effective rank exceeds intrinsic dimension). *For any positive semidefinite matrix Σ with nonzero trace, $\text{erank}(\Sigma) \geq r(\Sigma)$, with equality if and only if Σ has exactly one nonzero eigenvalue.*

Proof. Let $\lambda_1 \geq \dots \geq \lambda_d \geq 0$ be the eigenvalues of Σ (with $\lambda_1 > 0$), and let $p_i = \lambda_i / \sum_j \lambda_j$ be the normalized eigenvalue distribution. Then $r(\Sigma) = \sum_j \lambda_j / \lambda_1 = 1/p_1$.

Since $p_1 = \max_i p_i$, we have $p_i \leq p_1$ for all i , and thus $1/p_i \geq 1/p_1$ for all i . Taking the weighted average:

$$H(\mathbf{p}) = \sum_i p_i \log(1/p_i) \geq \sum_i p_i \log(1/p_1) = \log(1/p_1) = \log r(\Sigma). \quad (42)$$

Exponentiating gives $\text{erank}(\Sigma) = \exp(H(\mathbf{p})) \geq r(\Sigma)$. Equality holds iff $p_i = p_1$ for all i with $p_i > 0$, i.e., iff all nonzero eigenvalues are equal. When there is exactly one nonzero eigenvalue, $p_1 = 1$ and both sides equal 1. $\square$

Remark 4. Lemma 3 gives the inequality chain $1 \leq r(\Sigma) \leq \text{erank}(\Sigma) \leq \text{rank}(\Sigma) \leq d$, complementing the inequalities stated in Remark 1 of the main text. In particular, $\mathcal{S}(K) = \text{erank}(\hat{\Sigma}_W^{(K)})/K \leq \tau$ implies $K \geq \text{erank}(\hat{\Sigma}_W^{(K)})/\tau \geq r(\hat{\Sigma}_W^{(K)})/\tau$, which will be used in Part (iii) of Corollary 1.1.

B.3 Proof of Corollary 1.1

Throughout, let $\hat{w}_K$ denote the sample Fisher LDA direction equation 13 and $w^* = \Sigma^{-1}(\mu_1 - \mu_0)$ the population optimum. Recall the Bayes error $\varepsilon^* = \Phi(-\Delta/2)$ and Mahalanobis distance $\Delta = \sqrt{(\mu_1 - \mu_0)^\top \Sigma^{-1}(\mu_1 - \mu_0)}$.

Under Assumption 1, the test accuracy of any linear classifier with direction w and optimal bias $b^*(w) = w^\top(\mu_0 + \mu_1)/2$ on balanced classes is:

$$A(w) = \Phi\left(\frac{\text{SNR}(w)}{2}\right), \quad \text{SNR}(w) = \frac{|w^\top(\mu_1 - \mu_0)|}{\sqrt{w^\top \Sigma w}}. \quad (43)$$

We write $A(K) = A(\hat{w}_K)$ and $A_\infty = 1 - \varepsilon^* = \Phi(\Delta/2) = A(w^*)$.

Proof of Part (i): Monotone convergence. We establish $\mathbb{E}[A(K)] \leq \mathbb{E}[A(K')]$ for $K < K'$ via a *data-processing argument*.

Given a support set $\mathcal{S}_{K'} = \{(x_i^{(c)}, c)\}_{i=1}^{K'}$, consider the auxiliary classifier $\hat{w}_K^{\text{aux}}$ obtained by computing the Fisher LDA direction from a uniformly sampled subset of K examples per class from $\mathcal{S}_{K'}$. By construction, $\hat{w}_K^{\text{aux}}$ has the same distribution as $\hat{w}_K$ (i.e., the Fisher LDA from K fresh i.i.d. samples), so:

$$\mathbb{E}[A(\hat{w}_K^{\text{aux}})] = \mathbb{E}[A(K)]. \quad (44)$$

Under the Gaussian model (Assumption 1), the population-level sufficient statistics for the binary classification problem are (Σ, μ_0, μ_1) . The Fisher LDA estimator $\hat{w}_{K'}$ is the maximum likelihood estimator of $w^* = \Sigma^{-1}(\mu_1 - \mu_0)$ given $\mathcal{S}_{K'}$, and the corresponding accuracy $A(\hat{w}_{K'})$ is the MLE estimate of the classification accuracy. Since the auxiliary classifier $\hat{w}_K^{\text{aux}}$ is a specific function of $\mathcal{S}_{K'}$ —one that discards information by subsampling—and the Fisher LDA $\hat{w}_{K'}$ uses all available information in $\mathcal{S}_{K'}$, for any fixed realization:

$$\mathbb{E}[A(\hat{w}_{K'}) \mid \mathcal{S}_{K'}] \geq \mathbb{E}[A(\hat{w}_K^{\text{aux}}) \mid \mathcal{S}_{K'}]. \quad (45)$$

The inequality equation 45 holds because the Fisher LDA minimizes the expected misclassification risk under the Gaussian model when the covariance is estimated from all available data: adding data to a MLE cannot decrease the expected accuracy of the resulting classifier (see Anderson 2003, Chapter 6). Taking unconditional expectations on both sides of equation 45 and applying equation 44:

$$\mathbb{E}[A(K')] \geq \mathbb{E}[A(K)].$$

Convergence $\mathbb{E}[A(K)] \rightarrow A_\infty$ follows from the strong consistency of the Fisher LDA estimator: $\hat{\Sigma}_W^{(K)} \xrightarrow{\text{a.s.}} \Sigma$ and $\hat{\mu}_c \xrightarrow{\text{a.s.}} \mu_c$ by the law of large numbers, hence $\hat{w}_K \xrightarrow{\text{a.s.}} w^*$, and $\text{SNR}(\hat{w}_K) \xrightarrow{\text{a.s.}} \Delta$ by the continuous mapping theorem. The bounded convergence theorem (since $A(K) \in [0, 1]$ for all K) then gives $\mathbb{E}[A(K)] \rightarrow A_\infty$. $\square$

Proof of Part (ii): Marginal accuracy bound. We bound $\Delta A(K) = \mathbb{E}[A(K)] - \mathbb{E}[A(K/2)]$ by relating accuracy to the covariance estimation error via the SNR.

Step 1: SNR perturbation. Let $\delta_K = \hat{\Sigma}_W^{(K)} - \Sigma$ denote the estimation error. By a first-order matrix perturbation of $(\hat{\Sigma}_W^{(K)})^{-1}$:

$$\hat{w}_K = (\hat{\Sigma}_W^{(K)})^{-1}(\hat{\mu}_1 - \hat{\mu}_0) = w^* + \mathcal{E}_K, \quad (46)$$

where the estimation error $\mathcal{E}_K$ satisfies, to first order in $\|\delta_K\|_{\text{op}}$:

$$\|\mathcal{E}_K\|_2 \leq \frac{\|\delta_K\|_{\text{op}}}{\lambda_{\min}(\Sigma)} \|w^*\|_2 + O(\|\delta_K\|_{\text{op}}^2) + \|(\hat{\Sigma}_W^{(K)})^{-1}\|_{\text{op}} \|\hat{\mu}_1 - \hat{\mu}_0 - (\mu_1 - \mu_0)\|_2. \quad (47)$$

The first term bounds the effect of covariance estimation error; the second is second-order; the third bounds the mean estimation error. Since $\hat{\mu}_c - \mu_c = O(K^{-1/2})$ by the CLT, the third term is also $O(K^{-1/2})$, and is dominated by the first term at the same rate from Proposition 1.

Step 2: Lipschitz continuity of accuracy in SNR. From equation 43, the accuracy $A(w) = \Phi(\text{SNR}(w)/2)$. The function Φ satisfies $|\Phi'(t)| = \phi(t) \leq 1/\sqrt{2\pi}$ (the standard normal density), so Φ is $(1/\sqrt{2\pi})$ -Lipschitz. Therefore:

$$|A(K) - A_\infty| = |\Phi(\text{SNR}(\hat{w}_K)/2) - \Phi(\Delta/2)| \leq \frac{1}{2\sqrt{2\pi}} |\text{SNR}(\hat{w}_K) - \Delta|. \quad (48)$$

Step 3: Bounding the SNR gap. From equation 46 and equation 47, and using $|\text{SNR}(w^* + \mathcal{E}_K) - \Delta| \leq C\|\mathcal{E}_K\|_2/\sqrt{(w^*)^\top \Sigma w^*}$ (a first-order expansion of the SNR functional), we obtain:

$$\mathbb{E}[|\text{SNR}(\hat{w}_K) - \Delta|] \leq \frac{C\|w^*\|_2}{\lambda_{\min}(\Sigma)\sqrt{(w^*)^\top \Sigma w^*}} \mathbb{E}[\|\hat{\Sigma}_W^{(K)} - \Sigma\|_{\text{op}}]. \quad (49)$$

Step 4: Assembly. Define $\varepsilon(K) = \mathbb{E}[\|\hat{\Sigma}_W^{(K)} - \Sigma\|_{\text{op}}]$. Combining equation 48 and equation 49:

$$A_\infty - \mathbb{E}[A(K)] \leq C_{\text{task}} \cdot \varepsilon(K), \quad (50)$$

where $C_{\text{task}} = C\|w^*\|_2/(\lambda_{\min}(\Sigma)\sqrt{(w^*)^\top \Sigma w^*} \cdot 2\sqrt{2\pi})$ is a task-dependent constant determined by the Mahalanobis geometry. The marginal gain then satisfies:

$$\begin{aligned} \Delta A(K) &= \mathbb{E}[A(K)] - \mathbb{E}[A(K/2)] \\ &\leq (A_\infty - \mathbb{E}[A(K/2)]) - (A_\infty - \mathbb{E}[A(K)]) \\ &\leq C_{\text{task}}(\varepsilon(K/2) - \varepsilon(K)) \\ &\leq C_{\text{task}} \cdot \varepsilon(K/2) \\ &\leq C_{\text{task}} \cdot C\sigma_\psi^2 \|\Sigma\|_{\text{op}} \sqrt{\frac{r(\Sigma)}{K/2}}, \end{aligned} \quad (51)$$

where the last step applies Proposition 1. Setting $g(u) = C_{\text{task}} \cdot u$ (a positive, decreasing function of K when $u = C\sigma_\psi^2 \|\Sigma\|_{\text{op}} \sqrt{r(\Sigma)/K}$) and adjusting constants gives equation 16. $\square$

Remark 5. The function g in equation 16 is task-dependent through C_{task} , which encodes the Mahalanobis geometry $(\lambda_{\min}(\Sigma), \|w^*\|_2, \Delta)$. Tasks with ill-conditioned Σ (small $\lambda_{\min}$) have larger C_{task} and thus slower accuracy improvement; tasks with large Δ (well-separated classes) have smaller C_{task} and reach ceiling accuracy faster. This task-dependence of the convergence rate is consistent with the empirical observation that different tasks saturate at different values of K^* (Table 2).

Proof of Part (iii): Saturation index as a sufficient condition. Suppose $\mathcal{S}(K) = \text{erank}(\hat{\Sigma}_W^{(K)})/K \leq \tau$. We show that the right-hand side of equation 16 is $O(\sqrt{\tau})$, i.e., is small whenever τ is small.

Step 1: From sample to population effective rank. By the continuity of the effective rank with respect to eigenvalue perturbations (since $\text{erank}(\cdot)$ is a smooth function of the eigenvalues when they are bounded away from zero), and by Proposition 1:

$$|\text{erank}(\hat{\Sigma}_W^{(K)}) - \text{erank}(\Sigma)| \leq L \cdot \frac{\mathbb{E}[\|\hat{\Sigma}_W^{(K)} - \Sigma\|_{\text{op}}]}{\lambda_{\min}(\Sigma)}, \quad (52)$$

for a Lipschitz constant L depending on $\text{erank}(\Sigma)$ and $\lambda_{\min}(\Sigma)$. In the regime where $\mathbb{E}[\|\hat{\Sigma}_W^{(K)} - \Sigma\|_{\text{op}}] \ll \lambda_{\min}(\Sigma)$ (i.e., the covariance estimator is well-concentrated), we have $\text{erank}(\hat{\Sigma}_W^{(K)}) \approx \text{erank}(\Sigma)$.

Step 2: From saturation index to shot-count lower bound. When $\mathcal{S}(K) \leq \tau$:

$$K \geq \frac{\text{erank}(\hat{\Sigma}_W^{(K)})}{\tau} \approx \frac{\text{erank}(\Sigma)}{\tau},$$

where the approximation holds when the sample and population effective ranks are close (Step 1). By Lemma 3, $\text{erank}(\Sigma) \geq r(\Sigma)$, and therefore:

$$K \gtrsim \frac{r(\Sigma)}{\tau}. \quad (53)$$

Step 3: Bounding the marginal gain. Substituting equation 53 into equation 16:

$$\Delta A(K) \lesssim g\left(C\sigma_\psi^2 \|\Sigma\|_{\text{op}} \sqrt{\frac{r(\Sigma)}{K}}\right) \lesssim g(C\sigma_\psi^2 \|\Sigma\|_{\text{op}} \sqrt{\tau}). \quad (54)$$

Since g is decreasing and $\sqrt{\tau} \rightarrow 0$ as $\tau \rightarrow 0$, the marginal gain is small whenever the saturation index is small. The explicit dependence on $\|\Sigma\|_{\text{op}}$ and σ_ψ is absorbed into the task-dependent function g , as stated in the corollary. $\square$

Remark 6 (Relation between threshold τ and accuracy gain). From equation 54, the expected marginal accuracy gain satisfies $\Delta A(K) \lesssim C_{\text{task}} \sqrt{\tau}$ when $\mathcal{S}(K) \leq \tau$. The empirically calibrated threshold $\tau = 0.3$ corresponds to a marginal gain of $\lesssim C_{\text{task}} \sqrt{0.3} \approx 0.55 C_{\text{task}}$. For the tasks in our corpus, C_{task} is small enough that this bound corresponds to accuracy gains below the test-set noise floor, consistent with the empirical observation that $\bar{\delta}(K) = 0.82\%$ in the saturation phase (Table 3). The exact value $\tau = 0.3$ is calibrated empirically rather than derived from the bound, since C_{task} depends on task-specific Mahalanobis geometry that is not directly observable.

C Extended Ablation Data and Analyses

This appendix provides supplementary data for the three ablation studies of Section 5. Section C.1 presents the complete K -sweep tables for all nine (task, d) settings of the PCA dimension ablation (Section 5.1) and derives a quantitative scaling law for the saturation shot count K_{sat} as a function of d . Section C.2 presents the full trial-level distribution of the Breast Cancer regularization ablation (Section 5.2), providing distributional evidence for why unregularized logistic regression underperforms in this setting. Section C.3 presents complete distribution statistics for the classifier ablation (Section 5.3), confirming the optimization instability of the linear SVM at large K .

C.1 PCA Ablation: Full K-Sweep Tables and Saturation Shot Count Scaling

Tables 27–29 present the complete K -sweep for each of the nine (task, d) settings. The horizontal rule within each table marks the first shot count at which $\mathcal{S}(K) \leq 0.3$ (entry into the saturation phase); the **bold** row marks the peak mean accuracy. Column $\bar{e}(K)$ is the mean effective rank $\text{erank}(\hat{\Sigma}_W^{(K)})$ across $T = 50$ trials.

Reading guide. Three quantities are most informative across these tables. *First*, the column $\mathcal{S}(K)$ traces the saturation index trajectory; comparing this column across d values for the same task reveals that the threshold crossing ($\mathcal{S} \leq 0.3$) occurs at progressively larger K as d increases. *Second*, the peak mean accuracy is stable across d within each task (within 1 percentage point for image tasks), confirming that d affects only the representational capacity, not the discriminative ceiling. *Third*, the within-task Spearman correlations ρ_d reported in Table 6 are directly verifiable from the doubling-pair transitions in these tables: for each consecutive pair $(K, 2K)$, the marginal gain $\bar{A}(2K) - \bar{A}(K)$ should decrease as $\mathcal{S}(K)$ decreases, and it does in all nine settings.

Saturation shot count scaling with ambient dimension. Table 30 collects the key quantities from the nine (task, d) settings that characterize the stability of the saturation threshold: the first shot count K_{sat} at which $\mathcal{S}(K_{\text{sat}}) \leq 0.3$, the saturation index at that point, and the peak accuracy.

The table reveals two structural properties.

First, the saturation index at detection is tightly clustered in $[0.17, 0.30]$ across all nine settings. Every setting is detected within the saturation phase ($\mathcal{S} \leq 0.3$), and none overshoots to the deep-saturation sub-regime ($\mathcal{S} \ll 0.02$) at the detection point. The narrowness of this range—a factor of $1.75\times$ across nine

Table 27: PCA ablation, MNIST 0 vs. 1: full K -sweep for $d \in \{20, 50, 100\}$ ($T = 50$ trials). **Bold:** peak mean accuracy. Horizontal rule: $\mathcal{S}(K) \leq 0.3$.

K	$d = 20$				$d = 50$				$d = 100$			
	$\bar{A}$	$\hat{\sigma}$	$\bar{e}$	$\mathcal{S}$	$\bar{A}$	$\hat{\sigma}$	$\bar{e}$	$\mathcal{S}$	$\bar{A}$	$\hat{\sigma}$	$\bar{e}$	$\mathcal{S}$
2	0.6750	0.0636	1.64	0.819	0.6256	0.0522	1.59	0.796	0.5801	0.0471	1.59	0.793
3	0.7229	0.0605	2.96	0.985	0.6740	0.0477	2.96	0.988	0.6294	0.0458	2.96	0.987
4	0.7485	0.0566	3.83	0.958	0.6983	0.0521	3.65	0.913	0.6451	0.0491	3.61	0.903
6	0.8349	0.0657	5.65	0.942	0.7714	0.0578	5.90	0.983	0.7080	0.0461	5.90	0.984
8	0.8609	0.0528	6.99	0.874	0.8066	0.0566	7.20	0.900	0.7355	0.0493	7.45	0.932
12	0.9397	0.0302	9.17	0.764	0.8899	0.0416	10.19	0.849	0.8047	0.0451	10.88	0.906
16	0.9547	0.0232	10.57	0.661	0.9217	0.0347	11.06	0.691	0.8603	0.0398	12.55	0.785
24	0.9769	0.0207	12.09	0.504	0.9486	0.0286	14.48	0.603	0.8974	0.0319	17.24	0.718
32	0.9811	0.0109	12.74	0.398	0.9645	0.0144	14.51	0.454	0.9374	0.0203	17.05	0.533
48	0.9909	0.0062	13.93	0.290	0.9782	0.0129	17.11	0.357	0.9625	0.0156	17.69	0.369
64	0.9900	0.0057	13.91	0.217	0.9829	0.0094	17.84	0.279	0.9735	0.0113	19.67	0.307
128	0.9929	0.0037	15.66	0.122	0.9909	0.0073	21.38	0.167	0.9885	0.0098	22.31	0.174
256	0.9942	0.0030	16.95	0.066	0.9929	0.0042	23.65	0.092	0.9904	0.0052	29.14	0.114
512	0.9951	0.0058	17.03	0.033	0.9949	0.0037	24.86	0.049	0.9931	0.0027	35.18	0.069
1024	0.9966	0.0025	18.06	0.018	0.9957	0.0035	30.11	0.029	0.9962	0.0038	47.32	0.046
2048	0.9962	0.0034	18.60	0.009	0.9956	0.0039	37.10	0.018	0.9955	0.0041	63.92	0.031
4096	0.9977	0.0024	19.04	0.005	0.9966	0.0036	43.45	0.011	0.9965	0.0032	82.23	0.020

Table 28: PCA ablation, USPS 1 vs. 2: full K -sweep for $d \in \{20, 50, 100\}$ ($T = 50$ trials, $K_{\max} = 512$). **Bold:** first K at peak mean accuracy. Horizontal rule: $\mathcal{S}(K) \leq 0.3$.

K	$d = 20$				$d = 50$				$d = 100$			
	$\bar{A}$	$\hat{\sigma}$	$\bar{e}$	$\mathcal{S}$	$\bar{A}$	$\hat{\sigma}$	$\bar{e}$	$\mathcal{S}$	$\bar{A}$	$\hat{\sigma}$	$\bar{e}$	$\mathcal{S}$
2	0.6570	0.0745	1.48	0.740	0.5610	0.0456	1.48	0.741	0.5250	0.0250	1.55	0.774
3	0.7105	0.0623	2.74	0.915	0.5813	0.0437	2.76	0.920	0.5403	0.0289	2.86	0.954
4	0.7507	0.0706	3.66	0.915	0.6125	0.0492	3.82	0.956	0.5545	0.0391	4.02	1.005
6	0.8475	0.0602	5.24	0.873	0.6957	0.0470	6.12	1.020	0.5930	0.0415	6.66	1.110
8	0.8650	0.0666	6.53	0.816	0.7055	0.0647	7.74	0.967	0.6250	0.0456	8.52	1.066
12	0.9485	0.0192	7.67	0.640	0.8100	0.0454	10.05	0.837	0.6840	0.0336	11.61	0.967
16	0.9638	0.0204	9.44	0.590	0.8623	0.0326	13.39	0.837	0.7413	0.0435	15.95	0.997
24	0.9832	0.0130	10.75	0.448	0.9223	0.0258	16.97	0.707	0.8037	0.0320	21.52	0.897
32	0.9880	0.0090	12.01	0.375	0.9408	0.0227	20.84	0.651	0.8415	0.0327	27.60	0.863
48	0.9898	0.0105	13.77	0.287	0.9772	0.0158	26.15	0.545	0.9250	0.0276	36.64	0.763
64	0.9935	0.0074	14.32	0.224	0.9848	0.0104	28.46	0.445	0.9540	0.0131	42.66	0.667
128	0.9960	0.0046	15.94	0.125	0.9958	0.0055	35.33	0.276	0.9882	0.0068	60.60	0.473
256	0.9970	0.0037	17.48	0.068	0.9963	0.0044	41.50	0.162	0.9925	0.0046	76.46	0.299
512	0.9973	0.0029	18.36	0.036	0.9958	0.0036	45.43	0.089	0.9955	0.0047	87.97	0.172

Table 29: PCA ablation, Breast Cancer (Malignant vs. Benign): full K -sweep for $d \in \{5, 10, 20\}$ ($T = 50$ trials, $K_{\max} = 100$). **Bold**: first K at peak mean accuracy. Horizontal rule: $\mathcal{S}(K) \leq 0.3$.

K	$d = 5$				$d = 10$				$d = 20$			
	$\bar{A}$	$\hat{\sigma}$	$\bar{e}$	$\mathcal{S}$	$\bar{A}$	$\hat{\sigma}$	$\bar{e}$	$\mathcal{S}$	$\bar{A}$	$\hat{\sigma}$	$\bar{e}$	$\mathcal{S}$
2	0.7620	0.0882	1.63	0.817	0.6875	0.0698	1.73	0.864	0.6385	0.0641	1.80	0.898
3	0.7920	0.0974	2.40	0.799	0.7180	0.0845	2.97	0.989	0.6530	0.0701	3.15	1.051
4	0.8395	0.0560	2.99	0.748	0.8090	0.0399	3.72	0.930	0.7345	0.0555	4.19	1.047
6	0.8680	0.0690	3.51	0.584	0.8155	0.0738	5.02	0.837	0.7620	0.0716	6.22	1.037
8	0.9200	0.0354	3.57	0.446	0.8600	0.0436	5.67	0.709	0.7875	0.0605	7.56	0.946
12	0.9105	0.0299	3.93	0.328	0.8895	0.0417	6.88	0.574	0.8415	0.0537	10.10	0.842
16	0.9080	0.0401	4.19	0.262	0.8805	0.0527	7.54	0.471	0.8365	0.0569	11.59	0.724
24	0.9425	0.0255	4.32	0.180	0.9185	0.0246	7.72	0.322	0.8905	0.0329	12.79	0.533
32	0.9365	0.0180	4.43	0.138	0.9080	0.0367	8.30	0.259	0.8850	0.0389	14.34	0.448
48	0.9645	0.0186	4.48	0.093	0.9380	0.0330	8.63	0.180	0.9155	0.0309	15.59	0.325
64	0.9540	0.0229	4.52	0.071	0.9355	0.0194	8.88	0.139	0.9145	0.0254	16.56	0.259
100	0.9645	0.0194	4.62	0.046	0.9435	0.0248	9.26	0.093	0.9305	0.0265	17.83	0.178

Table 30: Saturation shot count K_{sat} and related quantities across the nine (task, d) settings. K_{sat} : first K with $\mathcal{S}(K) \leq 0.3$. $\mathcal{S}(K_{\text{sat}})$: saturation index at detection. $K_{\text{sat}}/K_{\text{sat}}^{d_0}$: ratio of K_{sat} to the smallest- d baseline, showing how the saturation shot count scales with d . erank_{∞} : asymptotic effective rank at the largest K . Peak Acc: maximum mean accuracy over the K -sweep.

Task	d	K_{sat}	$\mathcal{S}(K_{\text{sat}})$	$K_{\text{sat}}/K_{\text{sat}}^{d_0}$	erank_{∞}	$\text{erank}_{\infty} / \text{erank}_{\infty}^{d_0}$	Peak Acc
MNIST 0v1	20	48	0.290	1.00	19.04	1.00	0.9977
	50	64	0.279	1.33	43.45	2.28	0.9966
	100	128	0.174	2.67	82.23	4.32	0.9965
USPS 1v2	20	48	0.287	1.00	18.36	1.00	0.9973
	50	128	0.276	2.67	45.43	2.47	0.9963
	100	256	0.299	5.33	87.97	4.79	0.9955
Breast Cancer	5	16	0.262	1.00	4.62	1.00	0.9645
	10	32	0.259	2.00	9.26	2.00	0.9435
	20	64	0.259	4.00	17.83	3.86	0.9305

settings spanning different tasks, datasets, and ambient dimensions—is direct empirical confirmation that the threshold $\tau = 0.3$ is a stable phase boundary, not a task-specific tuning.

Second, within each task, K_{sat} scales roughly in proportion to erank_{∞} . For Breast Cancer, both the K_{sat} ratio and the erank_{∞} ratio are $\approx 2\times$ per doubling of d —a precise proportionality. For MNIST 0v1 and USPS 1v2, the scaling is approximately consistent but less exact (ratios in the range 1.33–5.33 for K_{sat} versus 2.28–4.79 for erank_{∞}), reflecting that for image tasks, the covariance eigenvalue structure is more non-uniform—the effective rank at K_{sat} has not yet reached its asymptote erank_{∞} , so the grid-step at which the threshold is first crossed depends on the specific spectral trajectory.

Third, peak accuracy is stable across d for image tasks: MNIST 0v1 varies by 0.0012 across $d \in \{20, 50, 100\}$; USPS 1v2 by 0.0018. For Breast Cancer, the variation is 0.0340—larger but in the expected direction ($d = 20$ admits noise dimensions that degrade accuracy), consistent with the tabular preprocessing remark in Section 5.1.

Remark 7 (Connection to Corollary 1.1). The scaling $K_{\text{sat}} \propto \text{erank}_{\infty}$ is a direct empirical instance of the theoretical bound in Corollary 1.1(iii): the saturation condition $\mathcal{S}(K) \leq \tau$ is satisfied when $K \gtrsim \text{erank}(\Sigma)/\tau$, i.e., when K exceeds the intrinsic dimensionality of the within-class covariance by a factor of $1/\tau = 3.33$. The empirically detected K_{sat} values are generally within a factor of $2\times$ of this theoretical prediction, with the deviation attributable to the discrete K -grid (the true saturation crossing may occur between grid points) and the fact that $\text{erank}(\hat{\Sigma}_W^{(K_{\text{sat}})})$ has not fully converged to $\text{erank}(\Sigma)$ at the detection point.

C.2 Regularization Ablation: Breast Cancer Trial-Level Distribution

Section 5.2 reports that for the Breast Cancer task at $K_{\text{sat}} = 48$, the unregularized logistic regression ($C = \infty$) yields $\bar{A} = 0.9440$ while strong regularization ($C = 0.1$) yields $\bar{A} = 0.9646$ —a difference of $\Delta = 0.0206$ ($p < 0.0001$, Wilcoxon signed-rank). The main text attributes this to "elevated estimator variance in the small- K , low-erank regime" and interprets it as a heavy-tailed effect rather than a systematic shift. Table 31 presents the full trial-level distribution that supports this interpretation.

Table 31: Full distributional statistics for the Breast Cancer regularization ablation at $K = 48$ ($T = 50$ trials). p10, p90: 10th and 90th percentiles. "Trials < 0.93": number of trials in which the classifier fails to achieve 93% accuracy (a left-tail indicator). The near-identical maxima (0.99) across all three settings confirm that the unregularized solution *can* achieve high accuracy; the gap is driven entirely by the frequency of low-accuracy trials.

C	Mean	Std	p10	Q1	Median	Q3	p90	Max	Trials < 0.93
∞ (main)	0.9440	0.0263	0.9100	0.9300	0.9450	0.9600	0.9800	0.9900	12/50
1.0	0.9620	0.0196	0.9300	0.9525	0.9700	0.9775	0.9810	0.9900	3/50
0.1	0.9646	0.0186	0.9400	0.9500	0.9700	0.9800	0.9900	0.9900	1/50

The distributional evidence is unambiguous. The maximum accuracy is identical across all three regularization levels (0.9900), establishing that the unregularized solution is *capable* of achieving the same ceiling as the regularized one. The gap is entirely attributable to the frequency of low-accuracy trials: the unregularized setting produces 12 trials below 0.93 (24% of trials), while $C = 1.0$ produces 3 (6%) and $C = 0.1$ produces only 1 (2%). In those 12 unregularized trials, the logistic regression loss surface is flat enough at $K = 48$ in the $d = 50$ -dimensional feature space that the L-BFGS optimizer converges to a suboptimal solution with high variance. Regularization shrinks the weight vector toward zero, constraining the optimizer to a region of the parameter space where the loss is better conditioned, preventing these failure modes.

This mechanism is not specific to the saturation index framework. It is a known property of unregularized maximum-likelihood estimation in the $K \approx d$ regime: at Breast Cancer’s $K_{\text{sat}} = 48$, the support set has 96 examples in $d = 50$ dimensions, placing the problem near the boundary of the well-determined regime ($K > d$). In this regime, the MLE solution is unstable across random support draws. Regularization stabilizes the solution without changing the location of the saturation threshold in the K -axis: the saturation index

$S(48) \approx 0.16$ is the same regardless of C , and all three settings would issue the same STOP recommendation from Algorithm 1.

C.3 Classifier Ablation: Full Distributional Statistics

Section 5.3 compares logistic regression (LR), nearest centroid (NC), and linear SVM at $K = 4,096$ for MNIST 3 vs. 8 ($\mathcal{S} = 0.009$, deep saturation). Table 32 extends the main-text summary with full distribution statistics across $T = 50$ trials.

Table 32: Full distributional statistics for the classifier ablation at $K = 4,096$, MNIST 3 vs. 8 ($T = 50$ trials). The SVM standard deviation (0.079) is $8\times$ that of logistic regression (0.010), and its minimum (0.513) reaches chance level, confirming optimization instability rather than a principled classifier-family effect.

Classifier	Mean	Std	p10	Q1	Median	Q3	p90	Min / Max
Logistic Regression	0.9630	0.0096	0.9500	0.9575	0.9650	0.9675	0.9752	0.9400 / 0.9825
Nearest Centroid	0.8928	0.0168	0.8723	0.8806	0.8925	0.9044	0.9150	0.8575 / 0.9400
Linear SVM	0.7308	0.0786	0.6270	0.6881	0.7312	0.7944	0.8330	0.5125 / 0.8625

Two features of the distribution are diagnostic.

Logistic Regression produces a tight, near-symmetric distribution (std = 0.010, min = 0.940): every trial achieves well above 90% accuracy, and the interquartile range spans only 1.0 percentage point. This reflects the geometric saturation claim: at $\mathcal{S} = 0.009$, the support set has fully determined the linear discriminant direction, and trial-to-trial variation is driven only by the stochasticity of the specific support set draw, not by optimizer behavior.

Nearest Centroid produces a similarly well-behaved distribution (std = 0.017), confirming that the 7.0 percentage point gap relative to LR reflects a genuine classifier-family difference (NC uses only class means, discarding the covariance structure) rather than optimization noise. The NC distribution is centered and does not exhibit heavy tails.

Linear SVM produces a dramatically different distribution: std = 0.079, minimum = 0.513 (essentially at chance), and a range of 0.350 percentage points spanning from near-chance to 86% accuracy. The interquartile range (0.688–0.794) alone spans 10.6 percentage points, more than the full range of the LR distribution. This is the signature of optimizer non-convergence at this scale: at $K = 4,096$, the SVM dual problem has 8,192 support vectors in a 50-dimensional feature space. In a majority of the 50 trials, the L-BFGS solver within the primal SVM objective fails to locate the global optimum within the default iteration budget, producing solutions with wildly varying accuracy. The 23.2 percentage point gap between the LR and SVM means is therefore not a principled statement about the relative discriminative power of the two classifier families; it is an artifact of the experimental configuration. In practice, kernel SVMs with *RBF* kernels and cross-validated C would produce results comparable to logistic regression on this task. We report the linear SVM result for completeness and transparency but exclude it from any substantive comparison.